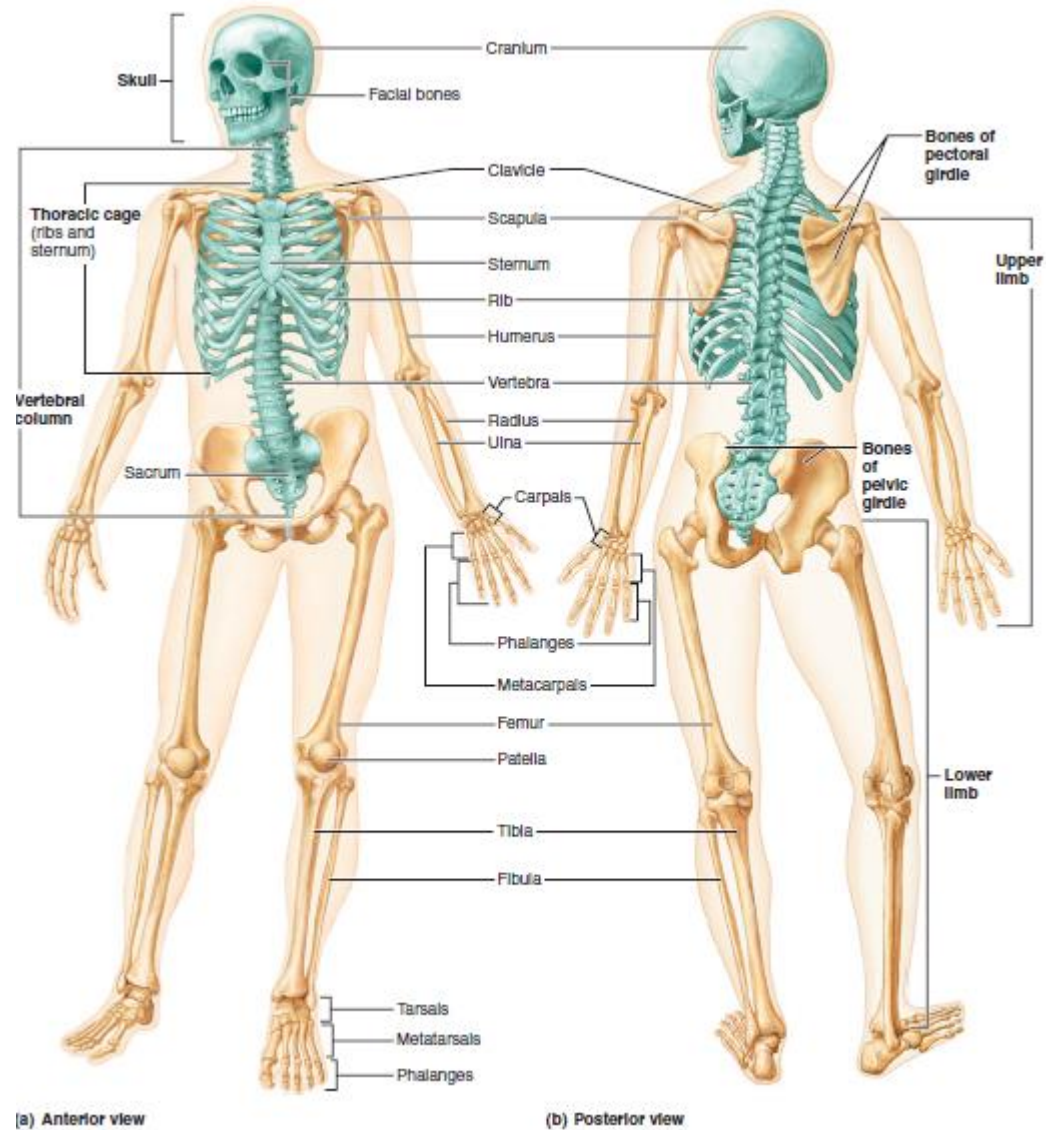


Musculoskeletal

and

Integumentary system module!

Musculoskeletal and Integumentary system module!



Skeletal muscle

Temporalis

Masseter

Sternocleidomastoid

Pectoralis major

Biceps brachii

Abdominal muscles

Sartorius

Quadriceps femoris

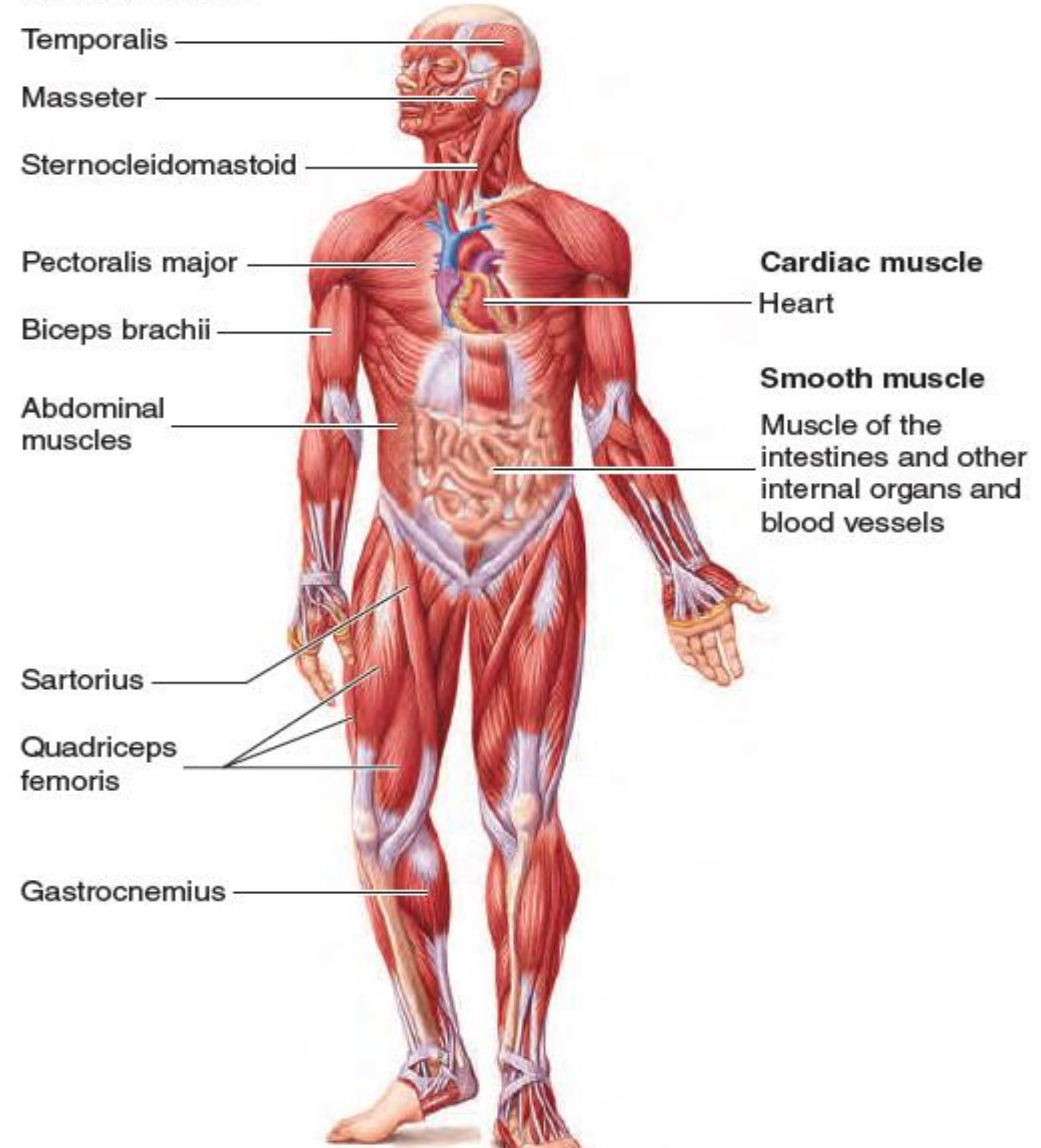
Gastrocnemius

Cardiac muscle

Heart

Smooth muscle

Muscle of the intestines and other internal organs and blood vessels



• **Anatomy of upper limb:**

- ✓ Introduction to the musculoskeletal system
- ✓ Introduction to the upper limb
- ✓ Bones of the upper limb
- ✓ Superficial structures of upper limb
- ✓ Axioappendicular Muscles
- ✓ Axilla
- ✓ Brachial plexus
- ✓ Arm
- ✓ Cubital fossa
- ✓ Forearm
- ✓ Hand
- ✓ Surface anatomy the upper limb
- ✓ Joints of the upper limb

Introduction to the musculoskeletal system

- The musculoskeletal system (locomotor system) is a human body system that provides our body with movement, stability, shape, and support.

❑ **It is subdivided into two broad systems:**

- ✓ **Muscular system**, which includes all types of voluntary muscles in the body.
- ✓ **Skeletal system**, whose main component is the bone.
- The integrity and function of the bones and joints is supported by the accessory structures of the skeletal system; articular cartilage, ligaments, and bursae

Introduction to the musculoskeletal system,,,

- The muscular system is an organ system composed of specialized contractile tissue called the muscle tissue.

□ There are three types of muscle tissue:

- I. Cardiac muscle, which forms the muscular layer of the heart (myocardium).
 - II. Smooth muscle, which comprises the walls of blood vessels and hollow organs.
 - III. Skeletal muscle, which attaches to the bones and provides voluntary movement.
- Based on their histological appearance, these types are classified into striated and non-striated muscles; with the skeletal and cardiac muscles being grouped as striated, while the smooth muscle is non-striated..

Introduction to the musculoskeletal system,,,

- The skeletal muscles are the main functional units of the muscular system. There are more than 600 muscles in the human body.
- ❑ **The skeletal muscles of the body are organized into 4 groups based on their location on the regions of the body:**
- I. Muscles of the head and neck**, which include the muscles of the facial expression, muscles of mastication, muscles of the orbit, muscles of the tongue, muscles of the pharynx, muscles of the larynx, and muscles of the neck
 - II. Muscles of the trunk**, which include the muscles of the back, anterior and lateral abdominal muscles, and muscles of the pelvic floor
 - III. Muscles of the upper limbs**, which include muscles of the shoulder, muscles of the arm, muscles of the forearm and muscles of the hand
 - IV. Muscles of the lower limbs**, which include hip and thigh muscles, leg muscles and foot muscles.

Introduction to the musculoskeletal system,,

- **The skeletal system** is composed of the bones, cartilages, and joints that form the internal framework of the body.
- **Although the term *skeleton* is derived from a Greek word meaning dried, but the skeletal** system consists of dynamic, living tissues that are able to grow, detect pain stimuli, adapt to stress, and undergo repair after injury.
- Bones provide support and shape to the body, attachment sites for muscle, and a storage depot for essential minerals.

Introduction to the musculoskeletal system,,,

- Cartilage is a major component of many joints.
- Cartilage and bone are also linked developmentally; most bones are first formed in cartilage tissue.
- Bone tissue, the main part of the skeletal system is a complex and dynamic living tissue, continually engages in a process called remodeling the construction of new bone tissue and breaking down of old bone tissue.
- ❖ **Two very strong tissues**, bone and cartilage, along with ligaments and tendons, constitutes the **skeletal system**.

Introduction to the musculoskeletal system,,,

- Knowledge of skeletal anatomy will be useful for medical professions.
- Many organs are named for their relationships to nearby bones, example **subclavian vessels, temporalis muscle, the ulnar nerve and radial artery, and the frontal, parietal, temporal, and occipital lobes of the brain.**
- Understanding how the muscles produce body movements also depends on knowledge of skeletal anatomy.

Introduction to the musculoskeletal system,,,

- Additionally, the positions, shapes, and processes of bones can serve as **landmarks for clinicians** in determining where to give an injection or record a pulse, what to look for in an X-ray, and how to perform physical therapy and other clinical procedures.
- Together, the bones, muscles, and joints form an integrated system called the **musculoskeletal system**.
- The branch of medical science concerned with the prevention or correction of disorders of the musculoskeletal system is called **orthopedics**.

Introduction to the musculoskeletal system,,,

The **Skeletal System** has many important functions:

- Providing support for the body
- Storing minerals (calcium, phosphate)
- Producing red blood cells
- Protecting the organs and tissues
- Allowing movement (the bones act as levers)



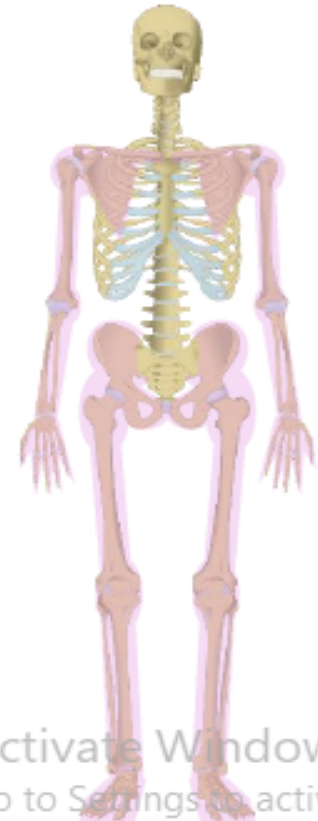
The skeleton can be subcategorized into two divisions:

The ***Axial Skeleton*** (left, in blue)

Includes: Bones of the skull, vertebrae, sternum, ribs, and sacrum

The ***Appendicular Skeleton*** (right, in pink)

Includes: Bones of the upper and lower limbs, scapula, clavicles, and hip bones,



Introduction to the musculoskeletal system,,,

- The adult human skeleton consists of 206 named bones, most of which are paired, with one member of each pair on the right and left sides of the body.
- The skeletons of infants and children have more than 206 bones because some of their bones (sacrum and coccyx) fuse later in life.
- **So it is traditional to list 206 bones in the average adult skeleton, although the actual number varies from person to person and decreases with age as some bones fuse.**
- Bones of the adult skeleton are grouped into two principal divisions:
 - ❖ The **axial skeleton** and the **appendicular skeleton** presents the 80 bones of the axial skeleton and the 126 bones of the appendicular skeleton.

Introduction to the musculoskeletal system...

The Bones of the Adult Skeletal System

DIVISION OF THE SKELETON	STRUCTURE	NUMBER OF BONES
--------------------------	-----------	-----------------

Axial skeleton



Skull	
Cranium	8
Face	14
Hyoid	1
Auditory ossicles	6
Vertebral column	26
Thorax	
Sternum	1
Ribs	24

Number of bones = 80

DIVISION OF THE SKELETON	STRUCTURE	NUMBER OF BONES
--------------------------	-----------	-----------------

Appendicular skeleton



Pectoral (shoulder) girdles	
Clavicle	2
Scapula	2
Upper limbs	
Humerus	2
Ulna	2
Radius	2
Carpals	16
Metacarpals	10
Phalanges	28
Pelvic (hip) girdle	
Hip, pelvic, or coxal bone	2
Lower limbs	
Femur	2
Patella	2
Fibula	2
Tibia	2
Tarsals	14
Metatarsals	10
Phalanges	28

Number of bones = 126

Total bones in an adult skeleton = 206

Named Bones in the Adult Human Skeleton

Bones		Number	Bones		Number
Axial Skeleton			Thoracic Cage		
Skull			Ribs		24
Braincase			Sternum (3 parts, sometimes considered 3 bones)		1
Paired	Parietal	2	TOTAL THORACIC CAGE		25
	Temporal	2	TOTAL AXIAL SKELETON		
Unpaired	Frontal	1			80
	Occipital	1	Appendicular Skeleton		
	Sphenoid	1	Pectoral Girdle		
	Ethmoid	1	Scapula		2
Face			Clavicle		2
Paired	Maxilla	2	Upper Limb		
	Zygomatic	2	Humerus		2
	Palatine	2	Ulna		2
	Nasal	2	Radius		2
	Lacrimal	2	Carpal bones		16
	Inferior nasal concha	2	Metacarpal bones		10
Unpaired	Mandible	1	Phalanges		28
	Vomer	1	TOTAL GIRDLE AND UPPER LIMB		
	TOTAL SKULL	22			64
Auditory Ossicles			Pelvic Girdle		
	Malleus	2	Coxal bone		2
	Incus	2	Lower Limb		
	Stapes	2	Femur		2
	TOTAL	6	Tibia		2
Hyoid			Fibula		2
		1	Patella		2
Vertebral Column			Tarsal bones		14
Cervical vertebrae		7	Metatarsal bones		10
Thoracic vertebrae		12	Phalanges		28
Lumbar vertebrae		5	TOTAL GIRDLE AND LOWER LIMB		
Sacrum		1			62
Coccyx		1	TOTAL APPENDICULAR SKELETON		
		1			126
	TOTAL VERTEBRAL COLUMN	26	TOTAL BONES		
					206

Introduction to the upper limb

❑ Upper limb consists of 4 segments

✚ Shoulder (pectoral girdle)

✚ Arm

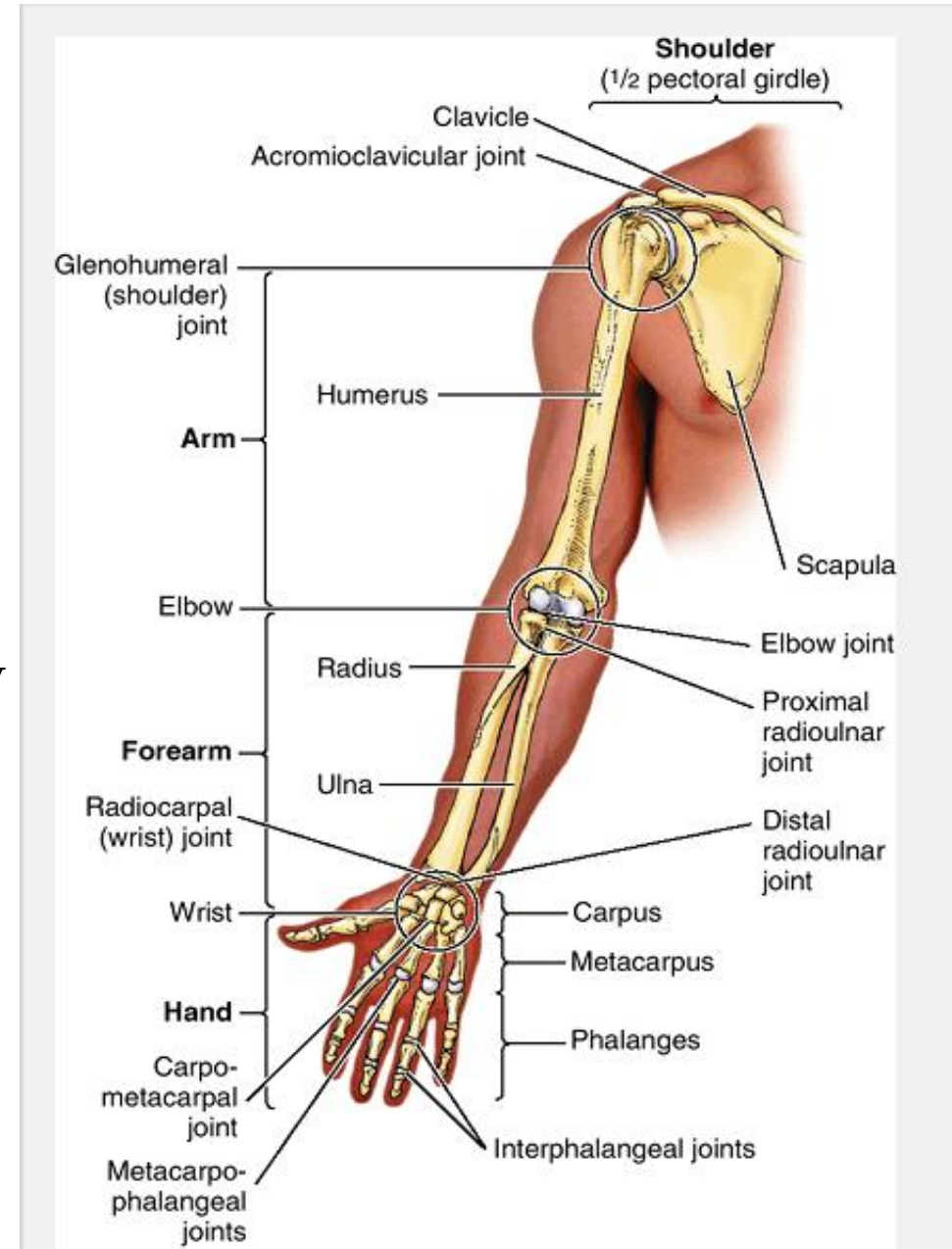
✚ Forearm

✚ Hand

❖ The upper limb is characterized by its mobility and ability to grasp, strike, and conduct fine motor skills (*manipulation*).

❑ The **pectoral girdle** (shoulder girdle) supports the arm and links it to the axial skeleton.

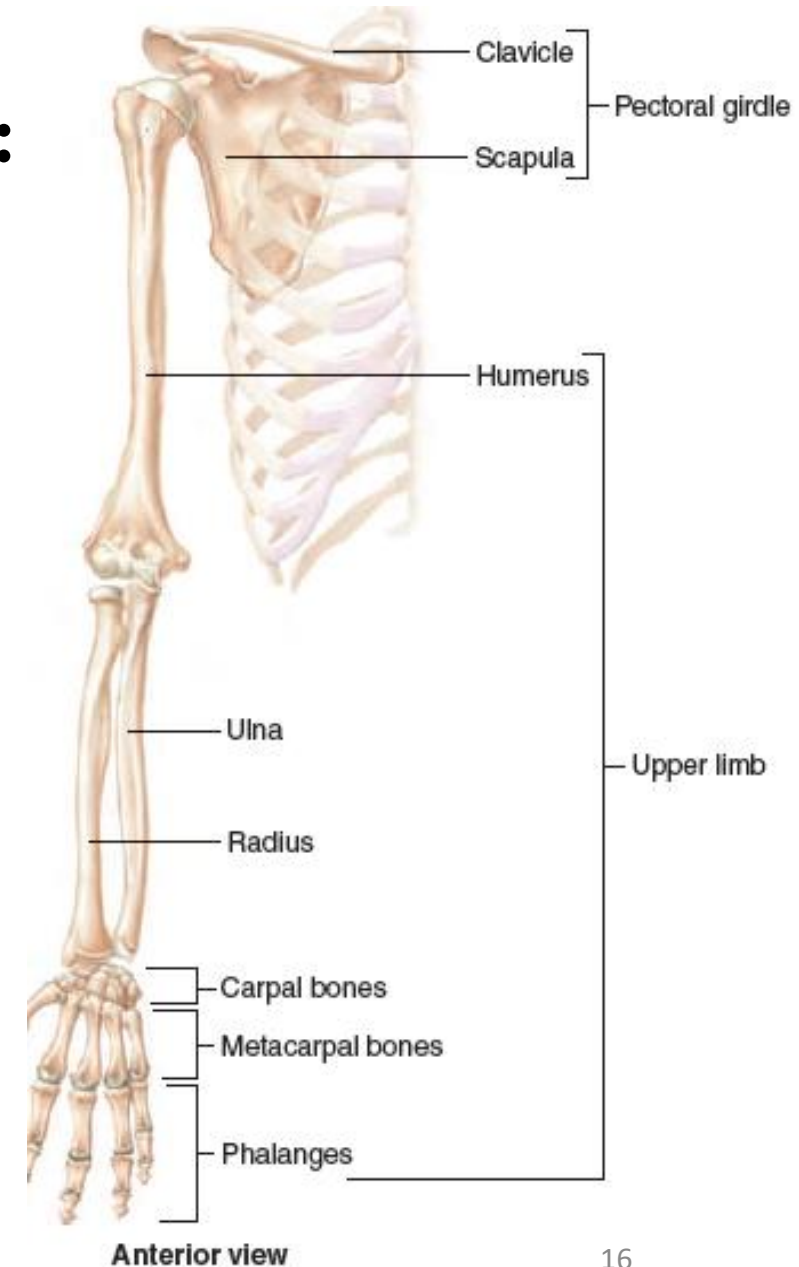
❑ It consists of 2 bones on each side of the body: the *clavicle* & *scapula*.



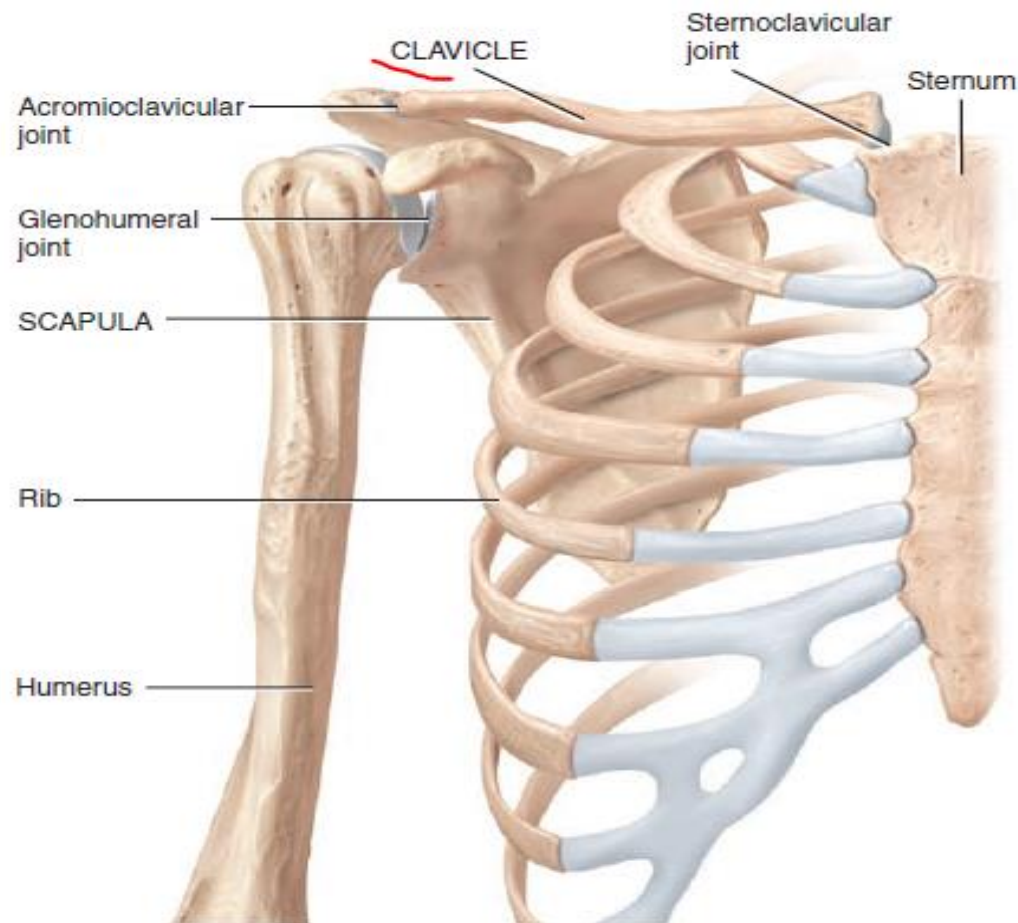
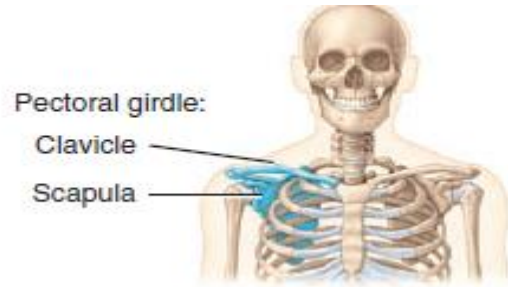
Bones of the upper limb

❑ The skeleton of each upper limb is composed of:

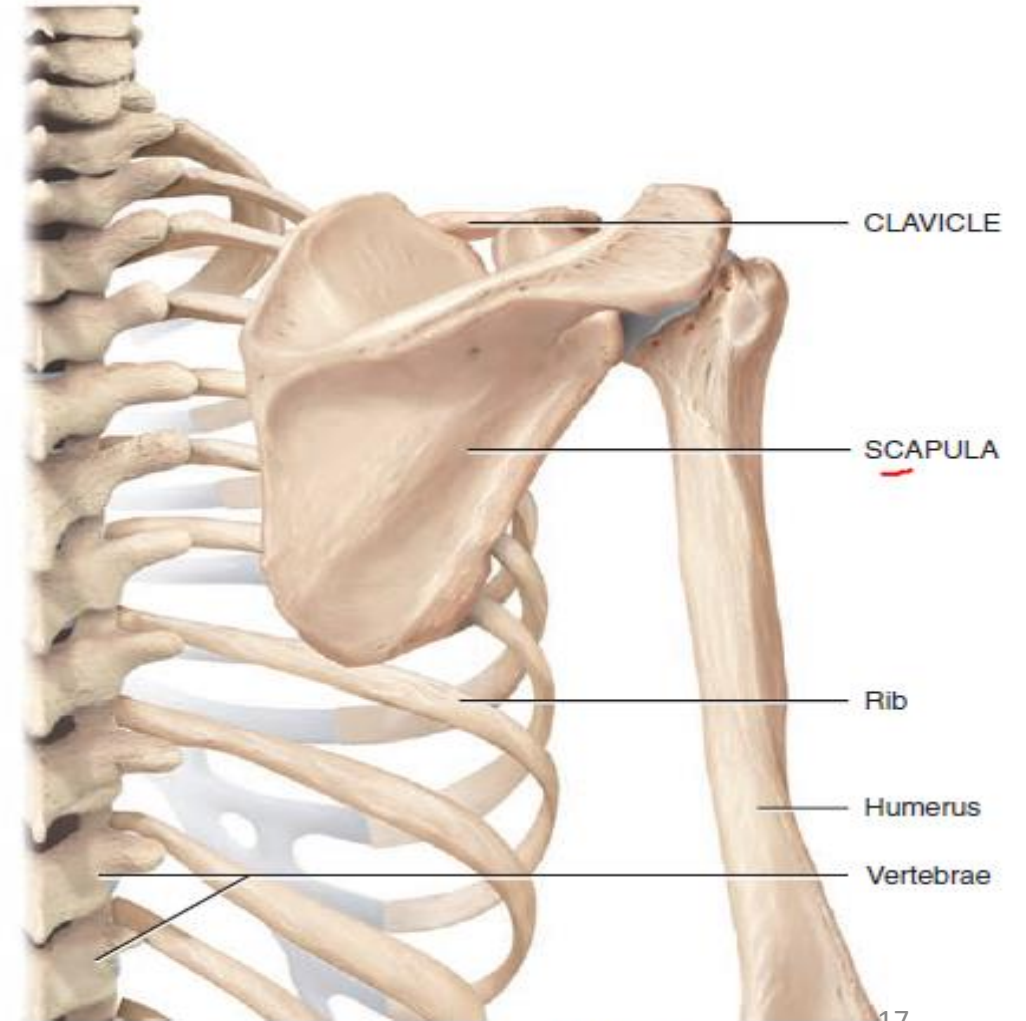
- ✓ clavicle and a scapula in the **pectoral** or **shoulder girdles**
- ✓ humerus in the arm;
- ✓ ulna and radius in the forearm;
- ✓ 8 carpals in the carpus (wrist),
- ✓ 5 metacarpals in the metacarpus (palm), and
- ✓ 14 phalanges (bones of the digits) in the hand.



Bones of the upper limb,,,



(a) Anterior view of pectoral girdle



(b) Posterior view of pectoral girdle

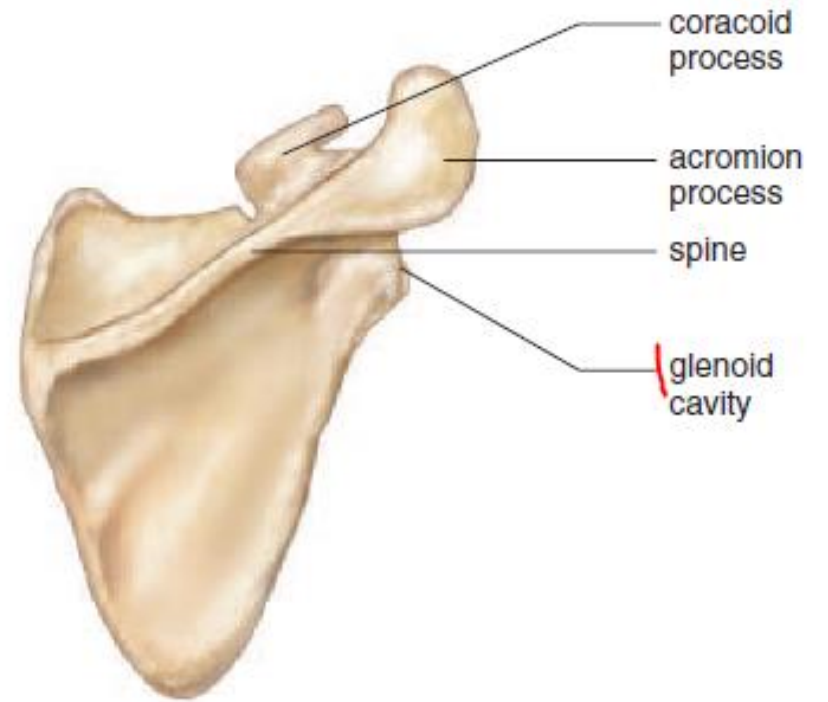
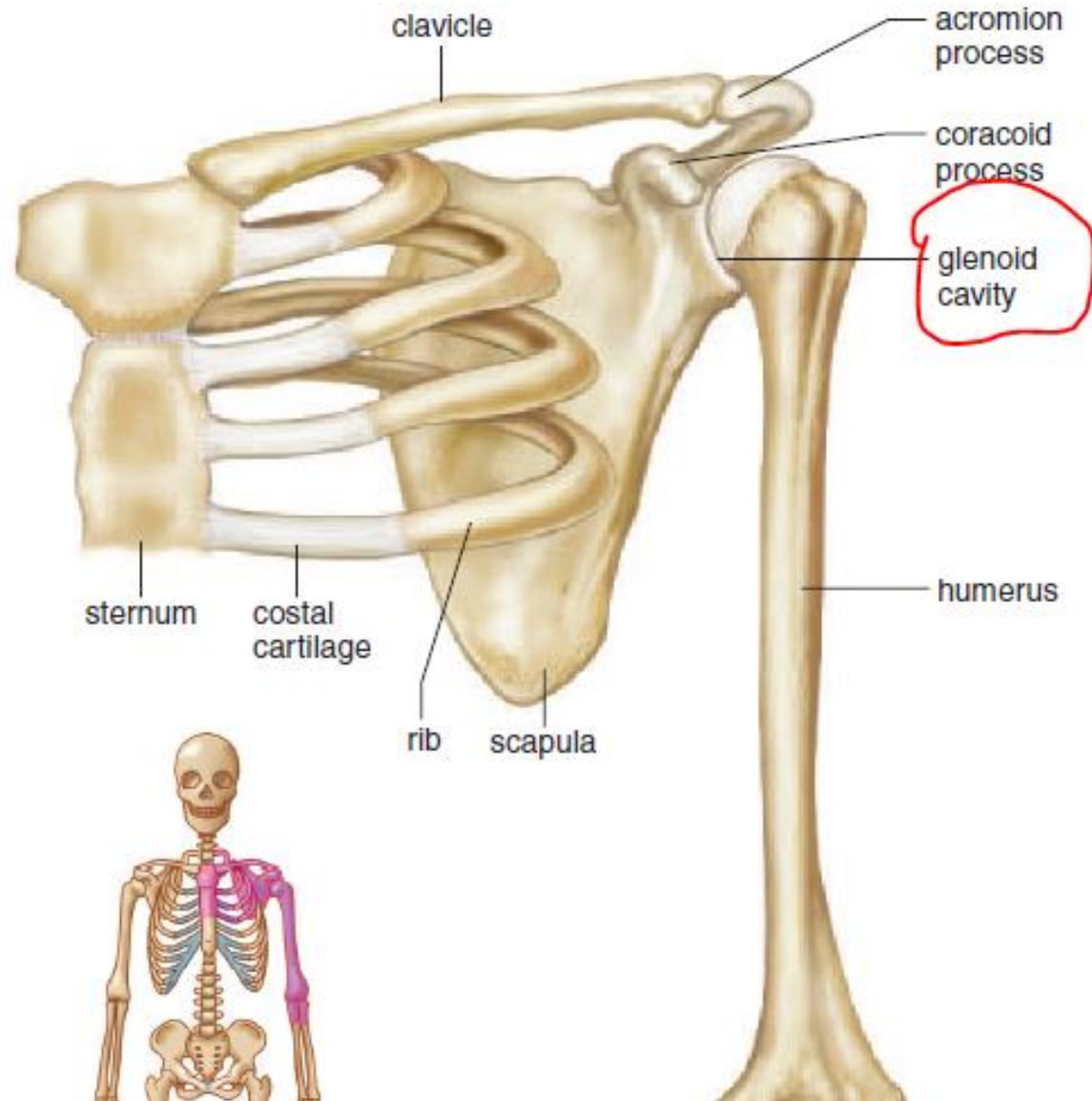
Bones of the upper limb,,

- The two scapulae do not articulate with the vertebral column posteriorly.
- They are held in position and stabilized by a group of large muscles that extend from the vertebral column and ribs to the scapula.
- Unlike to pelvic girdle which is complete ring of bone, the pectoral girdle **is incomplete posteriorly.**
- **This arrangement allows great flexibility but means that the pectoral girdle is prone to dislocation.**
- Besides attaching the upper limb to the trunk, the pectoral girdle provides attachment for many muscles that move the limb.

Bones of the upper limb,,,

- Anatomically there are two factors for great mobility of the upper limbs:
 1. only the clavicle attaches to the axial skeleton, the scapula can move quite freely across the thorax, allowing the arm to move with it.
 2. The socket of the shoulder joint the scapula's glenoid cavity is shallow, so it does not restrict the movement of the humerus.
- Although this arrangement is good for flexibility, it is bad for stability: Shoulder dislocations are fairly common.

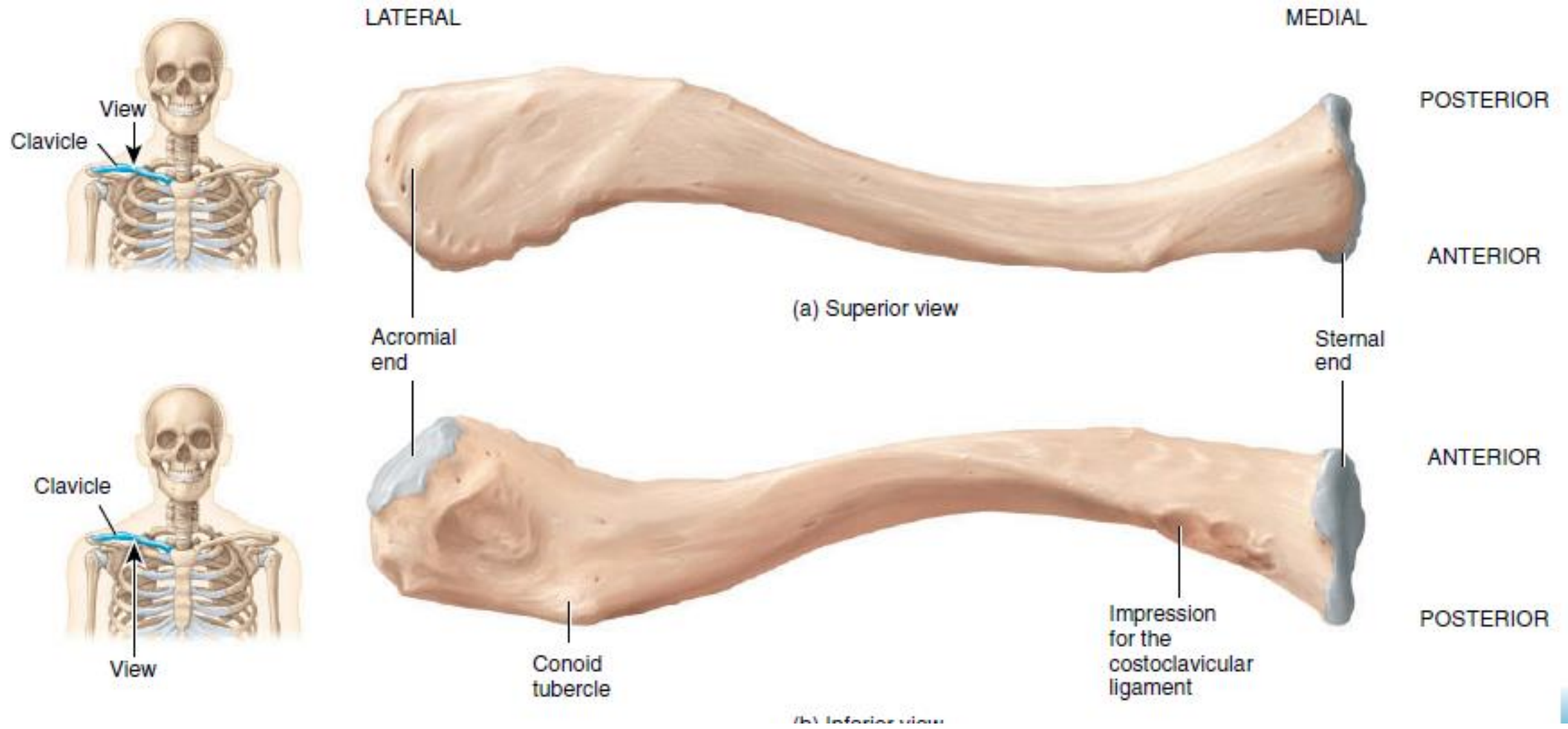
Bones of the upper limb,,



b. Scapula, posterior view

Clavicle (collar bone)

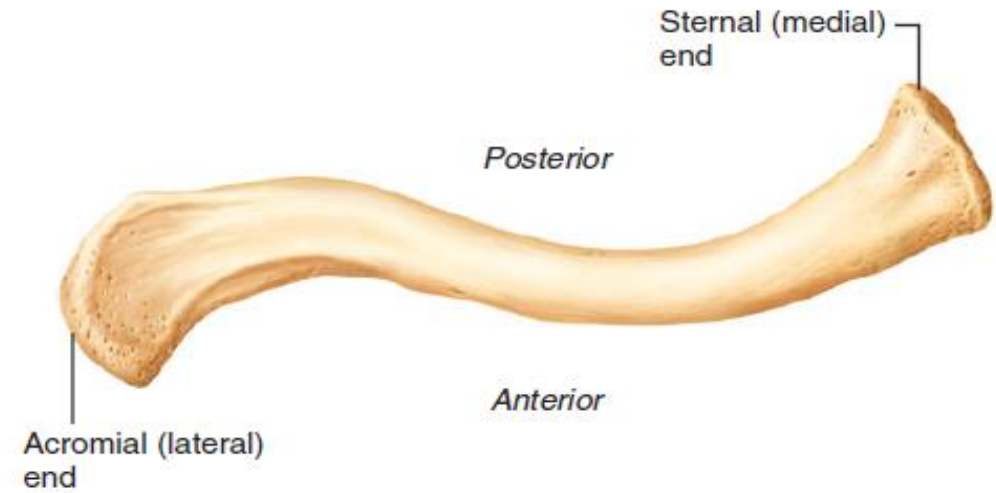
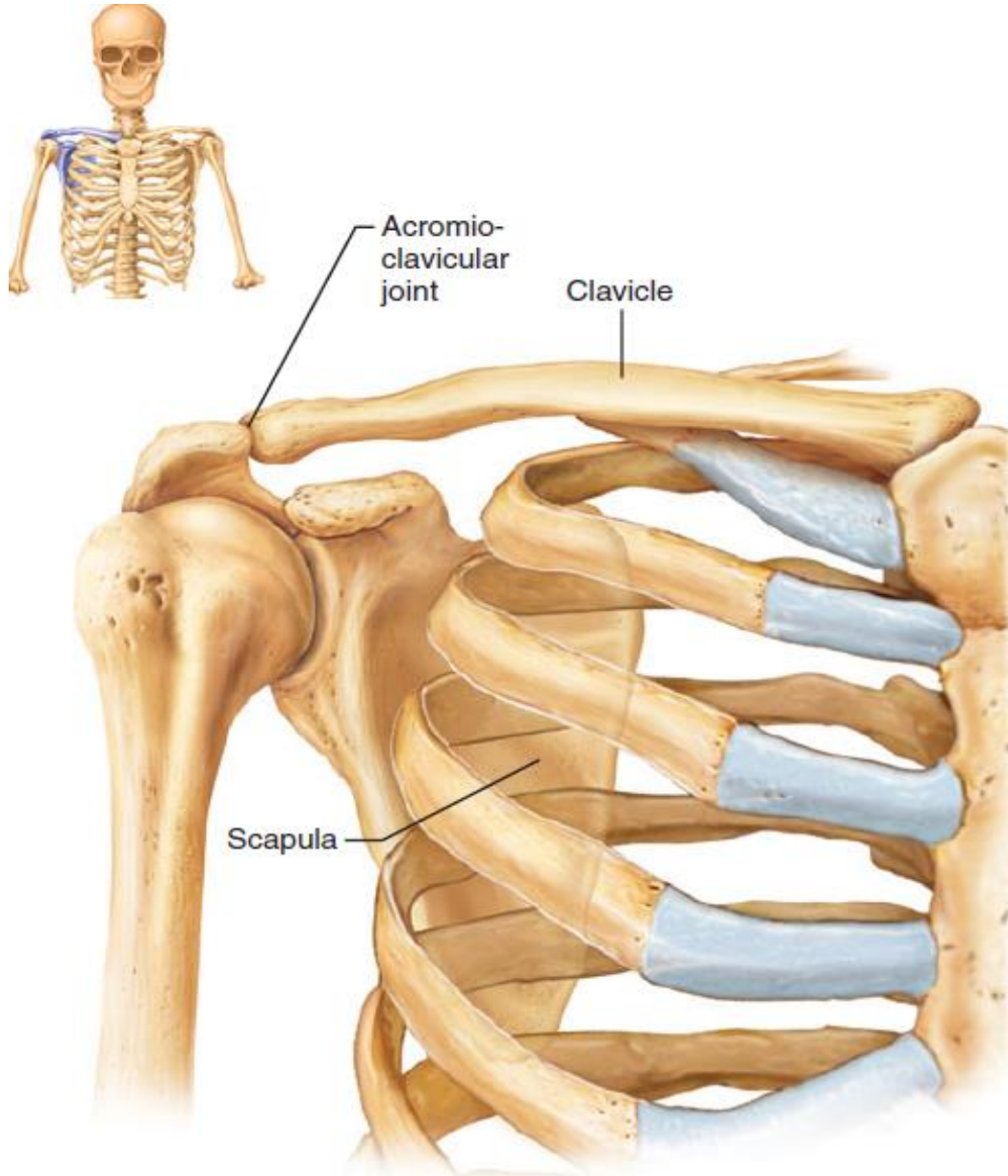
- Each slender, S-shaped **clavicle**(*collarbone*), lies horizontally across the anterior part of the thorax superior to the first rib



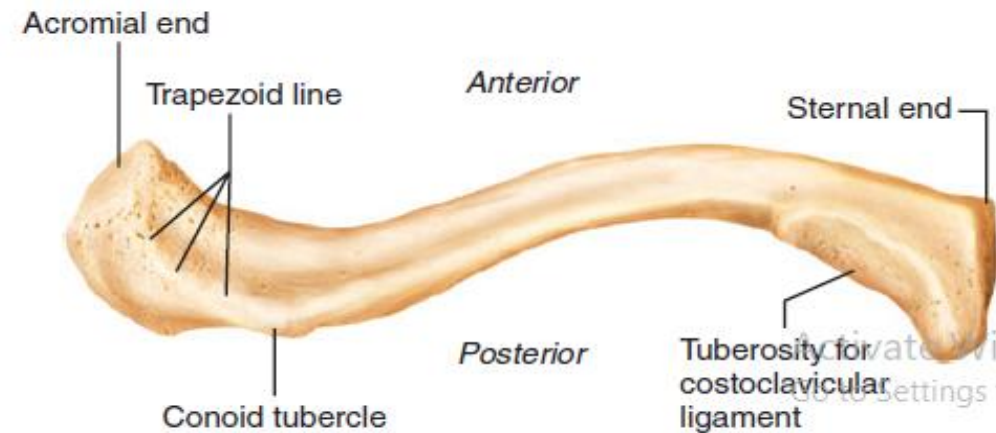
Clavicle (collar bone),,,

- The medial two thirds of the shaft of the clavicle are convex anteriorly, whereas the lateral third is flattened and concave anteriorly.
- **sternal end:** The medial end, called the *sternal end*, is rounded and articulates with the manubrium of the sternum to form the ***sternoclavicular joint***.
- **acromial end:** The broad, flat, lateral end, the *acromial end* , articulates with the acromion of the scapula to form the *acromioclavicular (AC) joint*
- **The *conoid tubercle*** on the inferior surface of the lateral end of the bone is a point of attachment for the conoid ligament, which attaches the clavicle and scapula.
- **Each S-shaped clavicle articulates with the acromion of a scapula laterally and with the sternum medially.**

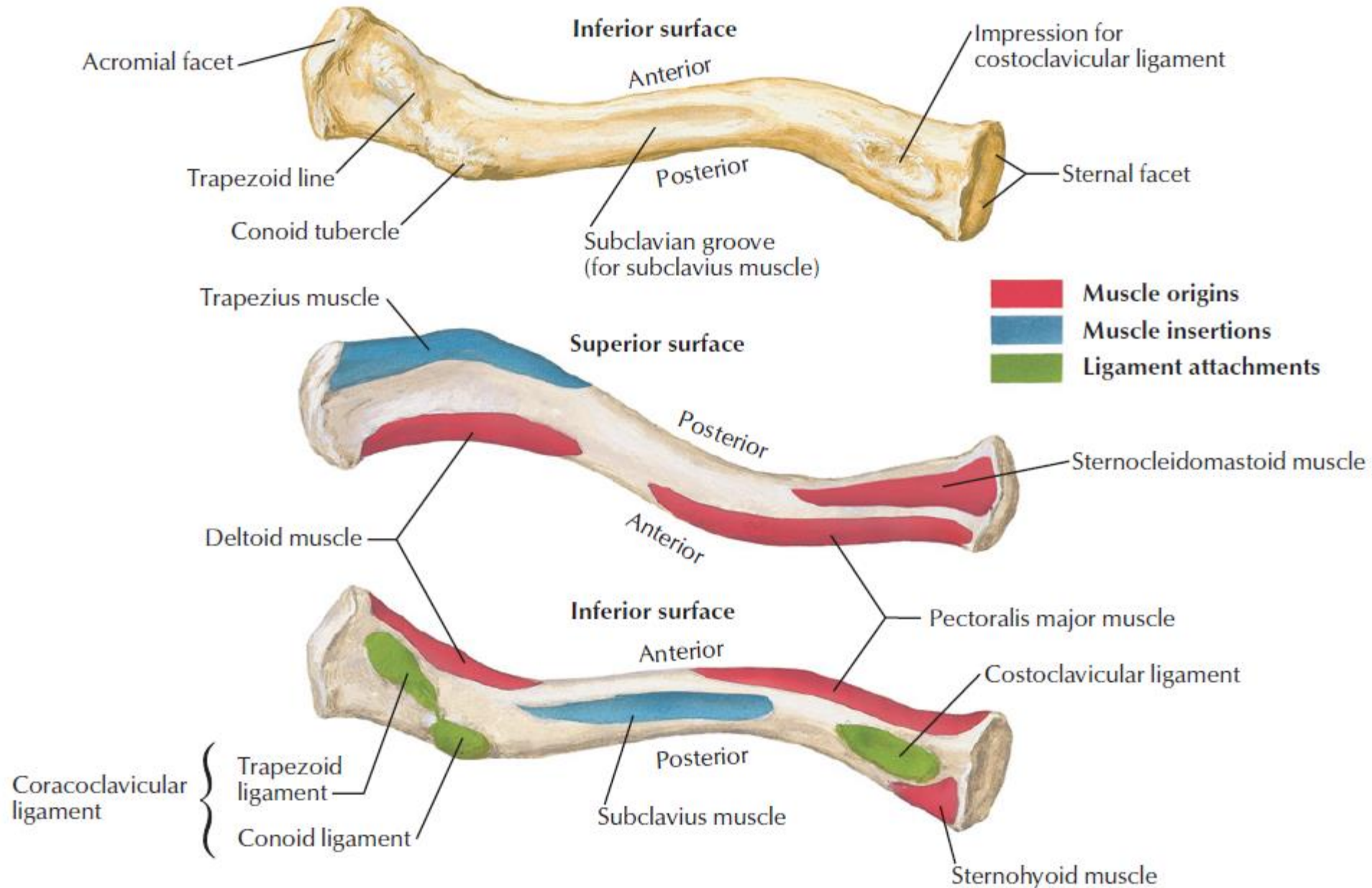
Clavicle (collar bone),,,



(b) Right clavicle, superior view



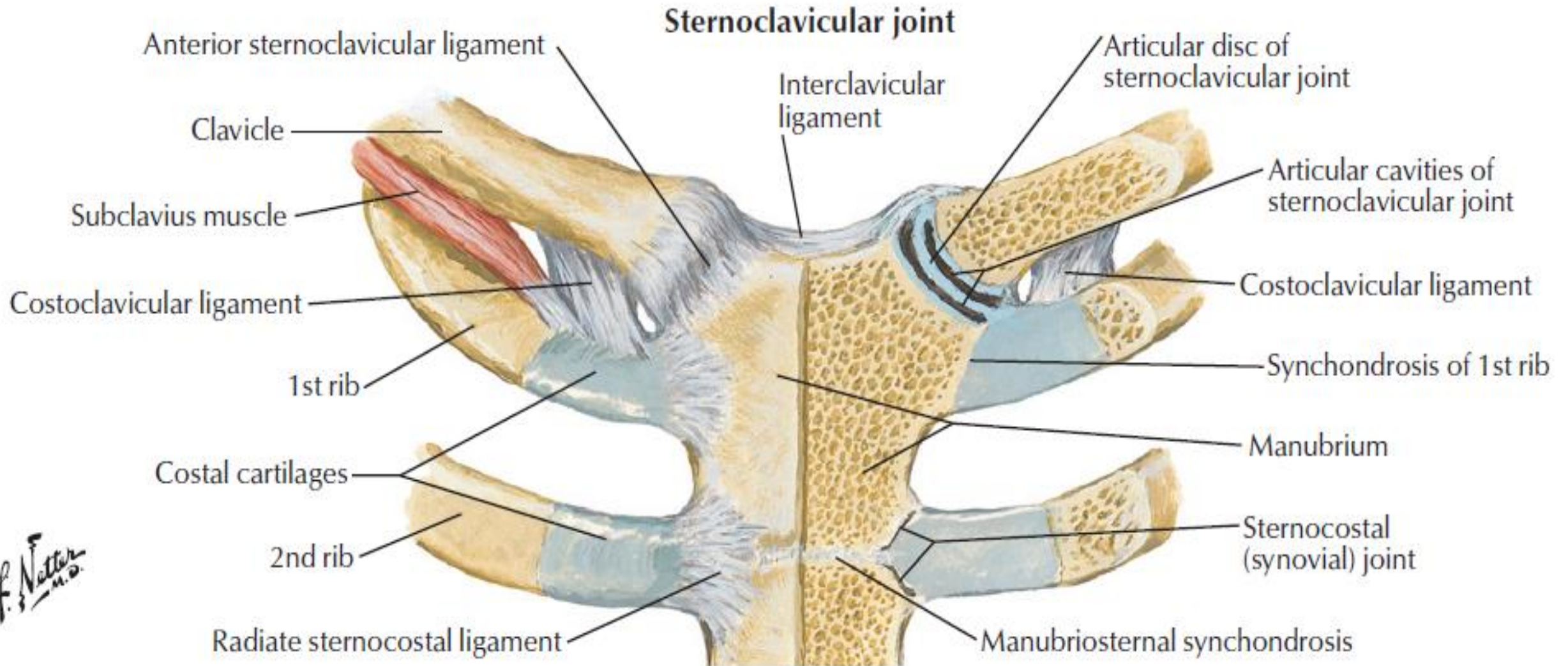
Clavicle (collar bone),,,



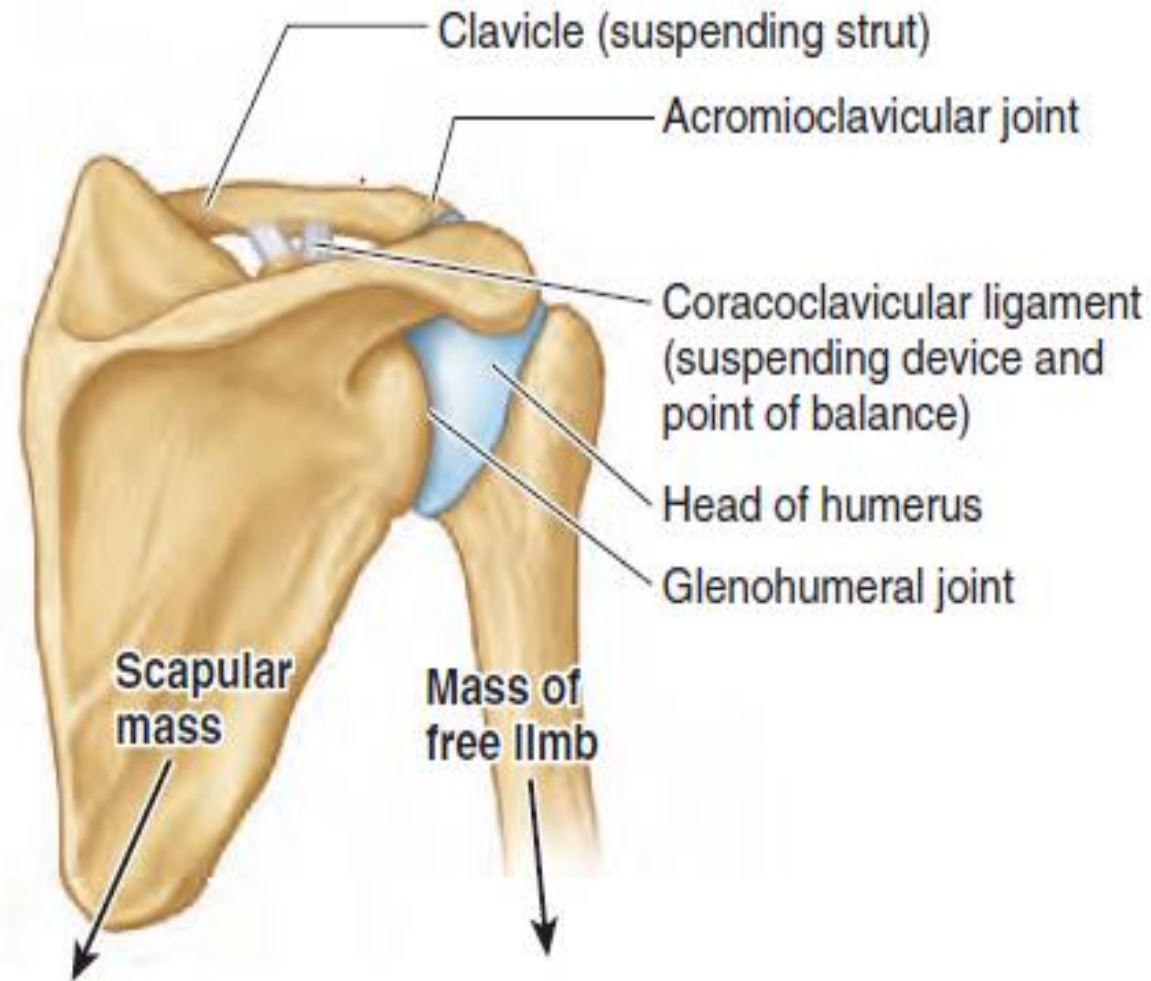
Clavicle (collar bone),,,

- The **superior surface of the clavicle**, lying just deep to the skin and platysma muscle in the subcutaneous tissue, is smooth.
- The **inferior surface of the clavicle** is rough because strong ligaments bind it to the 1st rib near its sternal end and suspend the scapula from its acromial end.
- The **conoid tubercle**, near the acromial end of the clavicle, gives attachment to the *conoid ligament*, the medial part of the *coracoclavicular ligament* by which the remainder of the upper limb is passively suspended from the clavicle.

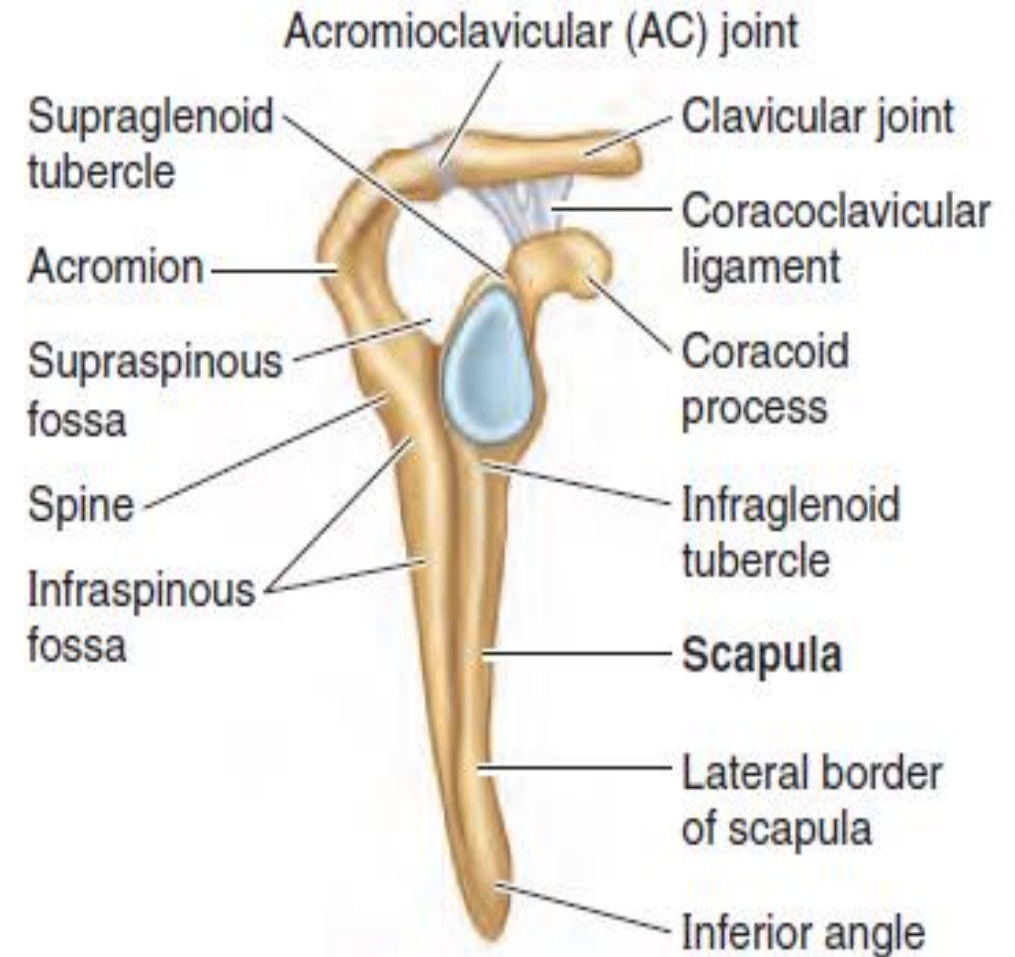
Clavicle (collar bone),,,



Clavicle (collar bone),,,

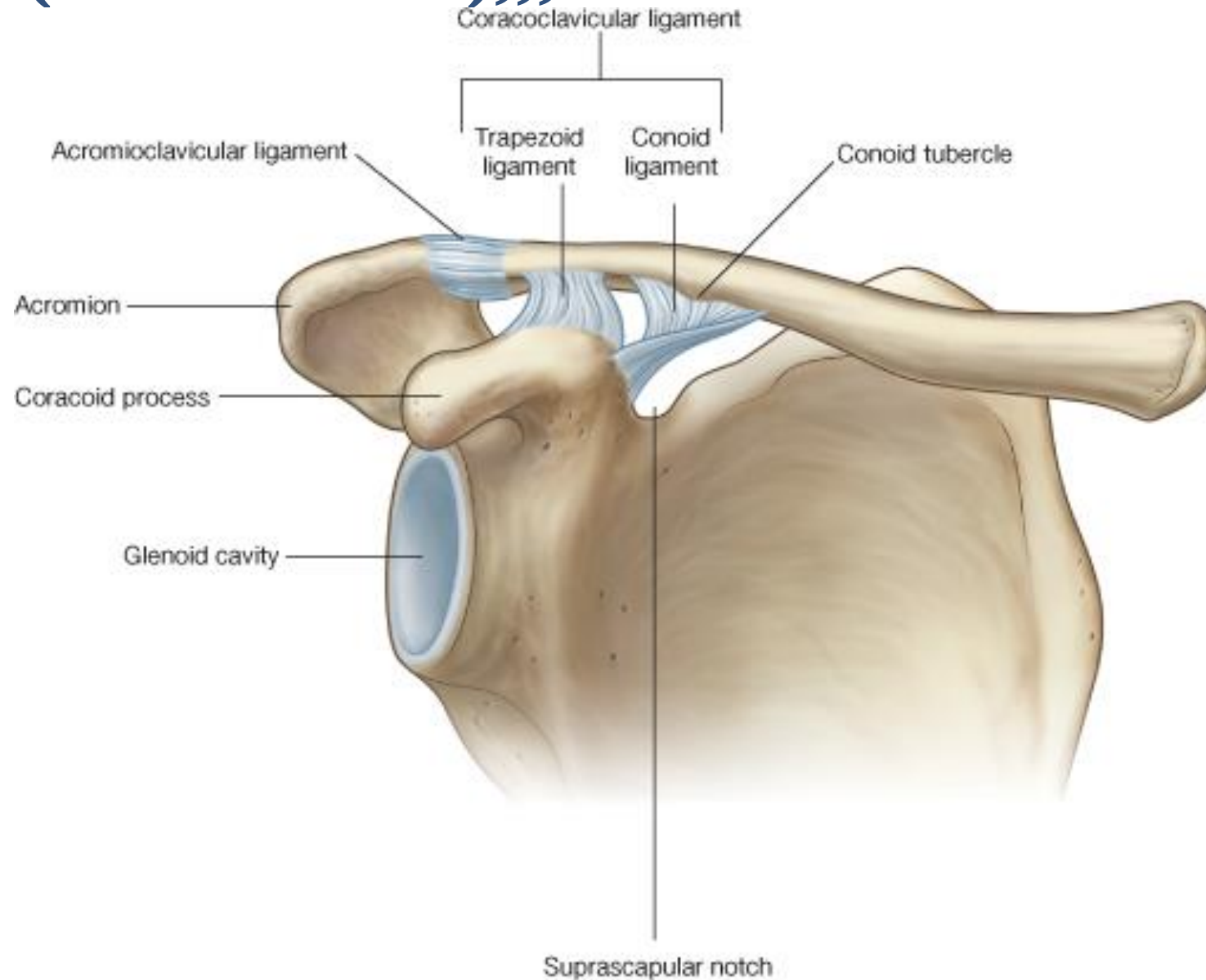


(C) Posterior view



Lateral view

Clavicle (collar bone),,,



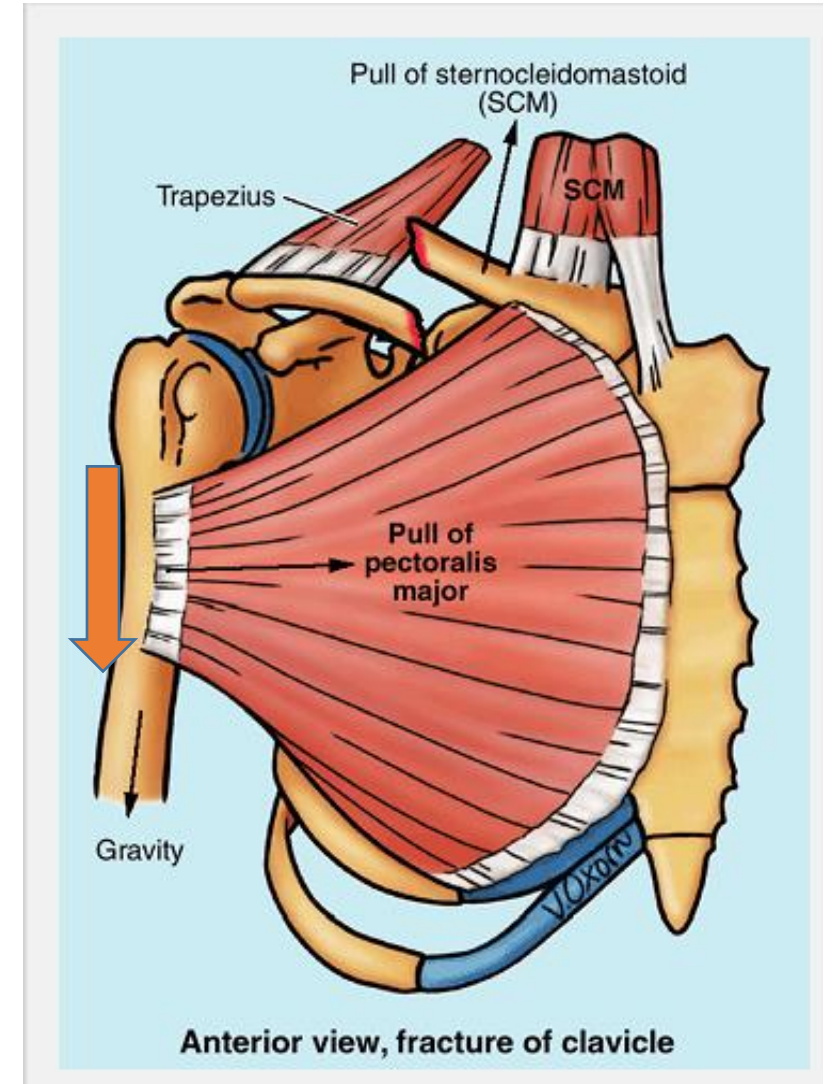
Clavicle: function

- **The clavicle:**

- ✓ Serves as a moveable, crane-like strut (rigid support) from which the scapula and free limb are suspended, keeping them away from the trunk so that the limb has maximum freedom of motion.
- ✓ Forms one of the bony boundaries of the *cervico-axillary canal* (passageway between the neck and arm), affording protection to the neurovascular bundle supplying the upper limb.
- ✓ Transmits shocks (traumatic impacts) from the upper limb to the axial skeleton.

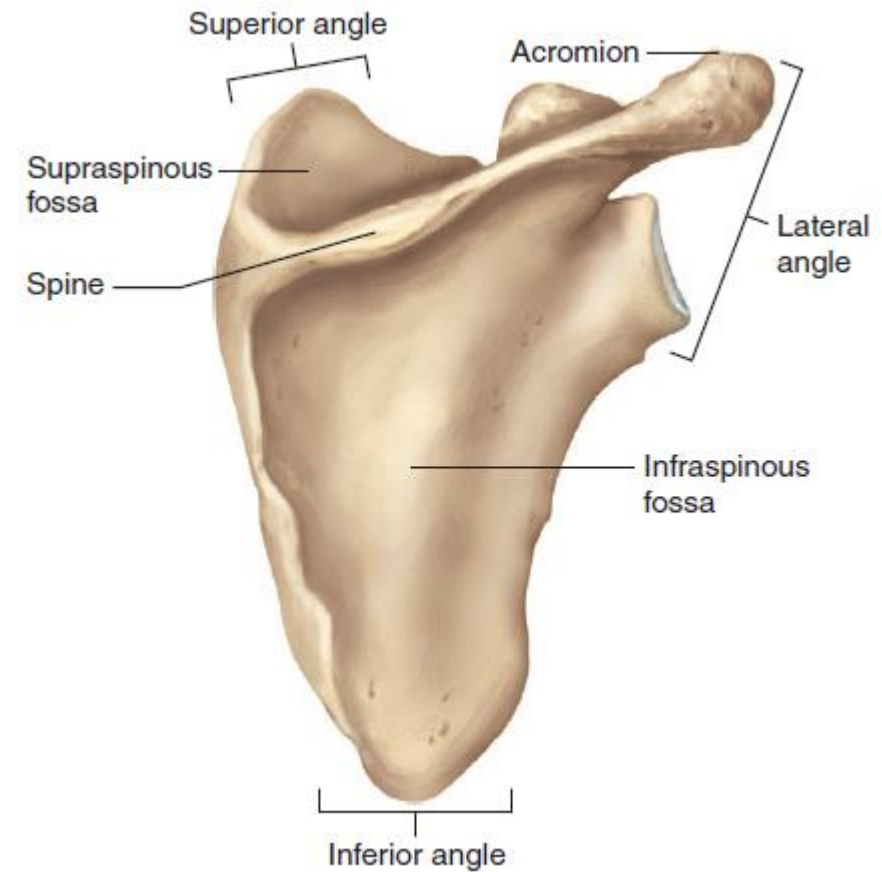
Fracture of clavicle

- The weakest part is at the junction of its middle and lateral thirds
- After fracture
 - sternocleidomastoid (SCM) muscle elevates the medial fragment of bone
 - trapezius muscle is unable to hold up the lateral fragment owing to the weight of the upper limb
 - results the **shoulder drops**



Scapula (shoulder blade)

- **Scapula** is triangular flat bone lies on the posterolateral aspect of the thorax, overlying the **2nd-7th** ribs
- Its only direct attachment to the thorax is by muscles; it glides across the rib cage as the arm and shoulder move.
- The three sides of the triangle are called the **superior**, **medial (vertebral)**, and **lateral (axillary) borders**, and its **three** angles are the **superior**, **inferior**, and **lateral angles**.

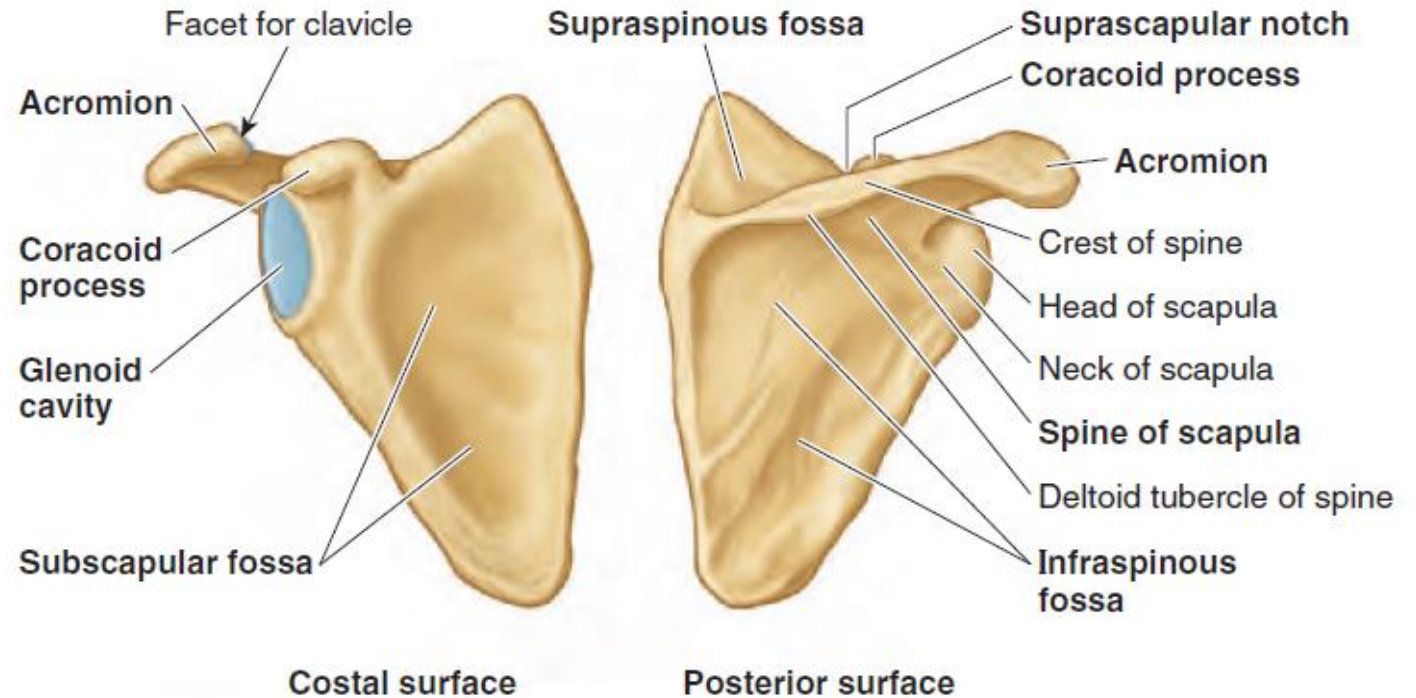


Scapula (shoulder blade)

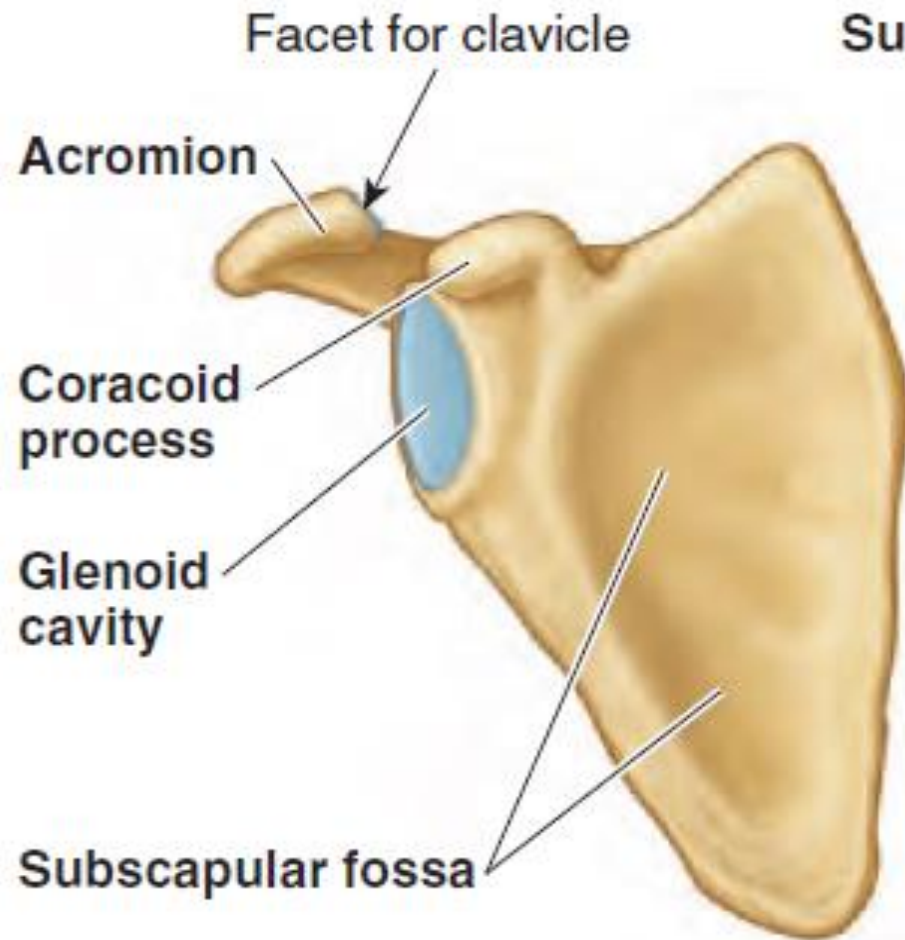
❑ The scapula has two surfaces:

- ✓ The convex **posterior surface** of the scapula is unevenly divided by a thick projecting ridge of bone, the **spine of the scapula**, into a small **supraspinous fossa** and a much larger **infraspinous fossa**.
- ✓ The concave **costal surface** of most of the scapula forms a large **subscapular fossa**.

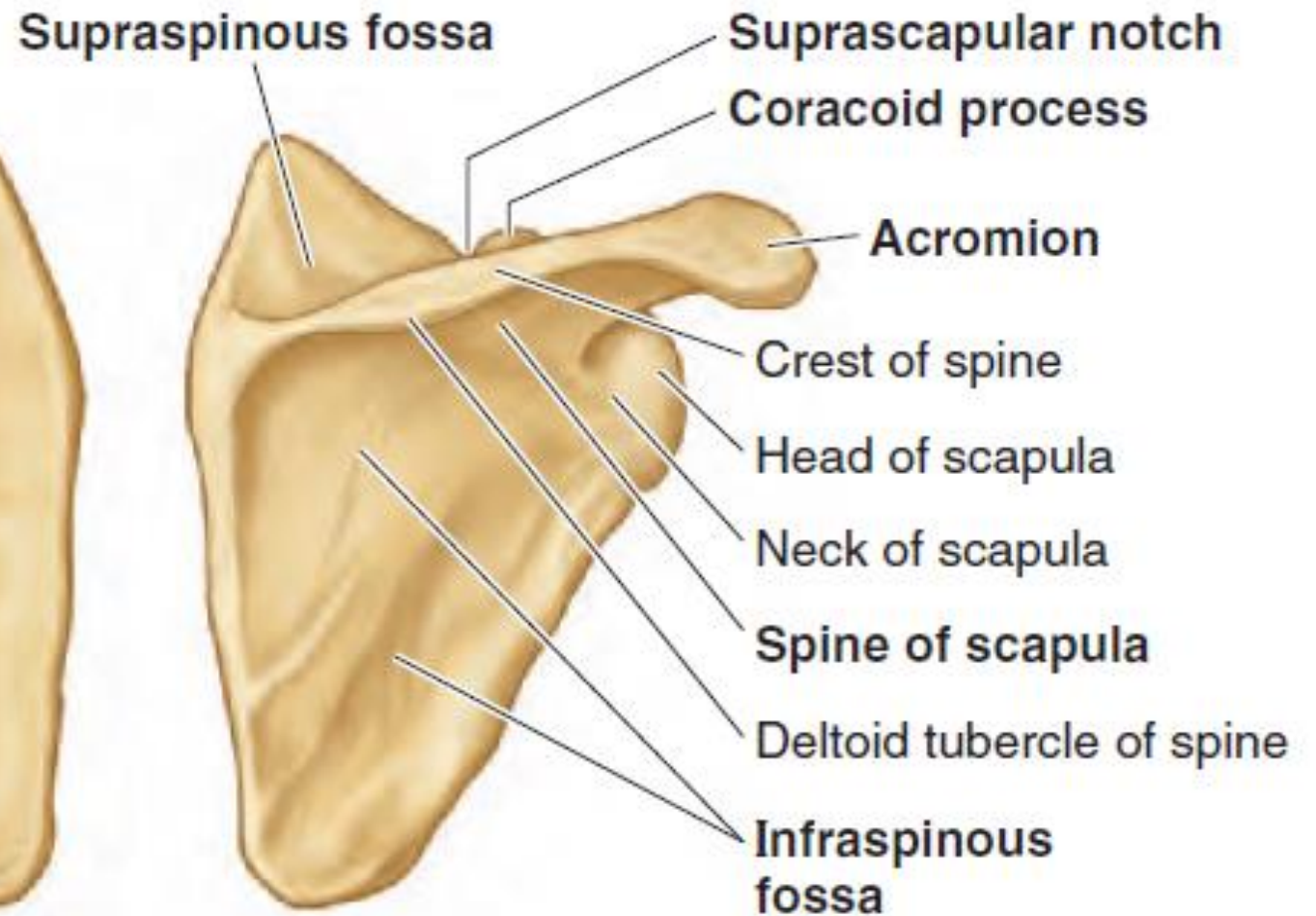
❖ The spine divided posterior surface into **supraspinous fossa**, and **infraspinous fossa**



Scapula (shoulder blade)...



Costal surface



Posterior surface

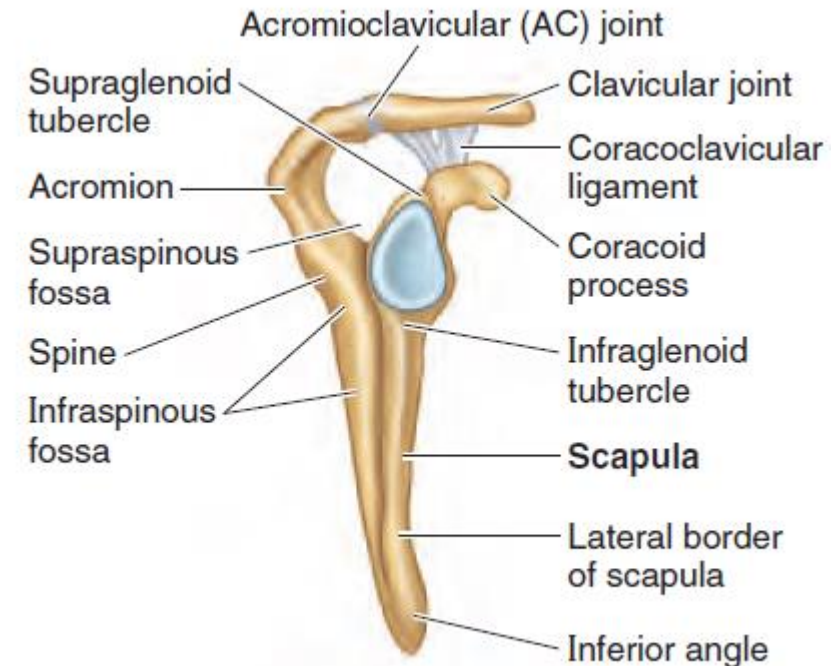
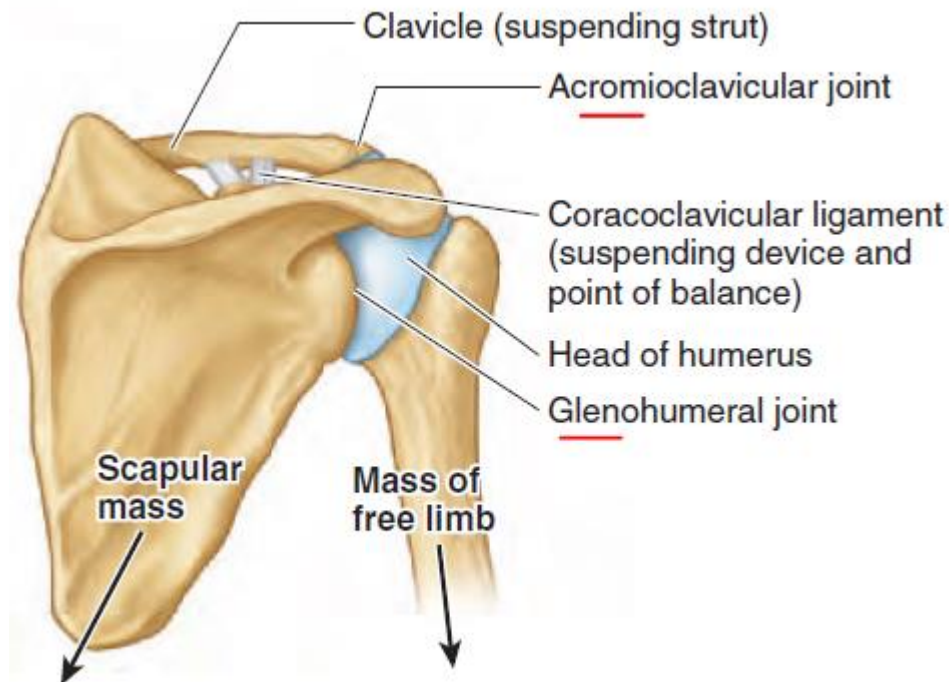
Scapula (shoulder blade),,,

❑ The most complex region of the scapula is its lateral angle, which has three main features:

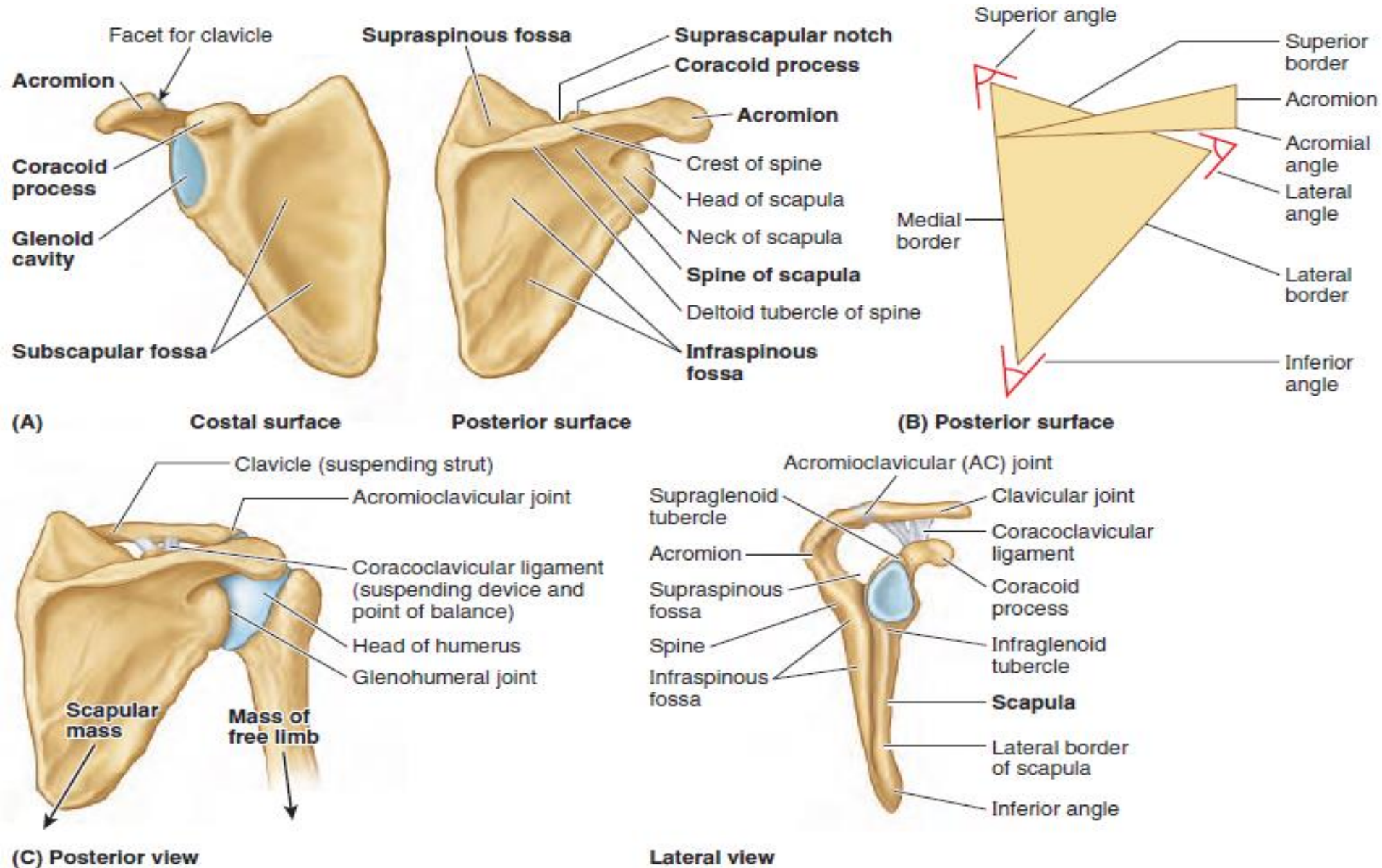
- ✓ The **acromion** is a platelike extension of the scapula which articulates with the clavicle, which forms the sole bridge from the appendicular to the axial skeleton.
- ✓ The **coracoid process** is shaped like a bent finger provides attachment for tendons of the *biceps brachii* and other muscles of the arm.
- ✓ The **glenoid cavity** is a shallow socket that articulates with the head of the humerus, forming the glenohumeral joint.

Scapula (shoulder blade),,,

- The two major joints are on the lateral sides of the scapula, **glenoid cavity**, which articulates with the head of the arm bone (humerus) to form the ***glenohumeral (shoulder) joint*** and the acromion articulates with the acromial end of the clavicle to form the ***acromioclavicular joint***.



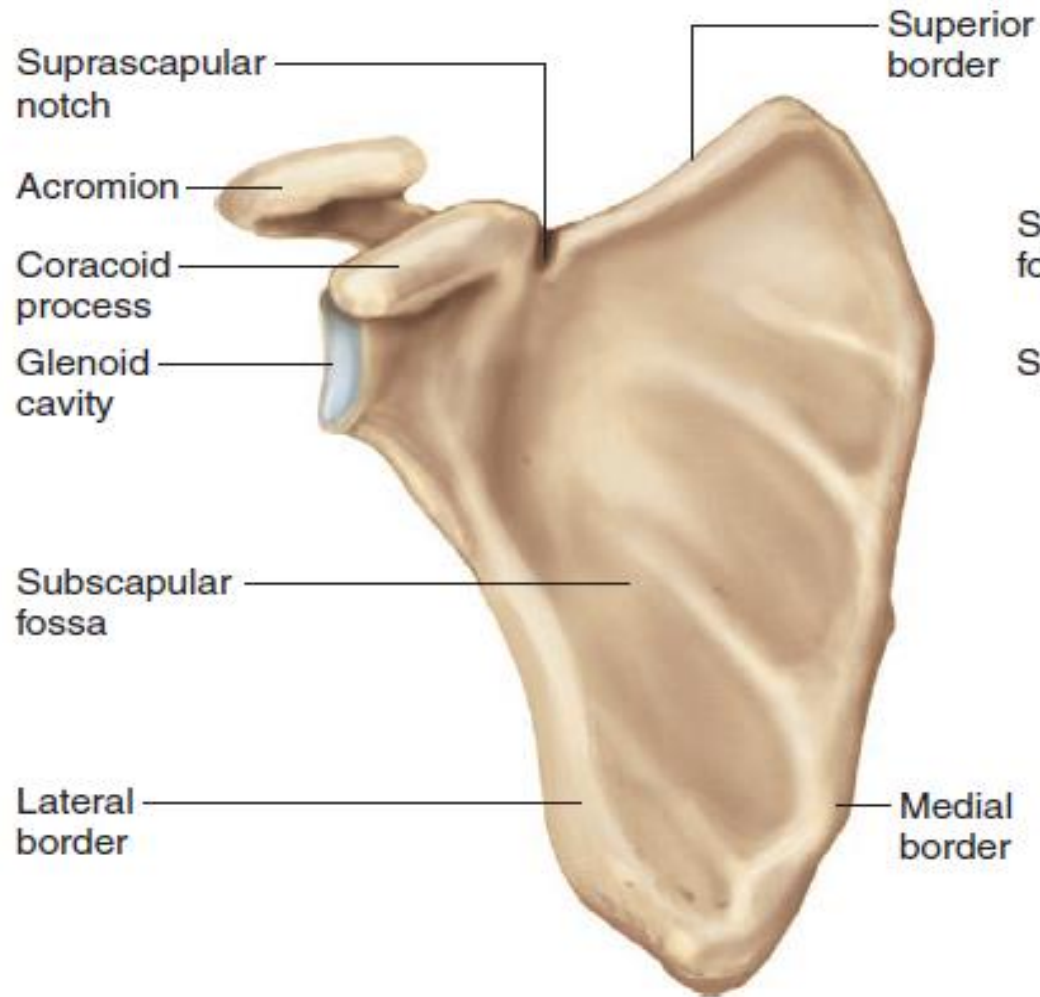
Scapula (shoulder blade),,,



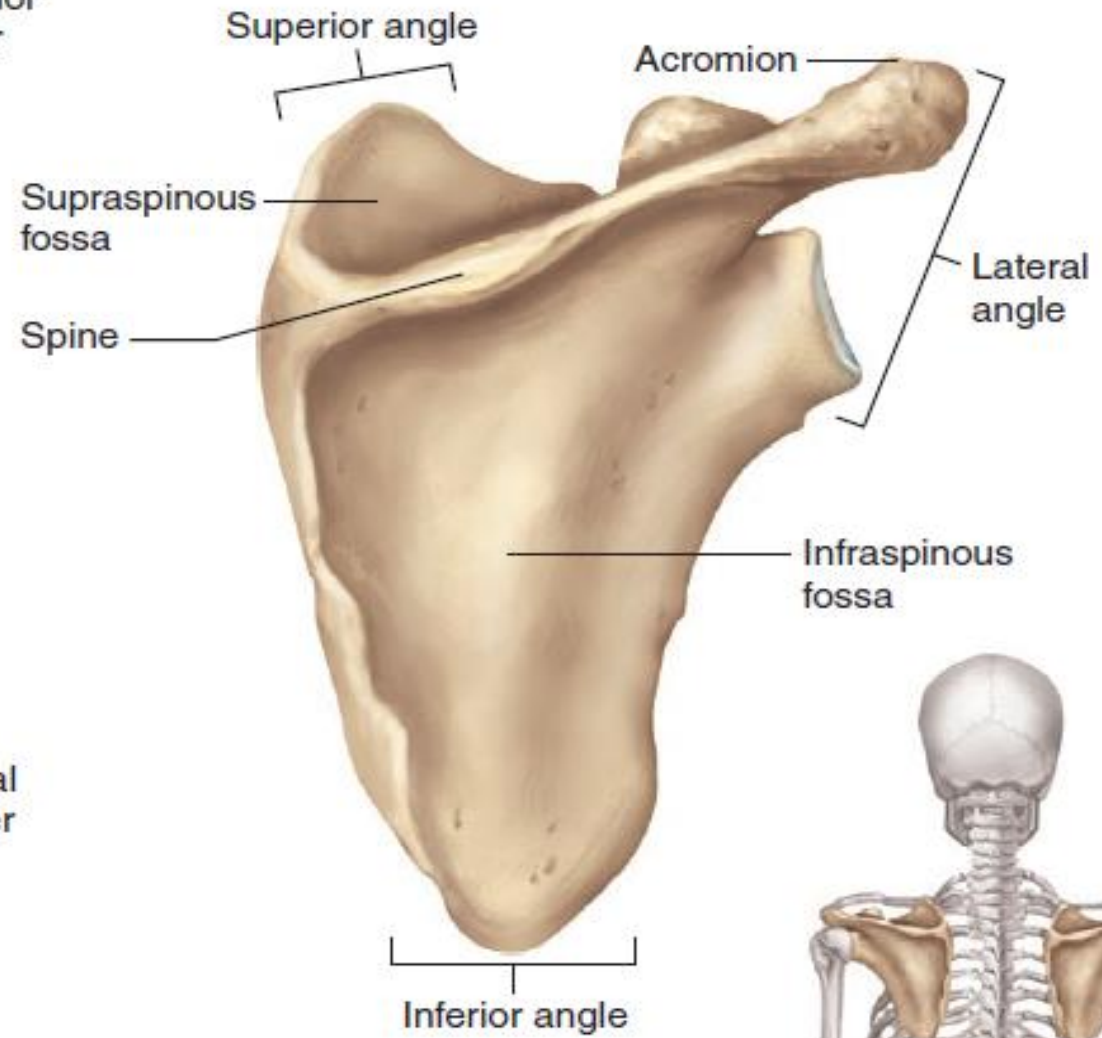
Scapula (shoulder blade),,,

- **Generally the scapula has:**
 - ✓ posterior and anterior **surfaces**
 - ✓ medial, lateral and superior **borders**
 - ✓ the **superior, inferior, and lateral angles.**
 - ✓ the **supraspinous fossa**, and the **infraspinous fossa**
 - ✓ *spinous , acromial ,and Coracoid* processes

Scapula (shoulder blade),,,



(a) Anterior view

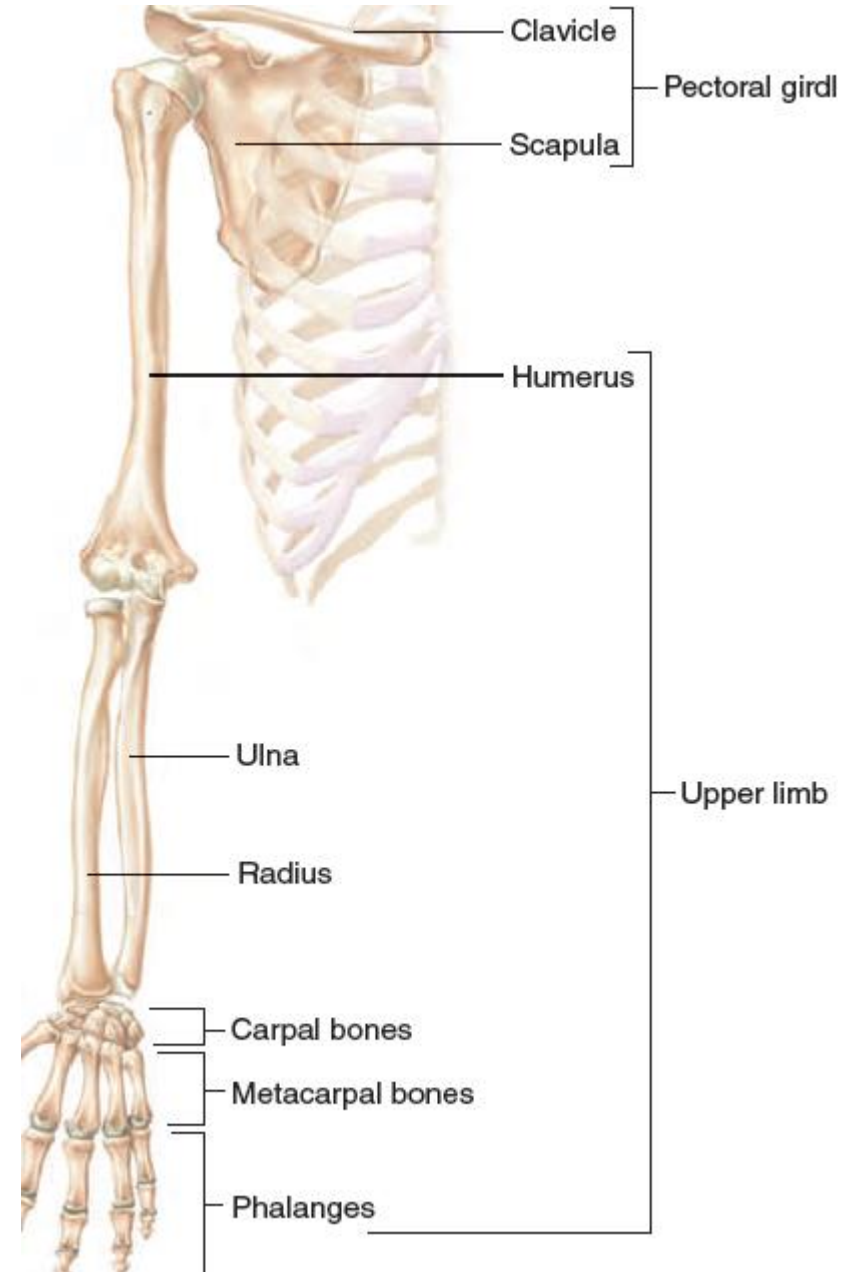


(b) Posterior view



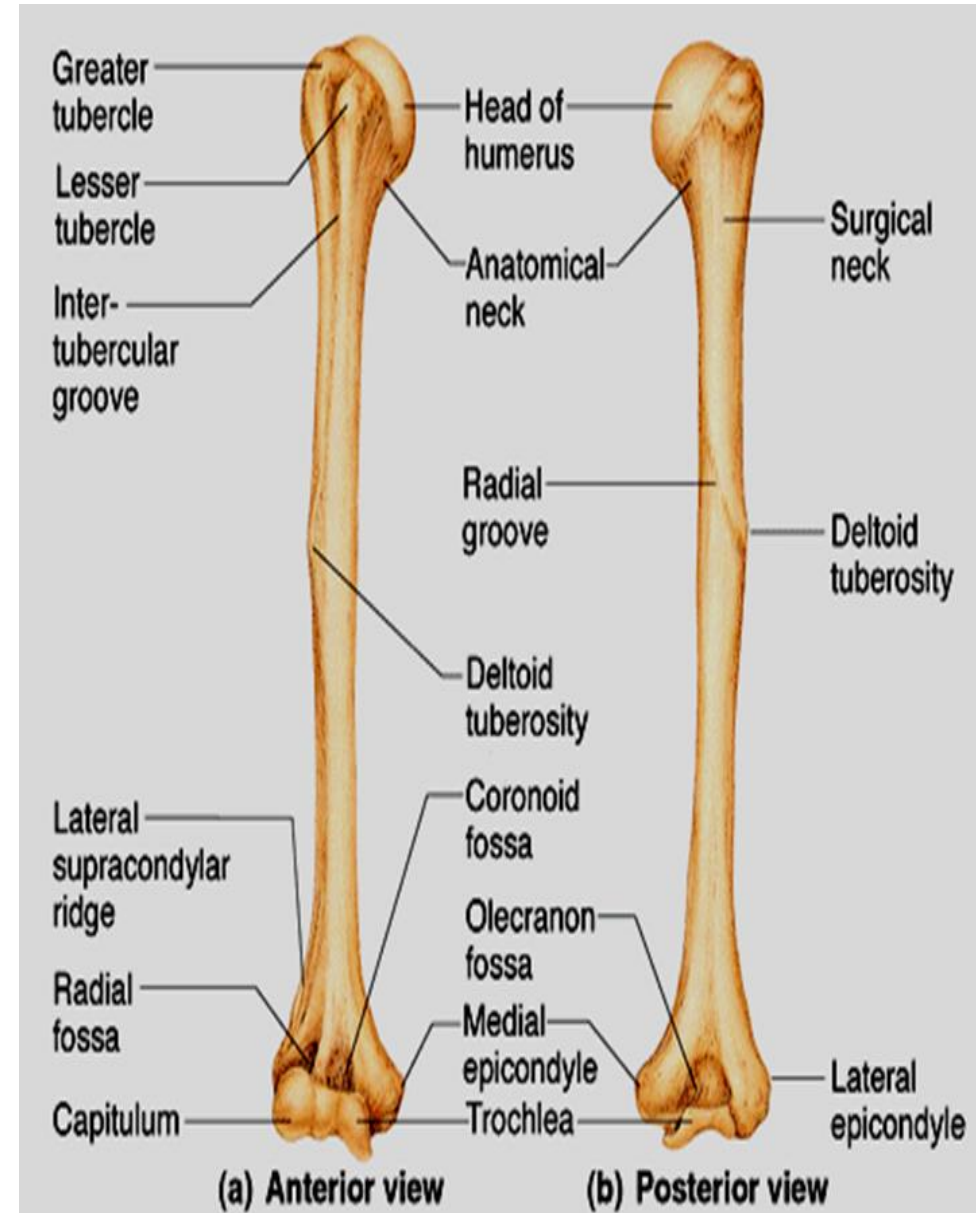
The free upper limb has :

- **Arm**
 - Humerus
- **Forearm**
 - Ulna
 - Radius
- **Hand**
 - Wrist (8 carpal bones)
 - Palm (5 metacarpal bones)
 - Fingers (14 phalanges)



Humerus

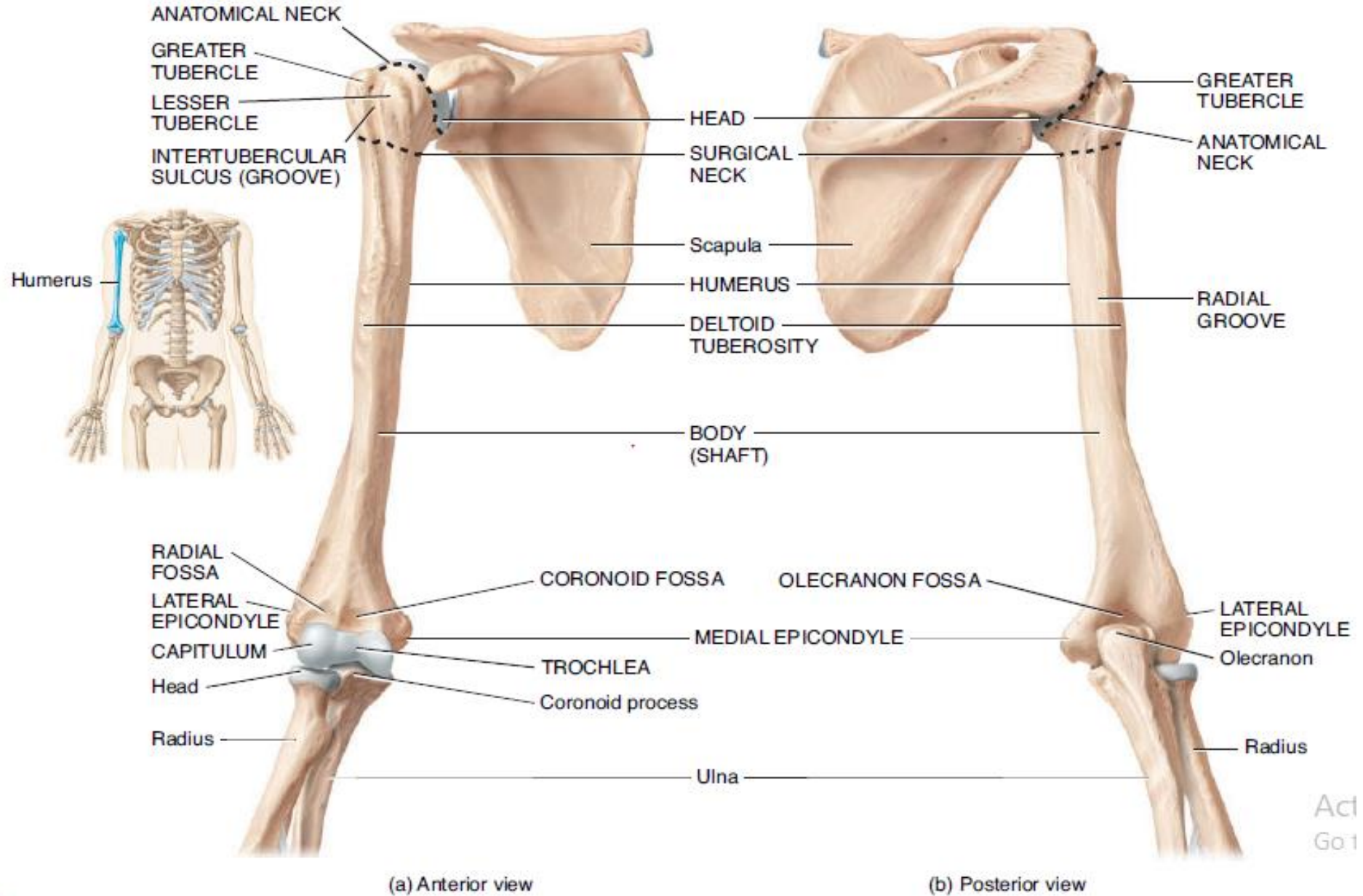
- The longest bone of upper extremity
- Consist of head, shaft (body), anatomical & surgical necks, tubercles (greater & lesser)
- **Head**
 - ball-shaped
 - articulate with the glenoid cavity of the scapula
- **Tubercles**
 - lesser and greater tubercle
 - **intertubercular groove** (bicipital groove) separates the two



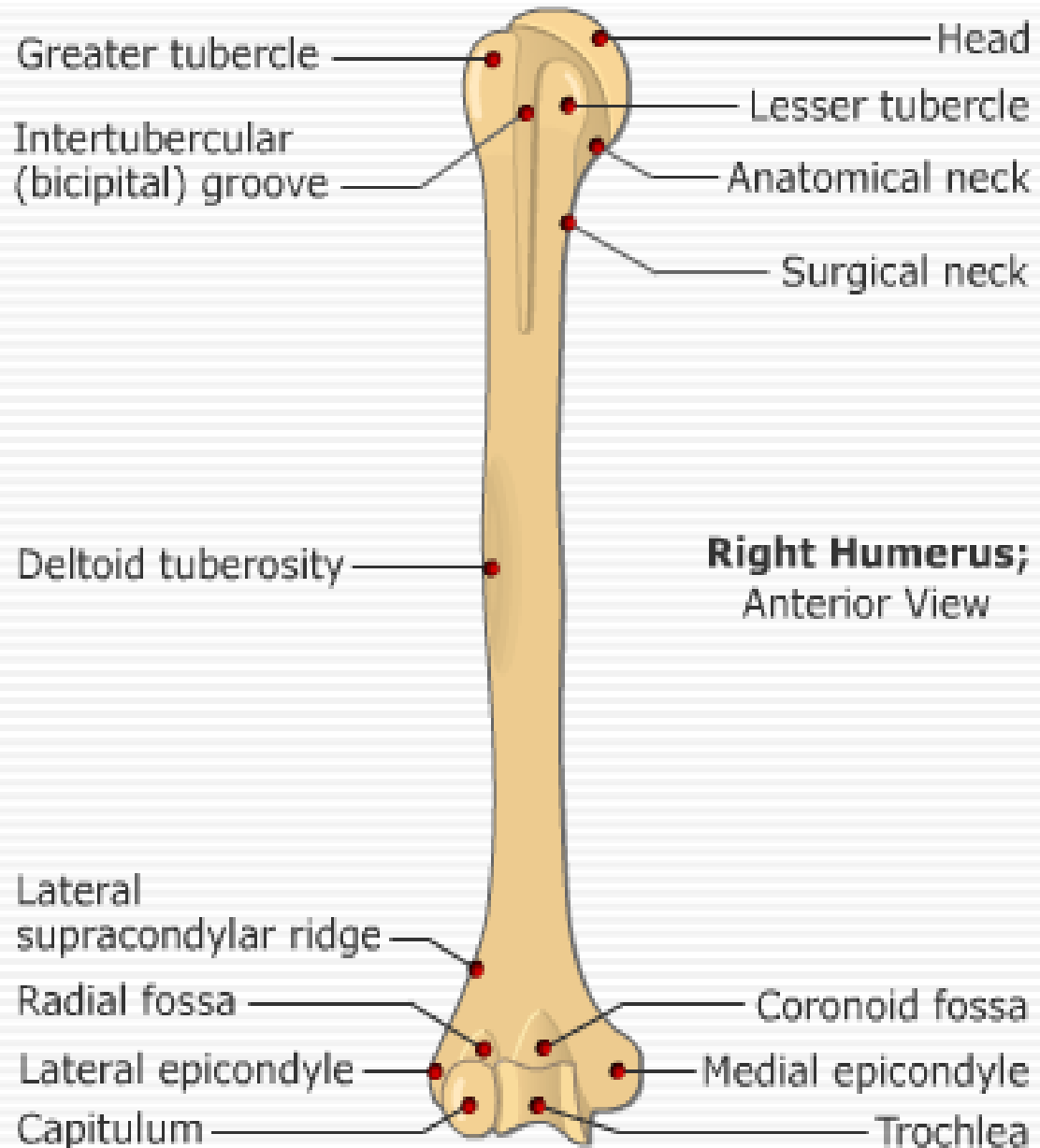
Humerus

- The **humerus** is the only bone of the Arm.
- The largest and longest bone in the upper limb, it articulates with the scapula at the glenohumeral joint, & the radius and ulna at the elbow joint.
- The proximal end of the humerus has a head, surgical and anatomical necks, and greater and lesser tubercles.
- The distal end of the humerus—including the trochlea, capitulum, olecranon, coronoid, and radial fossae—makes up the **condyle of the humerus**.

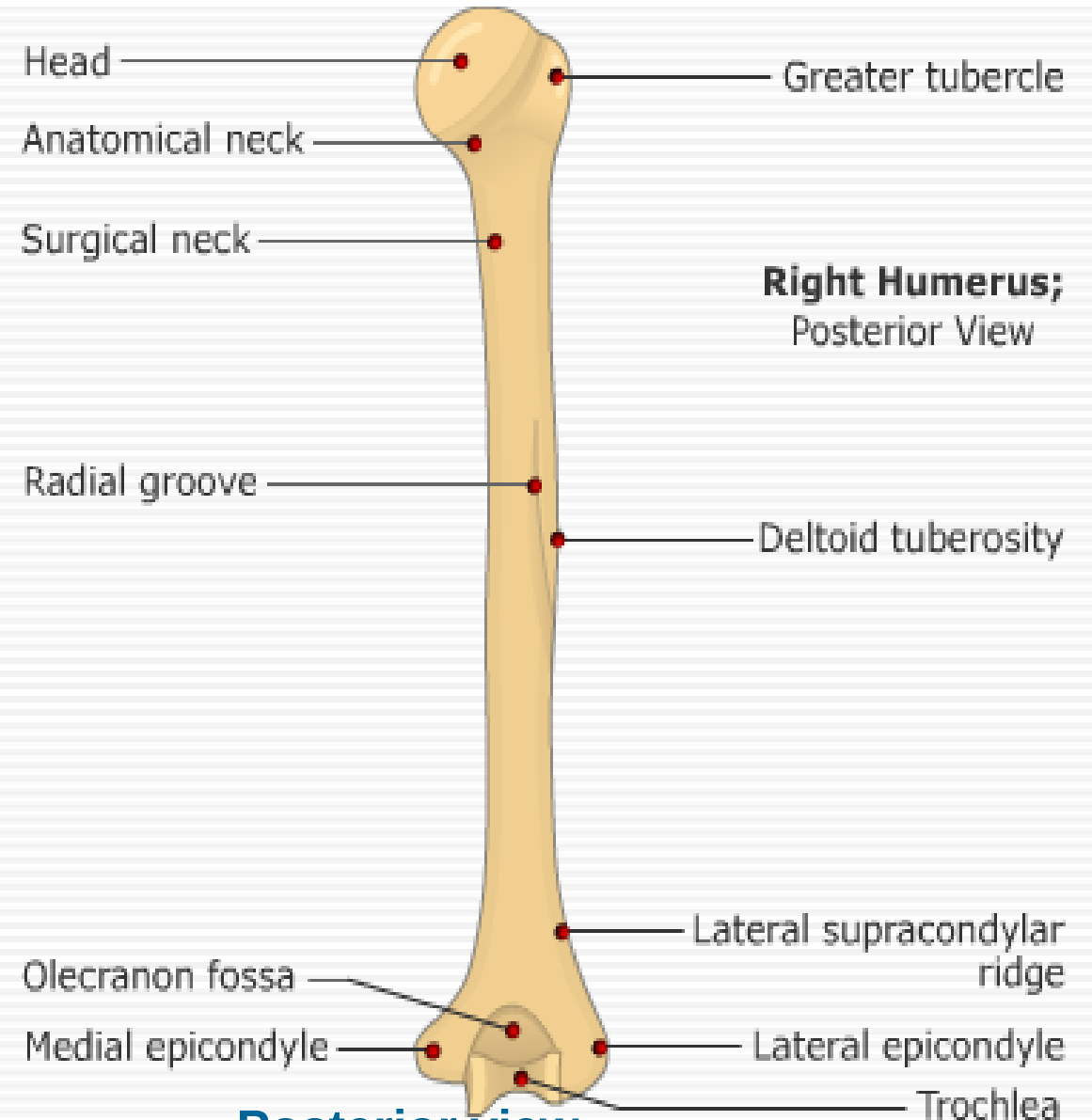
Humerus...



Humerus,,



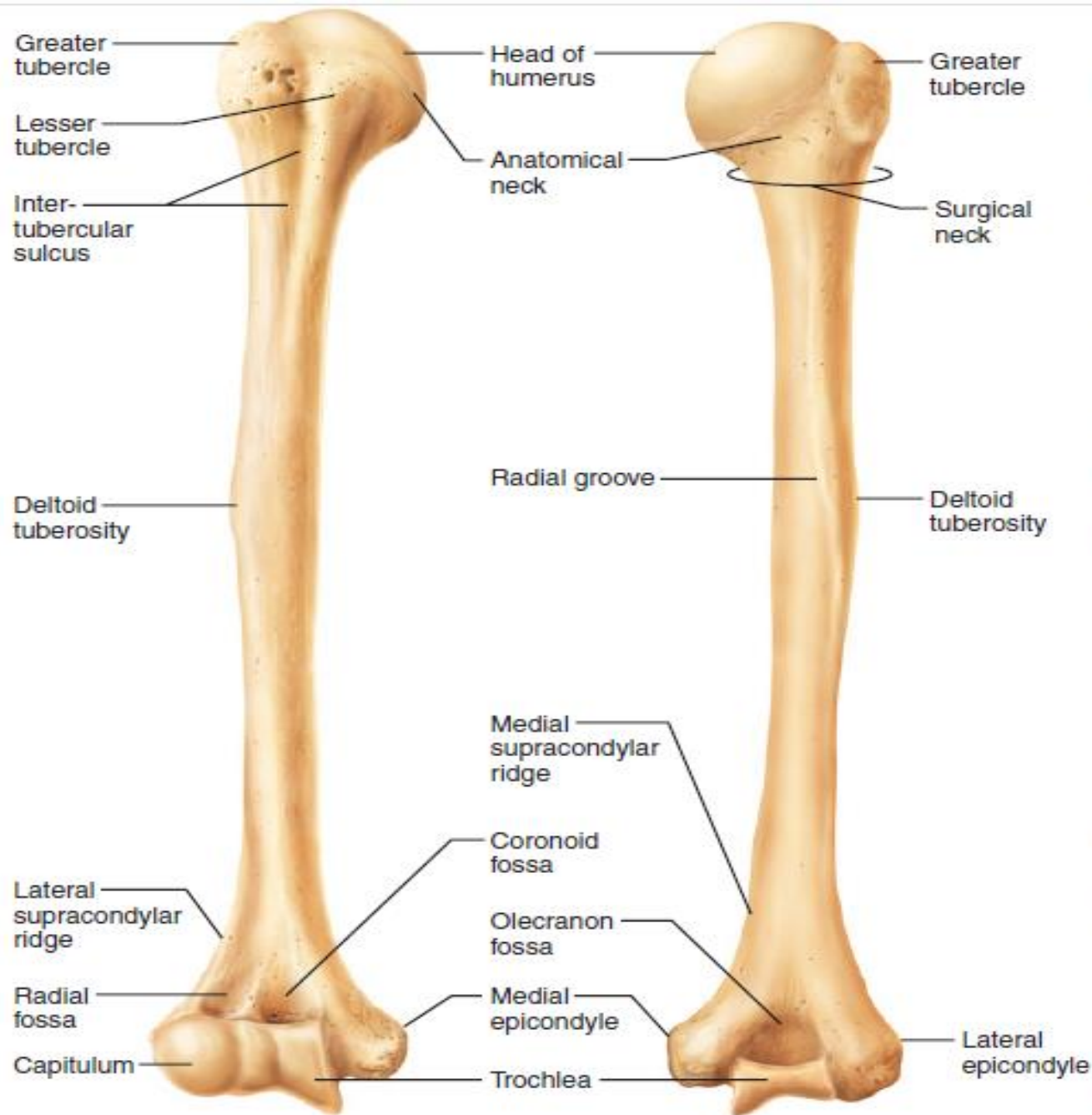
Anterior view



Posterior view

Humerus,,,

- The **surgical neck of the humerus**, a common site of fracture, is the narrow part distal to the head and tubercles.
- The **shaft of the humerus** has 2 prominent features: the **deltoid tuberosity** laterally, for attachment of the deltoid muscle, and the oblique **radial groove** (**groove for radial nerve**, spiral groove) posteriorly, in which the **radial nerve and profunda brachii artery** lie.
- The inferior end of the humeral shaft widens as the sharp medial and lateral **supra-epicondylar** (supracondylar) **ridges** form, and then end distally in the especially prominent **medial epicondyle** and the **lateral epicondyle**.

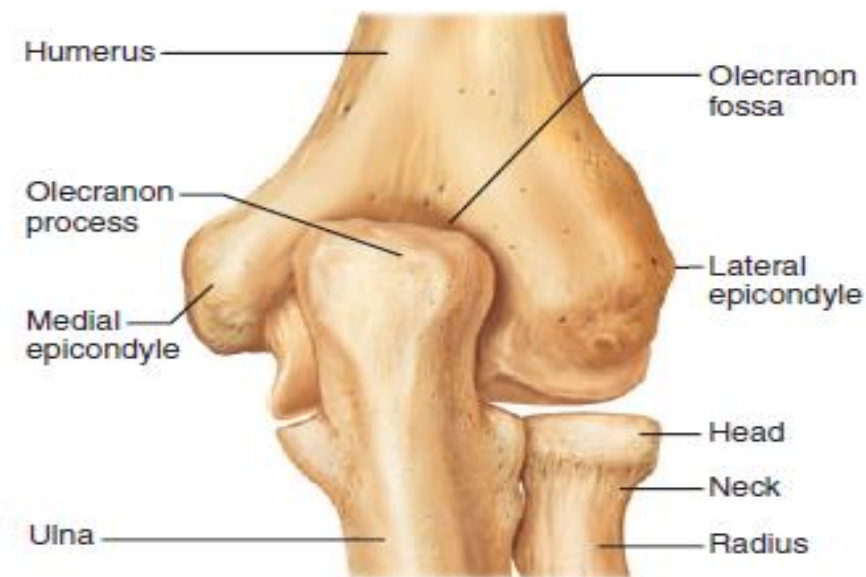


(a) Anterior view

(b) Posterior view



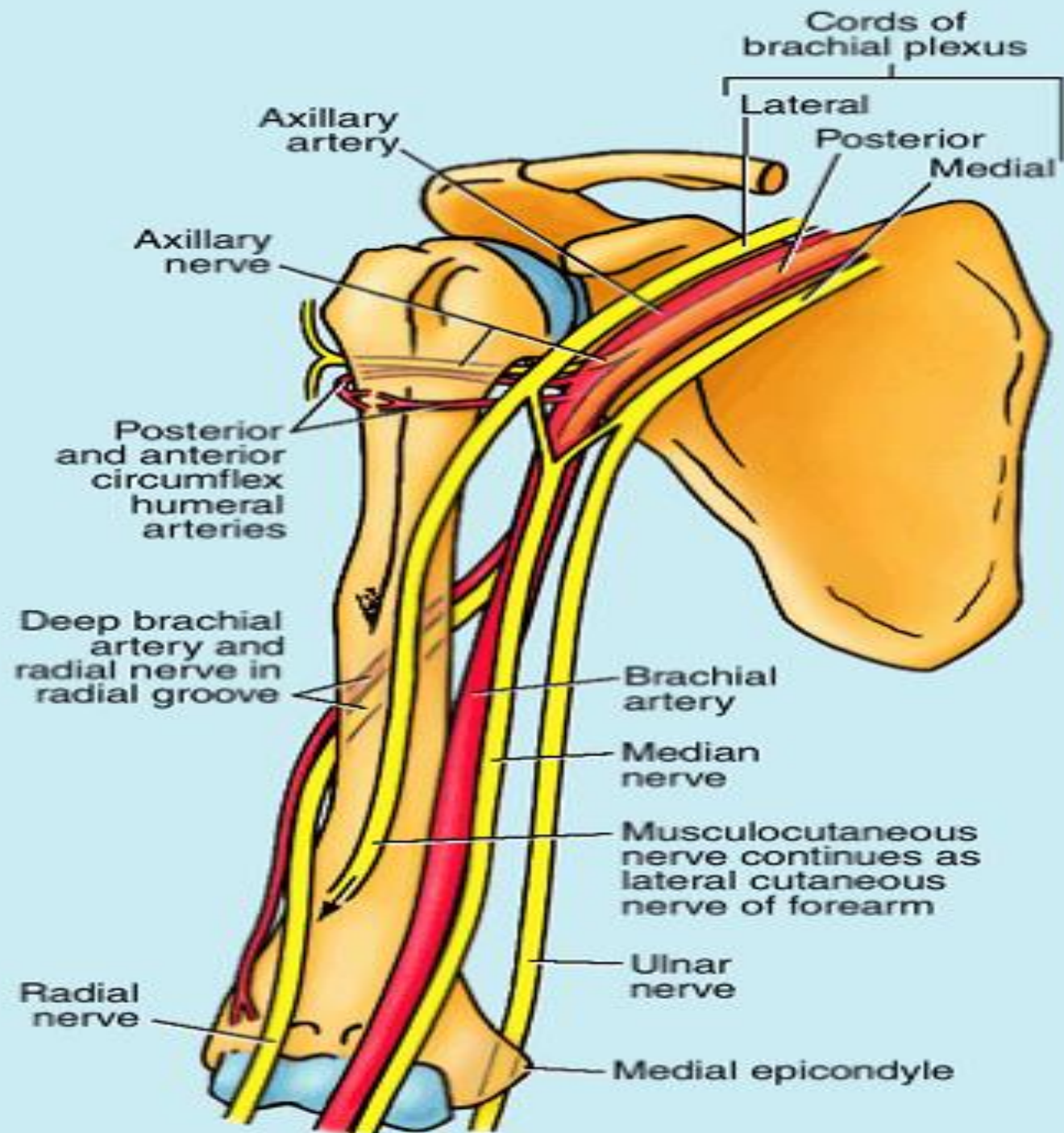
(c) Anterior view at the elbow region



(d) Posterior view of extended elbow

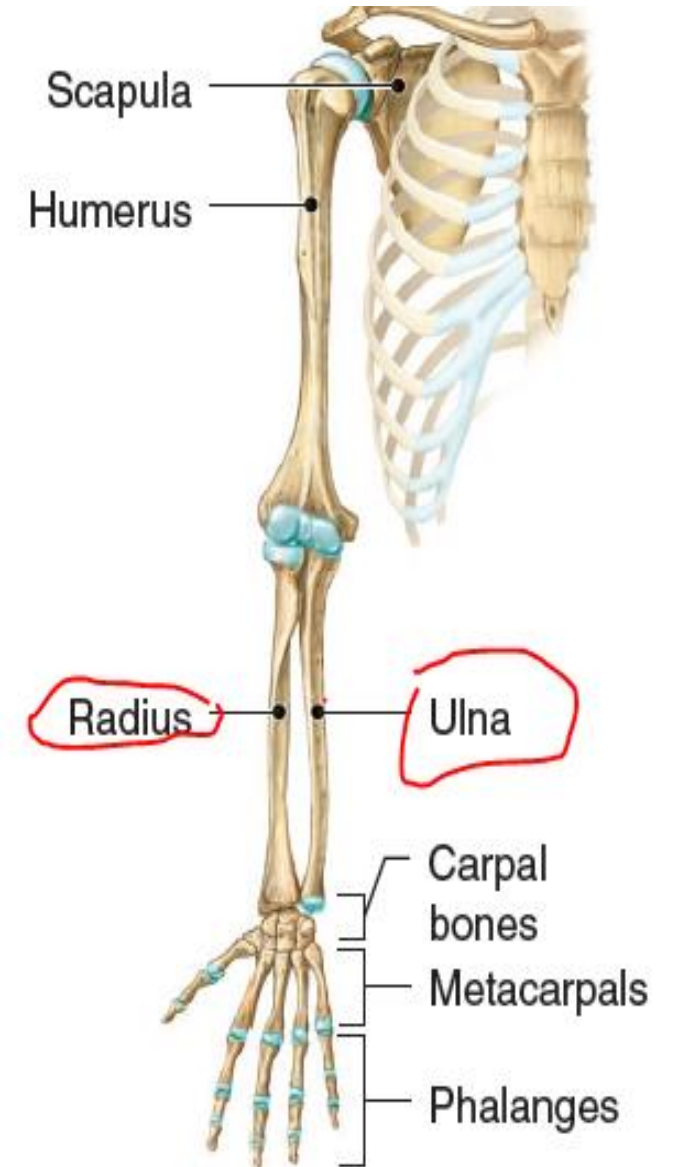
Fracture of the humerus

- Fractures of the **surgical neck** of the humerus
 - common in elderly people
 - usually result from a fall on the hand
- Because nerves are in contact with the humerus, they may be injured when the associated part of the humerus is fractured
 - **Surgical neck** – axillary nerve
 - **Radial groove** – radial nerve
 - **Distal humerus** – median nerve
 - **Medial epicondyle** – ulnar nerve

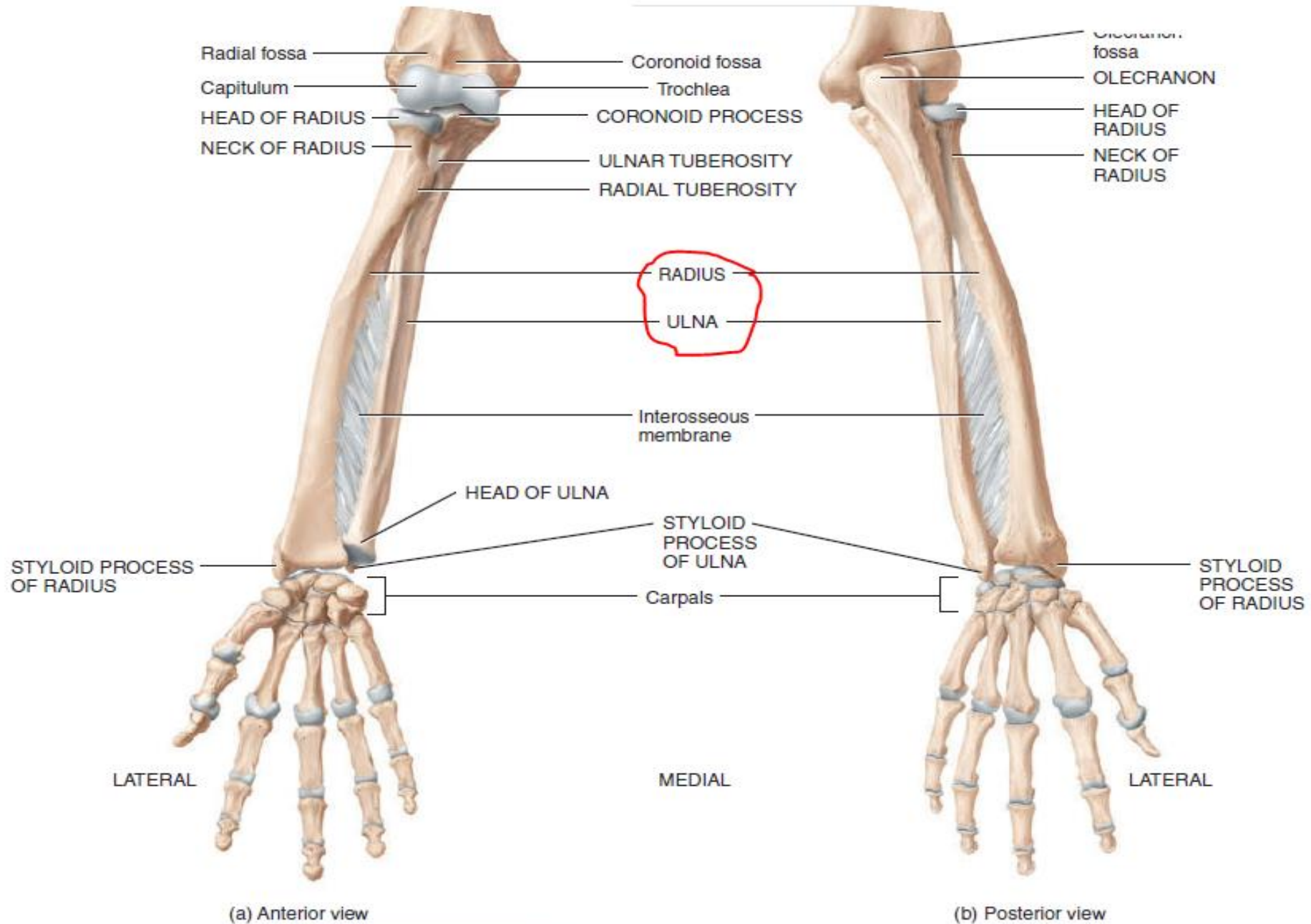


Bones of Forearm

- The **forearm** has 2 bones: the **ulna**, on the medial (little finger) side and the **radius** on the lateral (thumb) side.
- These 2 parallel long bones form the skeleton of the *forearm* or *antebrachium*
- The radius and ulna that articulate with the humerus proximally and the bones of the wrist distally.
- The radius and ulna also articulate with each other both proximally and distally at the small *radioulnar joints*.
- They are interconnected along their entire length by a flat ligament called the **interosseous membrane**.

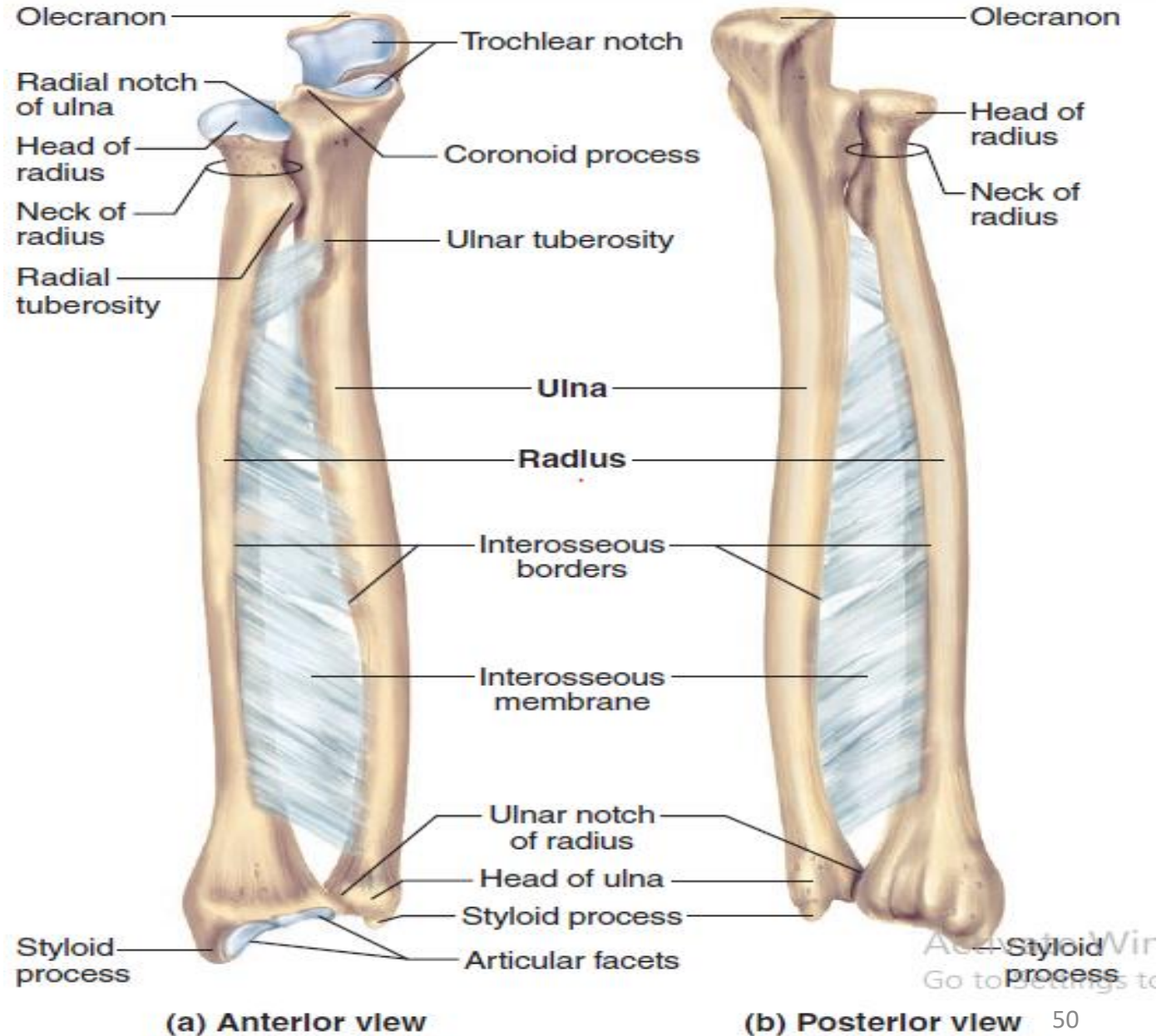


Bones of Forearm,,,

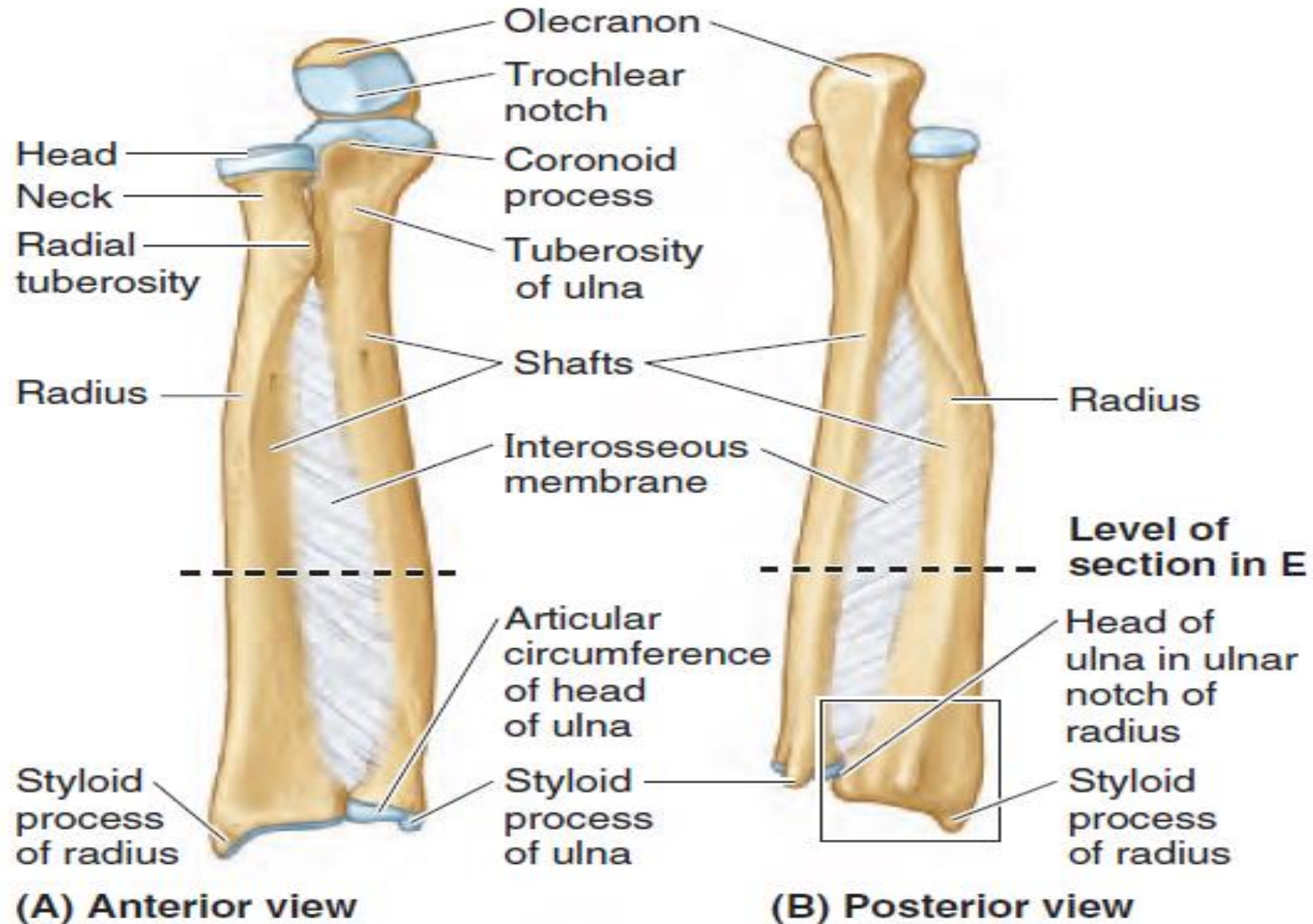


Bones of Forearm,,,

❑ The radius and ulna are attached along their shafts by a ligament called the **interosseous membrane**, which is attached to an angular ridge called the **interosseous margin** on each bone.



Bones of Forearm,,



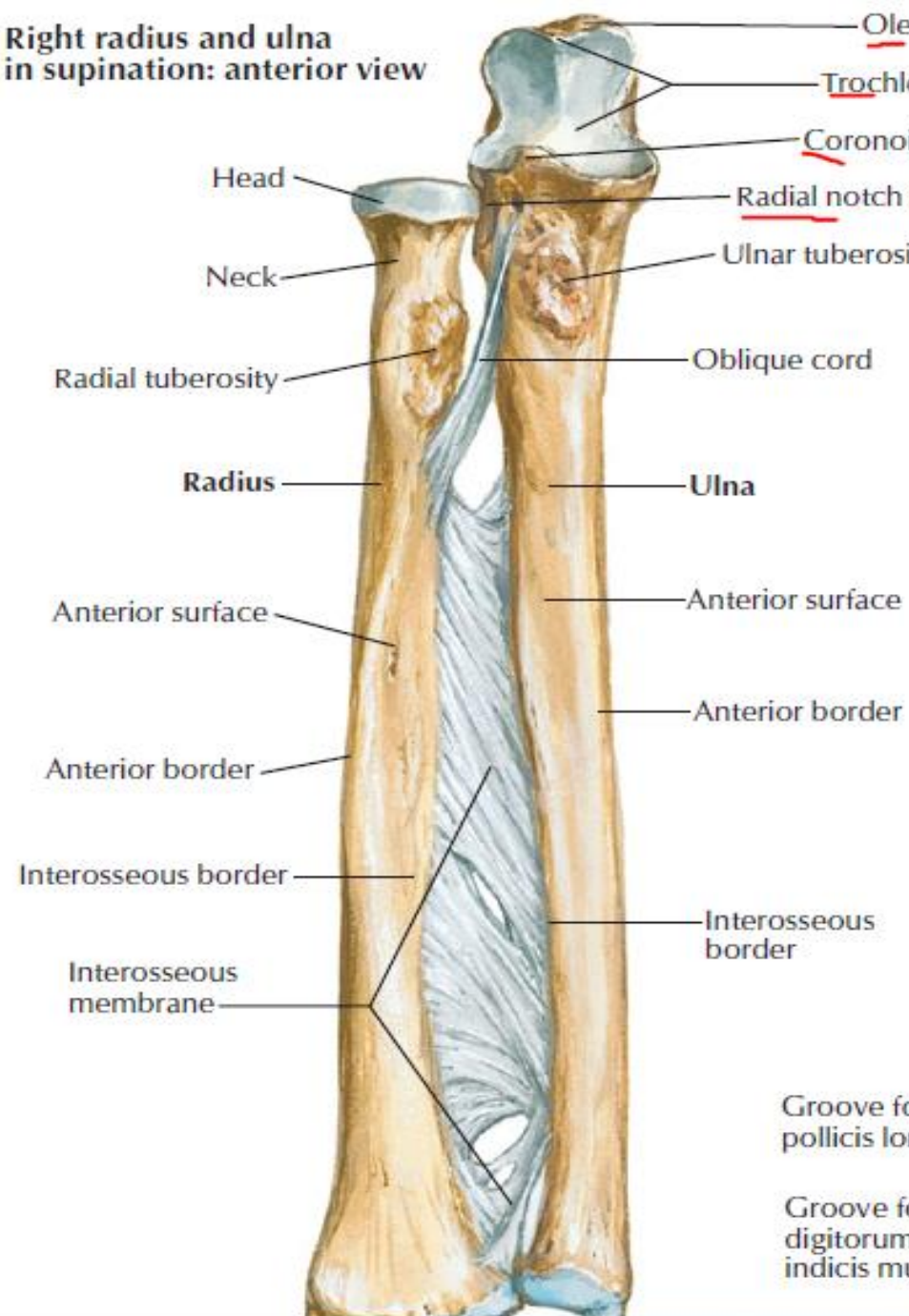
Ulna

- The **ulna** is the stabilizing bone of the forearm and is the medial and longer of the two forearm bones.
- The **ulna** is the main bone forming the elbow joint.
- Its more massive proximal end is specialized for articulation with the humerus proximally, and the head of the radius laterally.

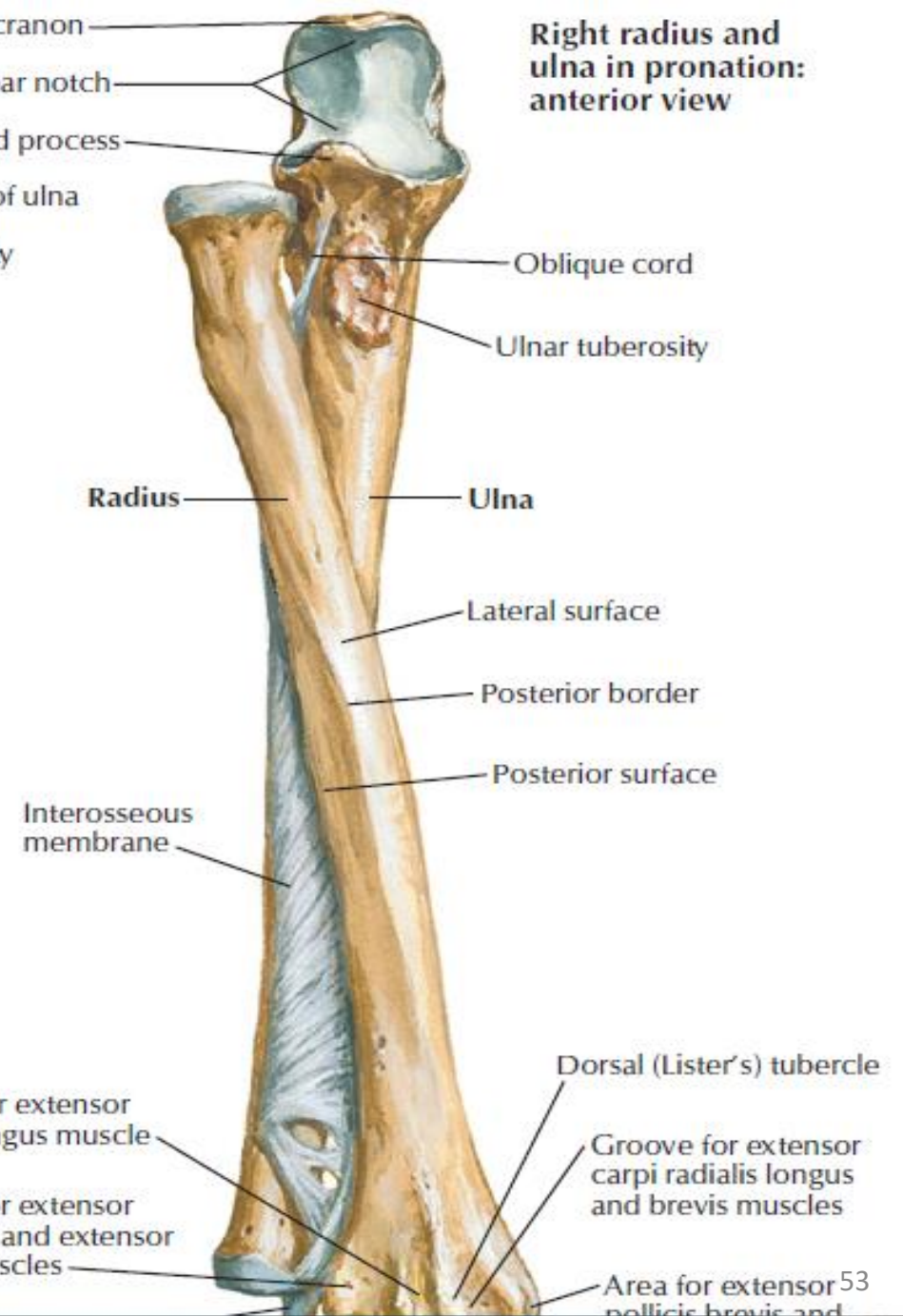
□ Proximally, the ulna has the following features:

- ✓ the **olecranon process**, posteriorly
- ✓ the **coronoid process**, which projects anteriorly
- ✓ The olecranon and coronoid processes form the walls of the **trochlear notch**,
- ✓ **Radial notch**; on the lateral side of the coronoid process articulates with the head of radius .

Right radius and ulna
in supination: anterior view



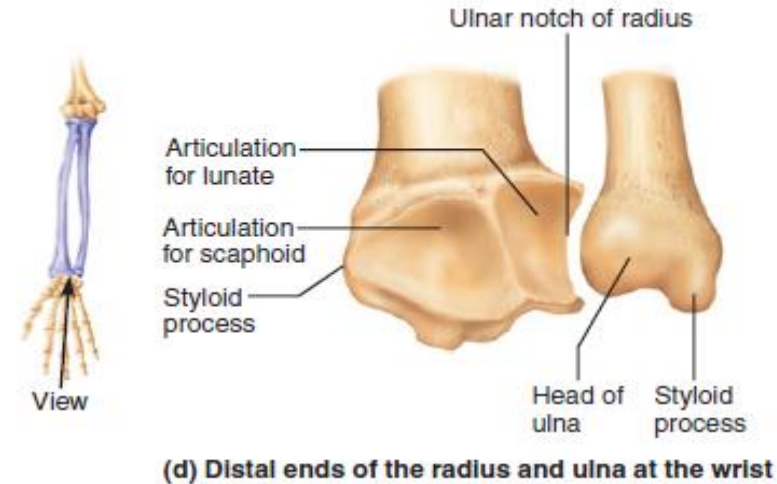
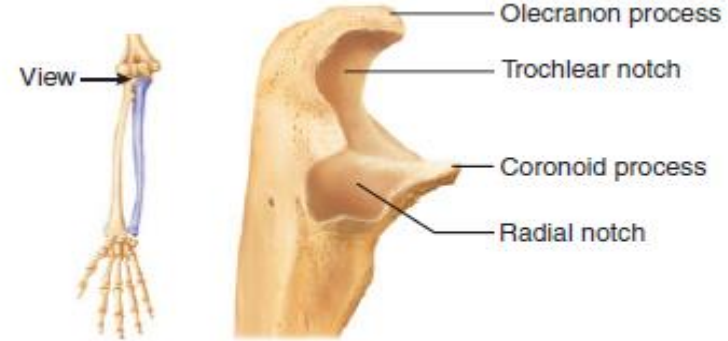
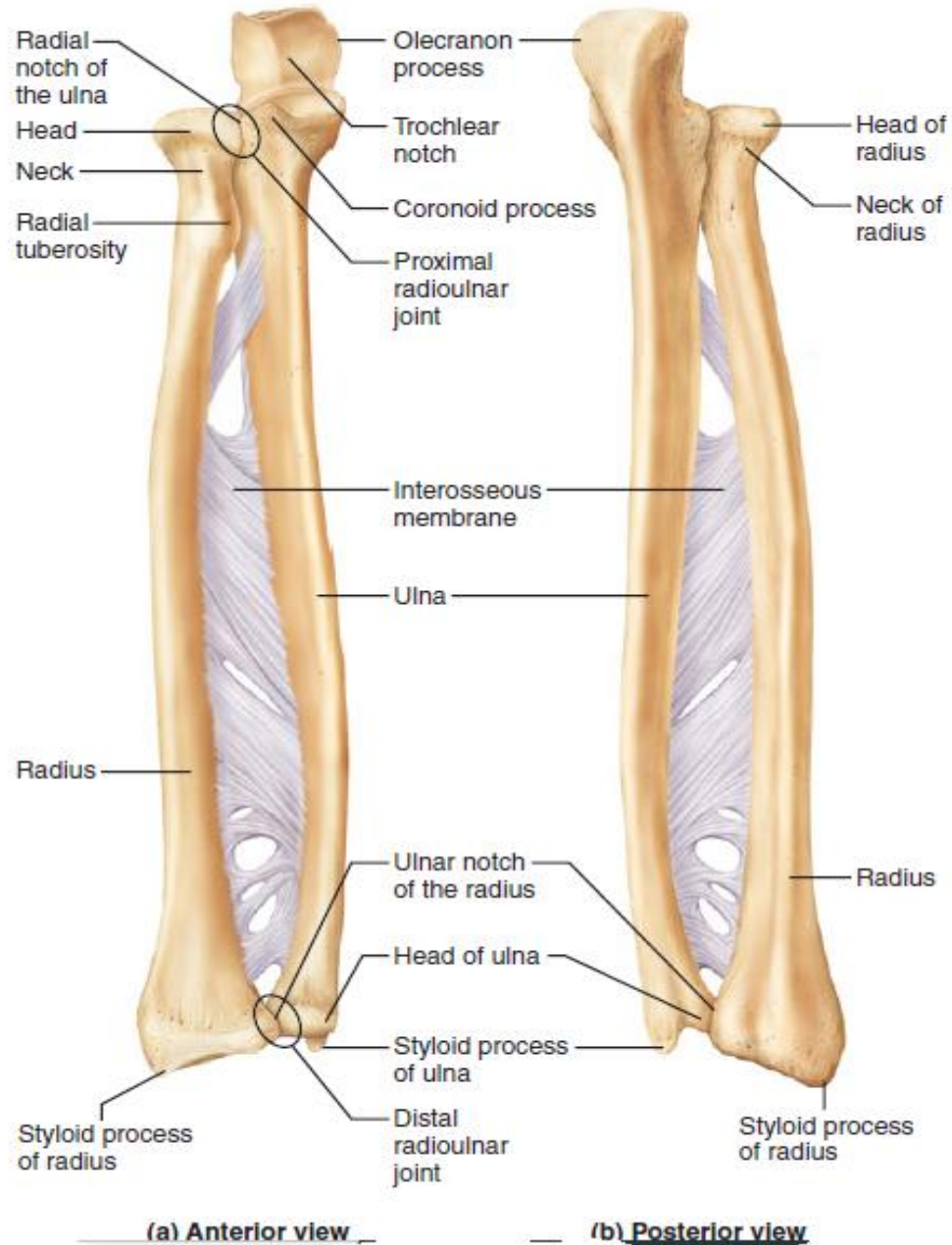
Right radius and
ulna in pronation:
anterior view



Ulna,,,

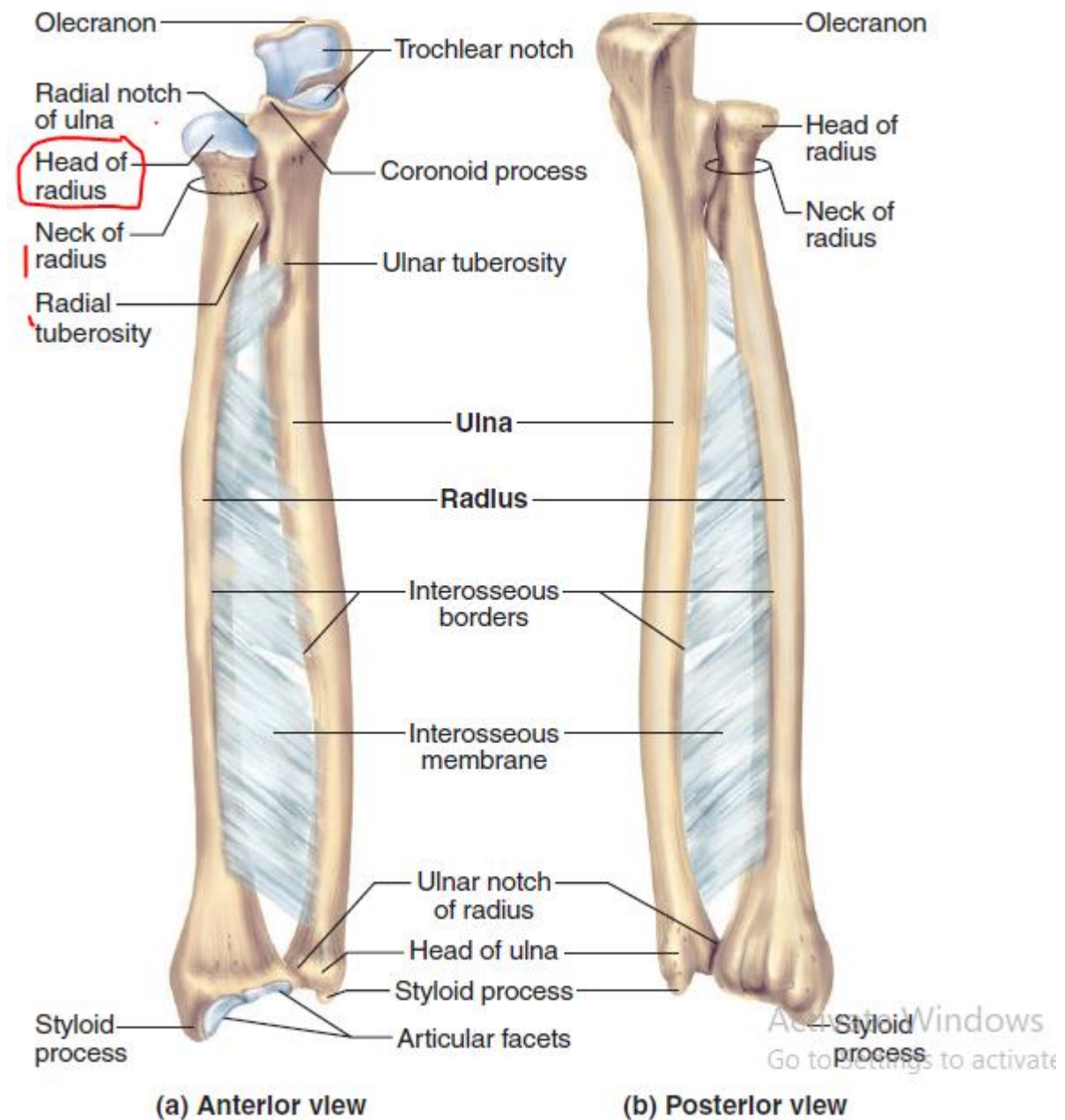
- The **shaft of the ulna** is thick and cylindrical proximally, but it tapers, diminishing in diameter, as it continues distally.
- At the narrow distal end of the ulna is a small but abrupt enlargement, the disc-like **head of the ulna** with a small, conical **ulnar styloid process**.
- The ulna does not participate in the wrist (radiocarpal) joint.

Ulna,,,



Radius

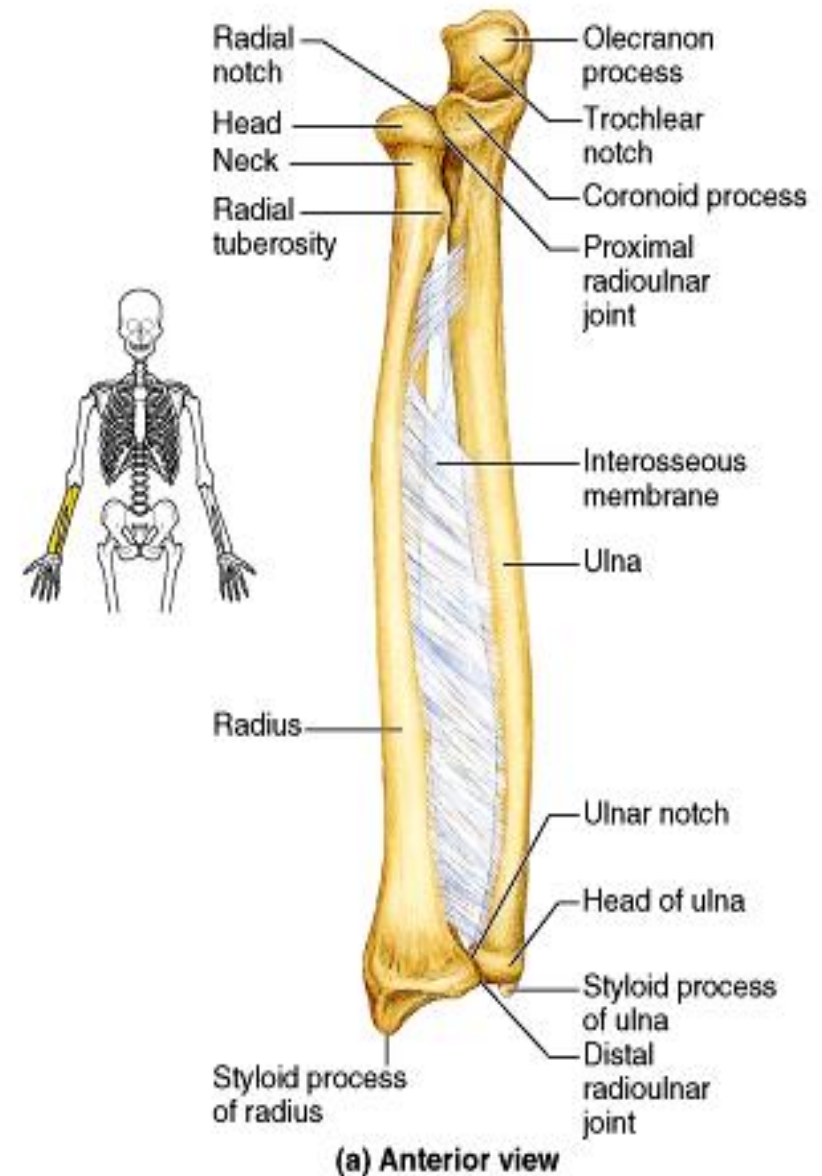
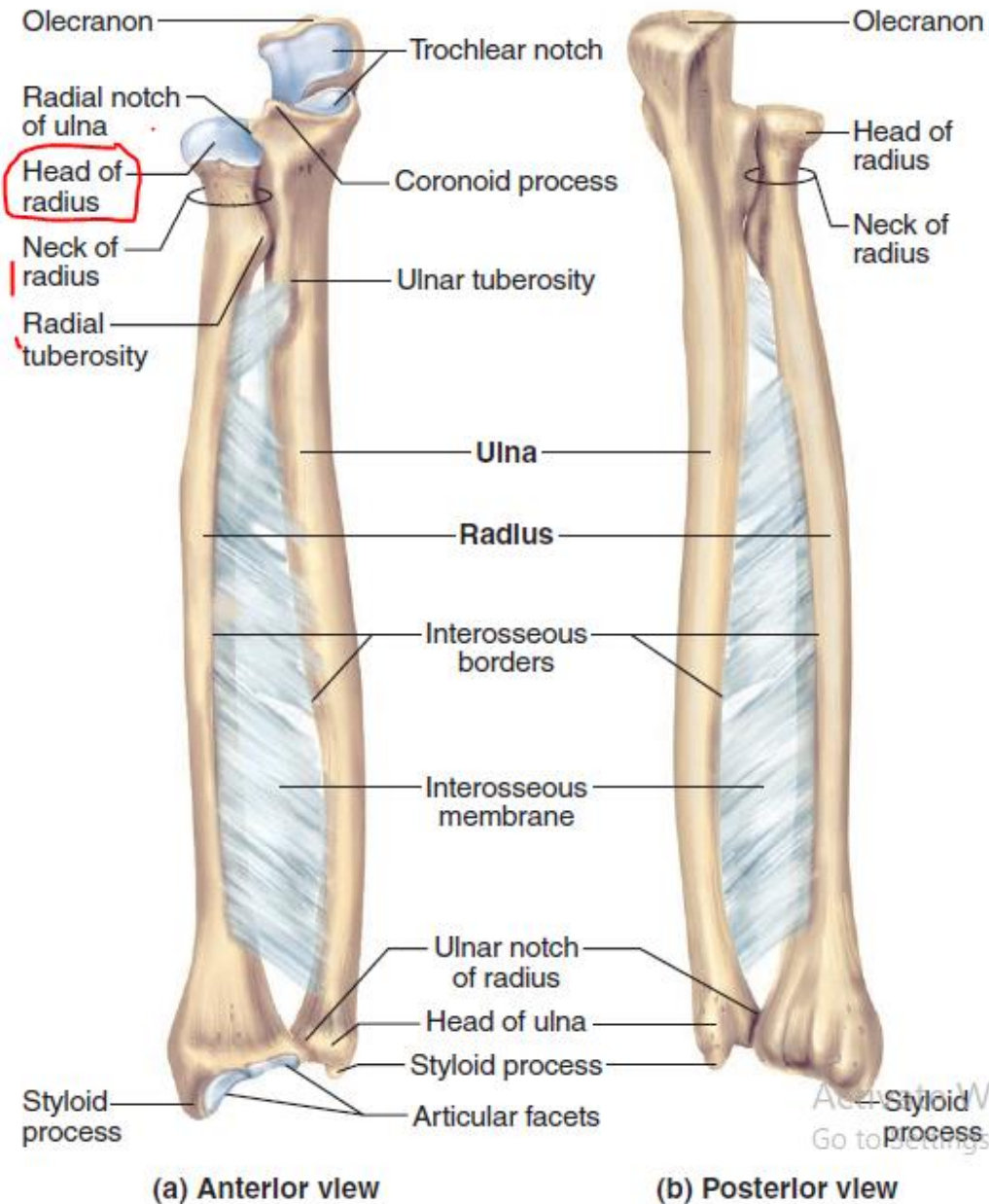
- The **radius** is the lateral and shorter of the two forearm bones.
- Its proximal end includes a short head, neck, and medially directed tuberosity.
- Proximally, the smooth superior aspect of the discoid **head of the radius** is concave for articulation with the capitulum of the humerus during flexion and extension of the elbow joint.



Radius,,,

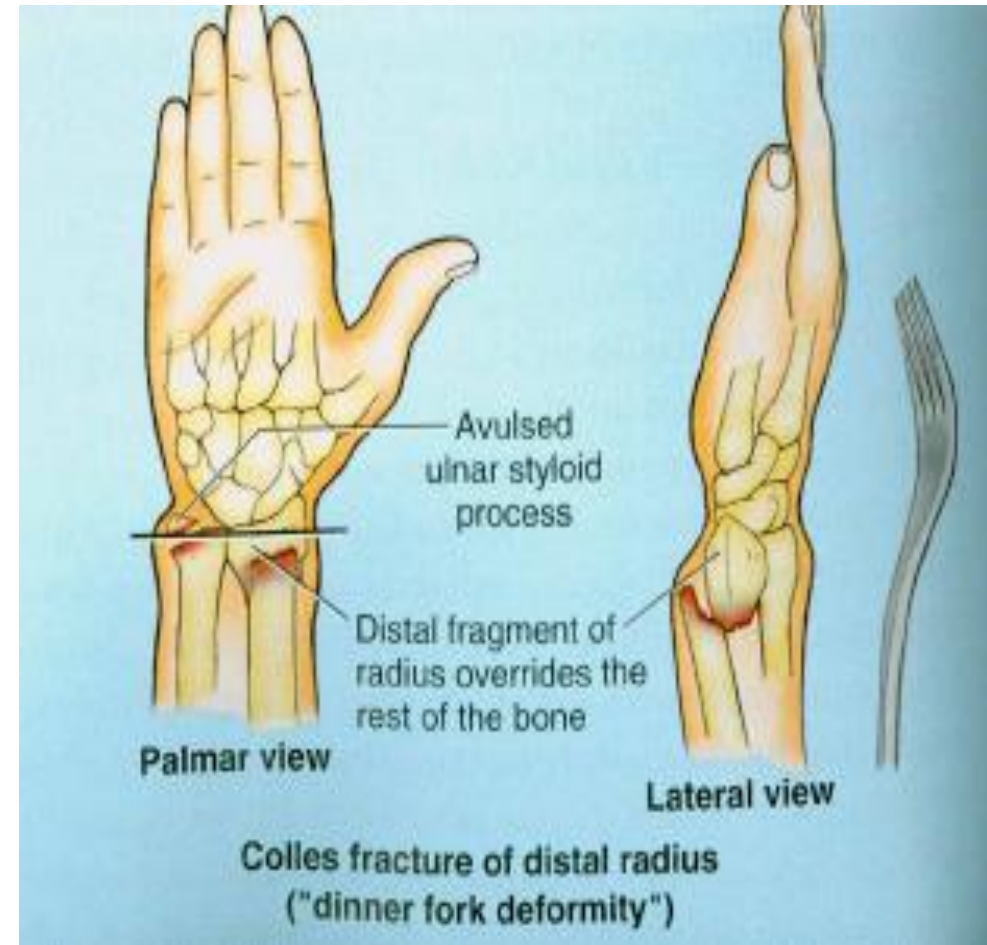
- The **neck of the radius** is a constriction distal to the head.
- The oval **radial tuberosity** is distal to the medial part of the neck, and demarcates the proximal end (head and neck) of the radius from the shaft.
- The **shaft of the radius**, in contrast to that of the ulna, gradually enlarges distally and the distal end of the radius is essentially four sided when sectioned.
- Its medial aspect forms a concavity, the **ulnar notch**, which accommodates the head of the ulna.
- Its lateral aspect becomes increasingly ridge-like, terminating distally in the **radial styloid process**.

Radius,,



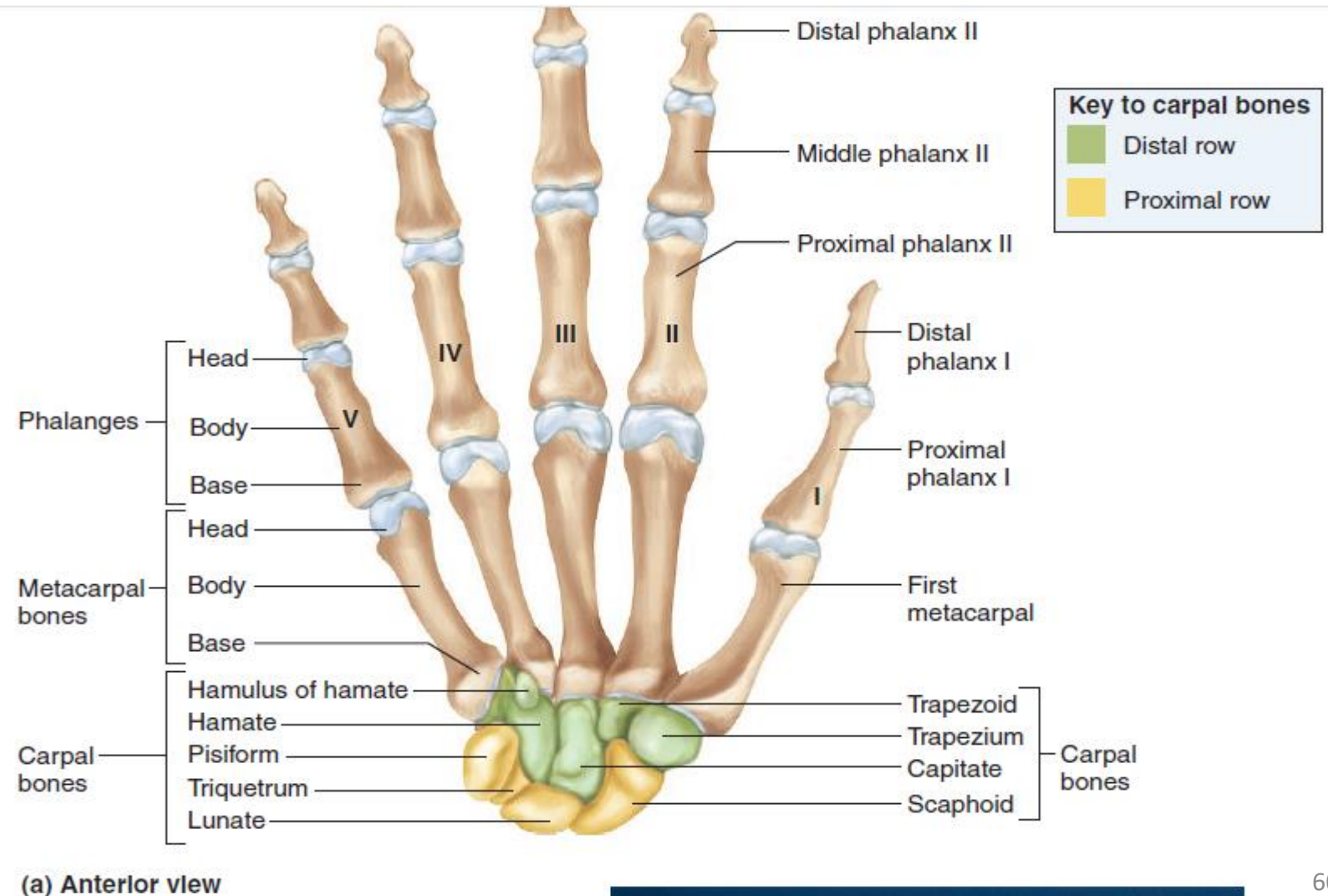
Fracture of ulna and radius

- **Colles fracture**
 - A complete fracture of the distal end of the **radius**
 - Results from forced dorsiflexion of the hand
 - The distal fragment of the radius is displaced dorsally and often broken into pieces

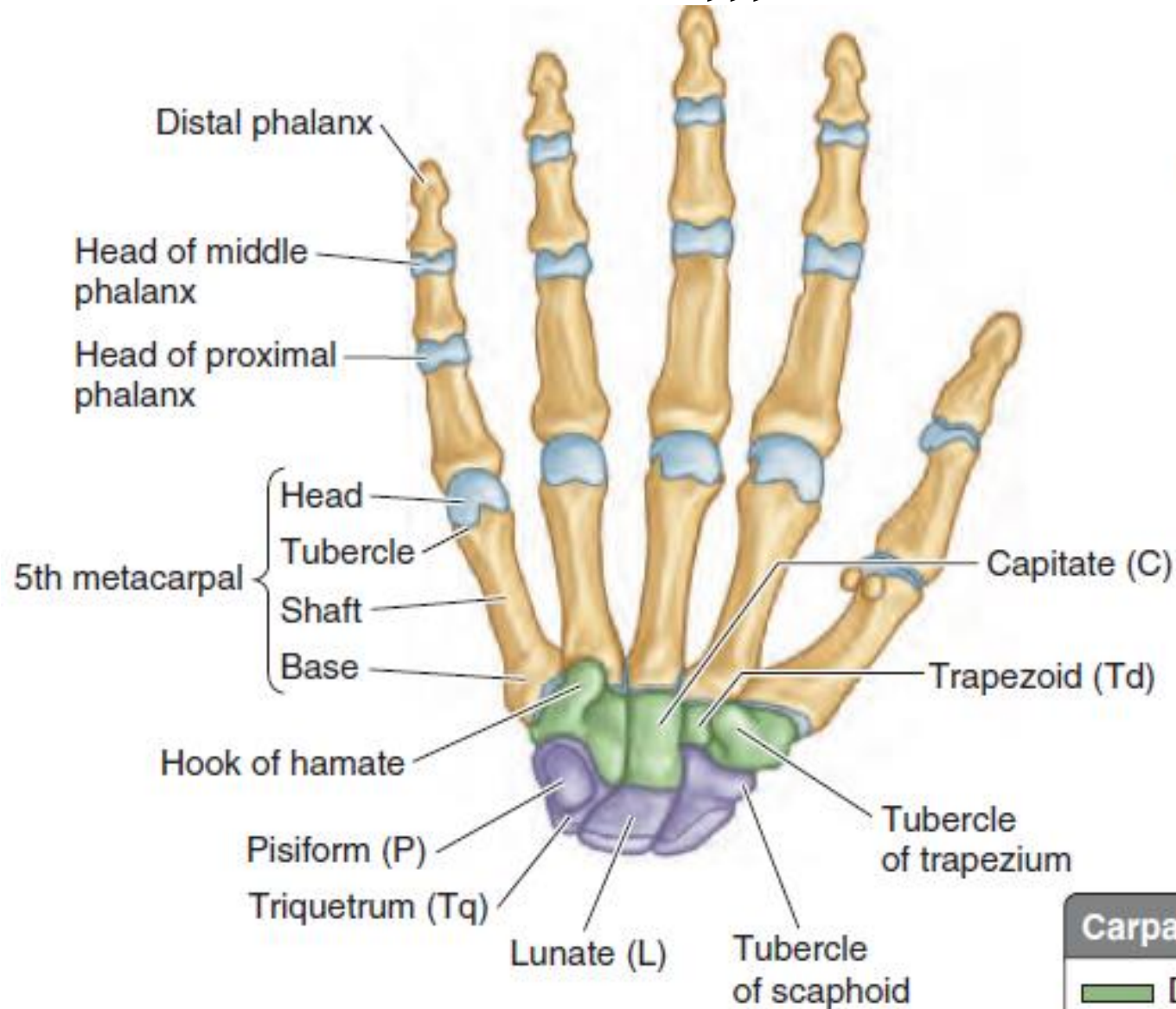


Bones of Hand

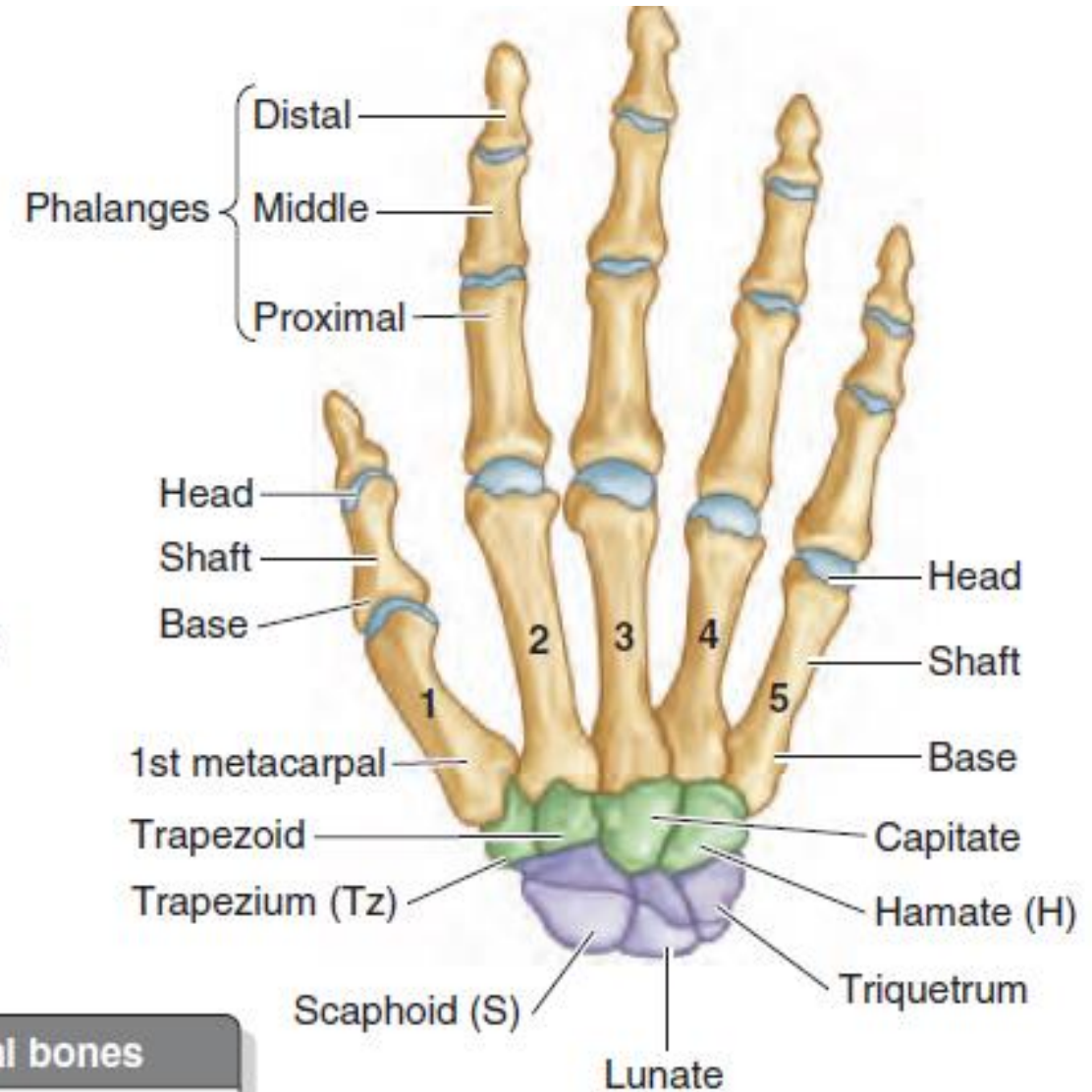
- The skeleton of the hand consists of the carpal bones, metacarpals, and phalanges



Bones of Hand,,



(A) Anterior (palmar) view



(B) Posterior (dorsal) view

Carpal bones	
	Distal row
	Proximal row

The carpal bones

- The **carpal bones**, or wrist bones, consist of 8 small bones that are arranged in two transverse rows of four bones each.
- They are joined by ligaments that allow limited gliding movement.
- The carpal bones of the proximal row, starting at the lateral (thumb) side, are the **scaphoid, lunate, triquetrum**, and **pisiform**.
- Unlike the other carpal bones, the pisiform is a sesamoid bone; it is not present at birth but develops around the age of 9 to 12 years within the tendon of the *flexor carpi ulnaris muscle*.
- The bones of the distal row, again starting on the lateral side, are the **trapezium, trapezoid, capitate**, and **hamate**.

The carpal bones,,,

- ❖ Proximal row articulate with radius and articular disc of wrist joint proximally and distal row of carpals distally.
- ❖ Distal row articulate with proximal row of carpals proximally and metacarpals distally

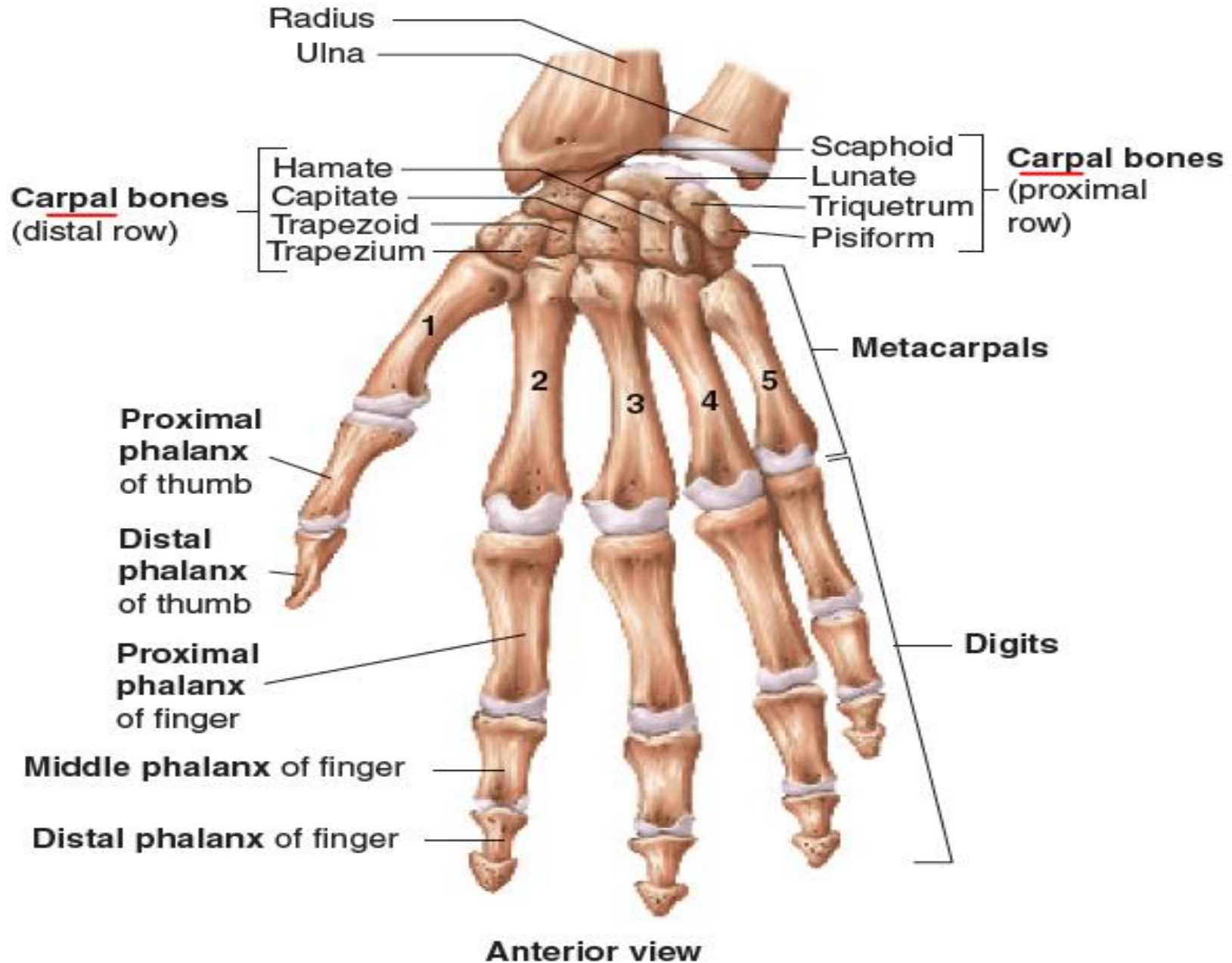
❑ Proximal row from lateral to medial:

- ✓ Scaphoid: boat-shaped with scaphoid tubercle
- ✓ Lunate: moon-shaped
- ✓ Triquetrum: pyramidal
- ✓ Pisiform: pea-shaped; lies on palmar surface of triquetrum

❑ Distal row from lateral to medial:

- ✓ Trapezium: four sided
- ✓ Trapezoid: wedge-shaped
- ✓ Capitate: head like; largest
- ✓ Hamate: wedge-shaped, has hooked process (hook of hamate)

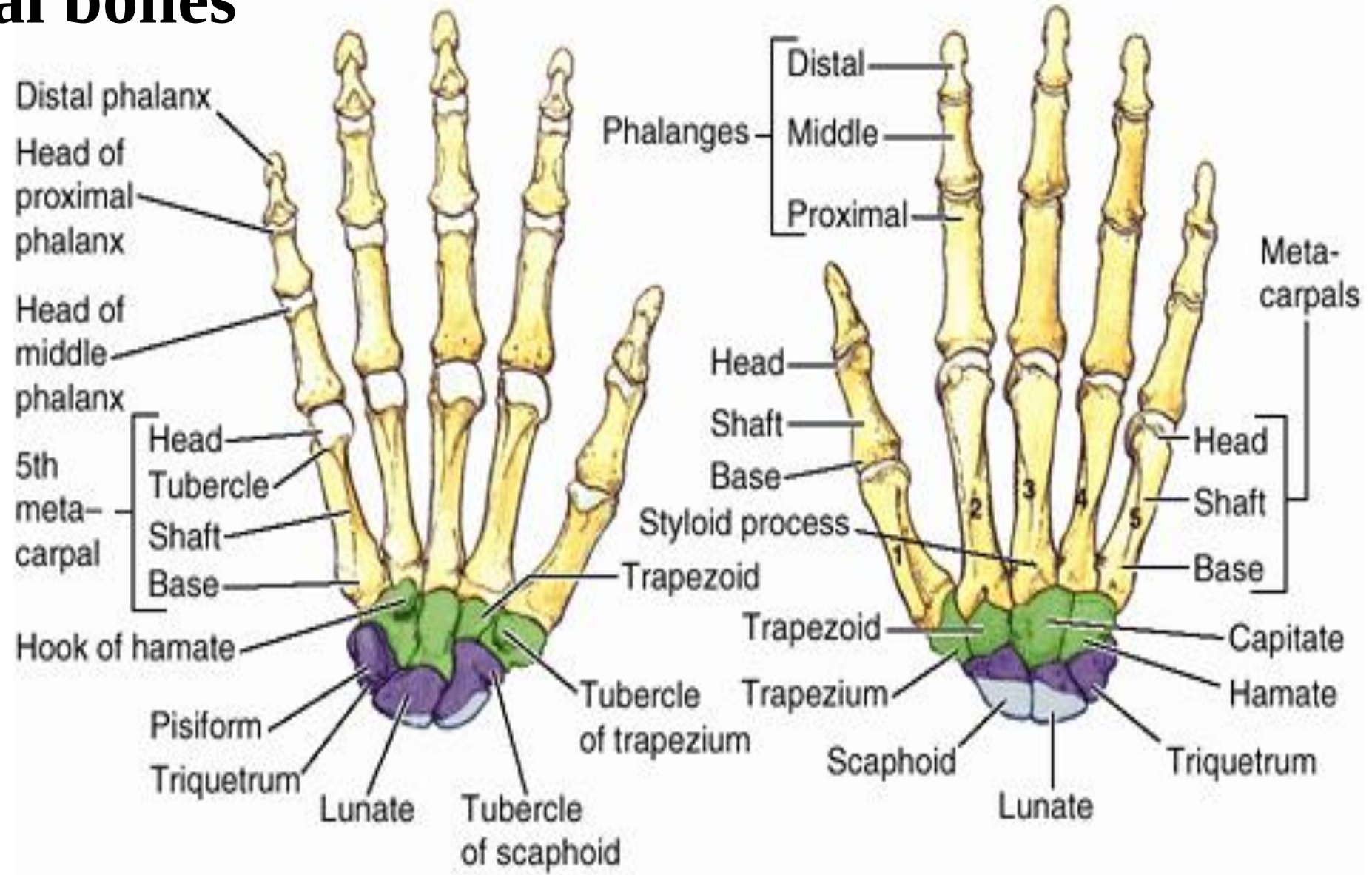
The carpal bones,,,



Metacarpal bones

- The **metacarpus** forms the skeleton of the palm of the hand between the carpus and the phalanges.
- Composed of **five** metacarpal bones
- Each metacarpal consists of a base, shaft, and head.
- The proximal **bases of the meta carpals** articulate with the carpal bones, and the distal **heads of the metacarpals** articulate with the proximal phalanges, and form the **knuckles** of the hand.
- The 1st metacarpal (of the thumb) is the thickest and shortest of these bones.
- The 3rd metacarpal is distinguished by a **styloid process** on the lateral side of its base

Metacarpal bones

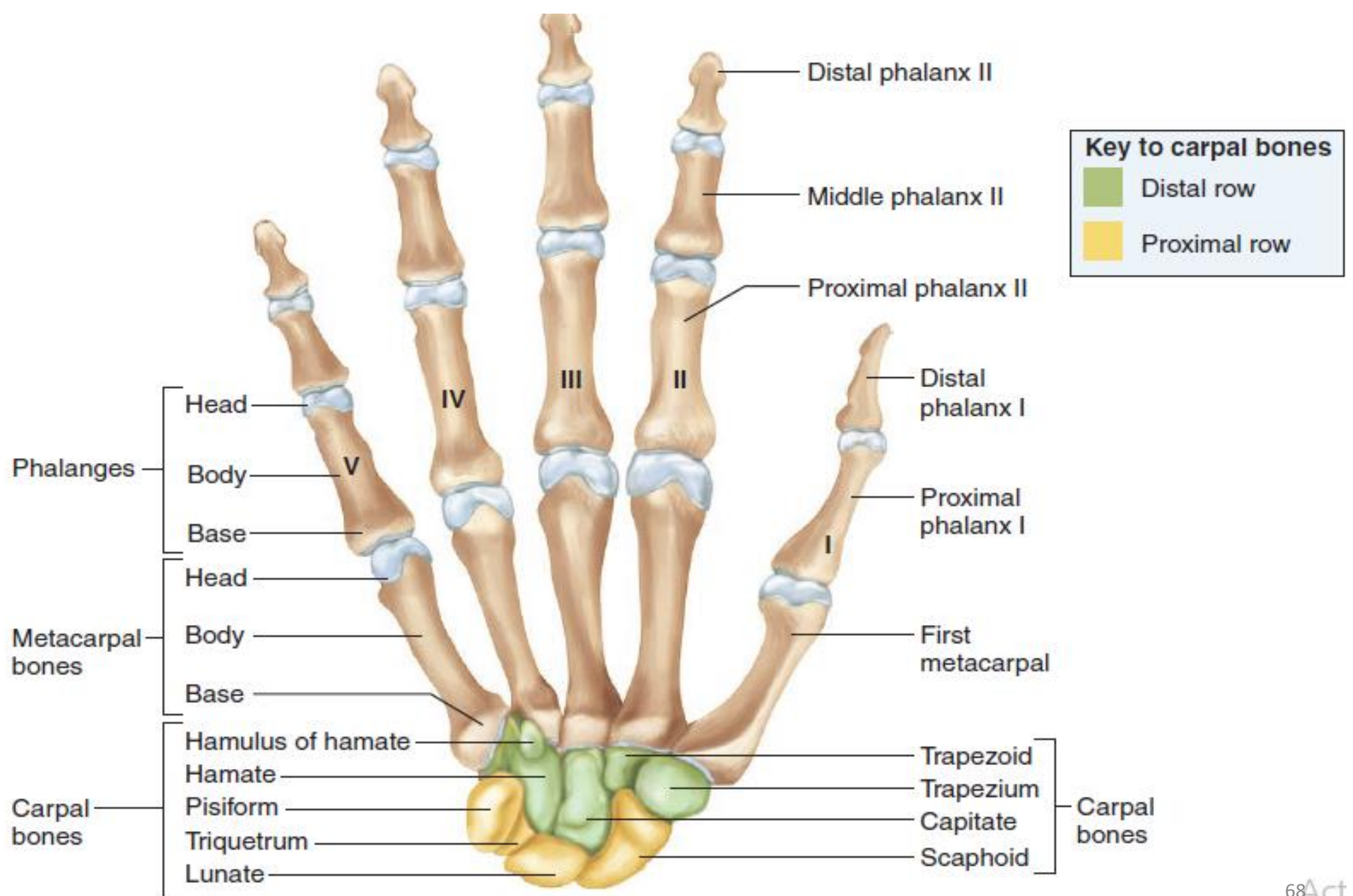


(A) Anterior (palmar) view

(B) Posterior (dorsal) view

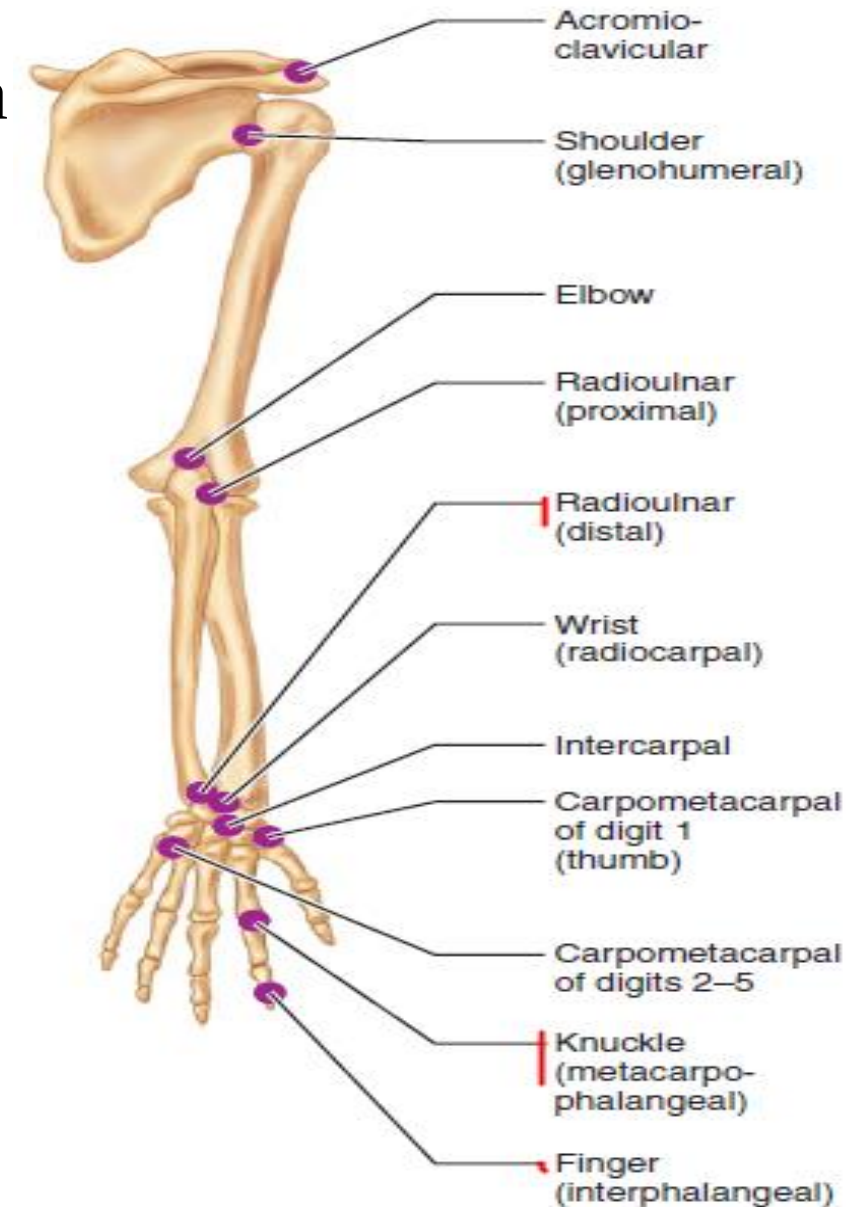
The Phalanges

- The bones of the fingers are called **phalanges**.
- Each digit (finger) has three **phalanges** except for the first (the thumb), which has only two; however, the phalanges of the first digit are stouter than those in the other fingers.
- Composed of **five 14** bones
- Each phalanx has a **base** proximally, a **shaft** (body), and a **head** distally.
- The proximal phalanges are the largest, the middle ones are intermediate in size, and the distal ones are the smallest.
- The terminal phalanges are flattened and expanded at their distal ends, which underlie the nail beds



Joints of the upper limb

- A **joint**, also called an **articulation** or **arthrosis**, is a point of contact between two bones, between bone and cartilage, or between bone and teeth.
- **Arthrology** is the scientific study of joints.
- The study of musculoskeletal movement is **kinesiology**.
- Joints are commonly named according to the bones that are united at the joint. For example, the *atlanto occipital joint*, *glenohumeral joint*, *radioulnar joint*



Joints of the upper limb,,

- Joints are **classified structurally**, based on their anatomical characteristics, and **functionally**, based on the type of movement they permit.
- The *structural classification* is based on the material that binds the bones together and on the presence or absence of a joint cavity

❑ Structurally, joints are classified as:

- ✓ **Fibrous joints:** There is no synovial cavity, and the bones are held together by dense irregular connective tissue that is rich in collagen fibers.
- ✓ **Cartilaginous joints:** There is no synovial cavity & the bones are held together by cartilage.
- ✓ **Synovial joints:** The bones forming the joint have a synovial cavity and are united by the dense irregular connective tissue of an articular capsule, and often by accessory ligaments.

Joints of the upper limb,,,

❑ The functional classification of joints relates to the degree of movement they permit.

❑ A joint is usually considered movable, but that is not always true.

- Functionally, joints are classified as one of the following types:

I. **Synarthrosis:** An immovable joint. The plural is *synarthroses*.

II. **Amphiarthrosis:** A slightly movable joint. The plural is *amphiarthroses*.

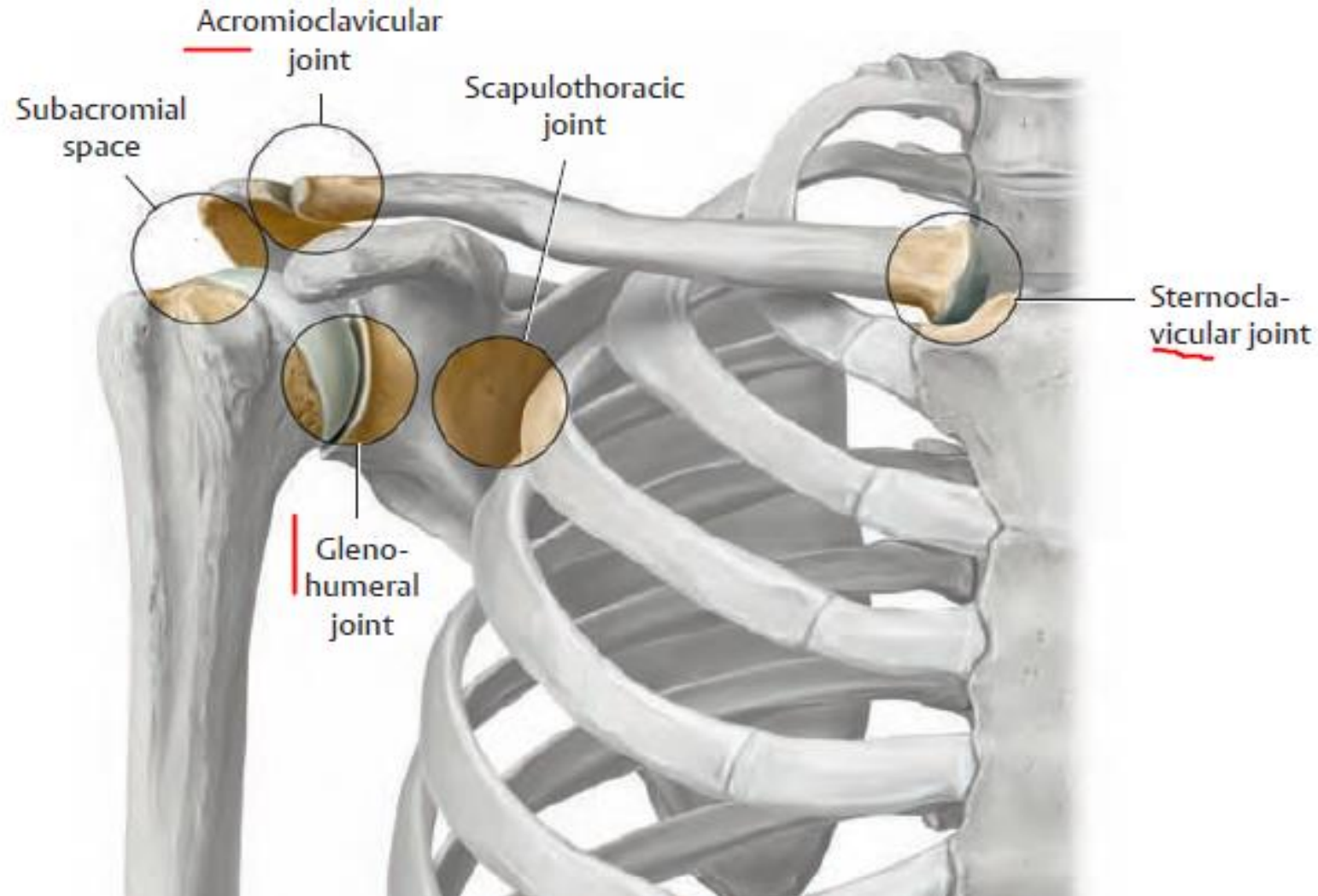
III. **Diarthrosis:** A freely movable joint. The plural is *diarthroses*.

- All diarthroses are synovial joints.

- They have a variety of shapes and permit several different types of movements.

The shoulder joint

- The shoulder joint complex is composed of three joints; **the sternoclavicular, acromioclavicular & glenohumeral** joints, usually all moving simultaneously.

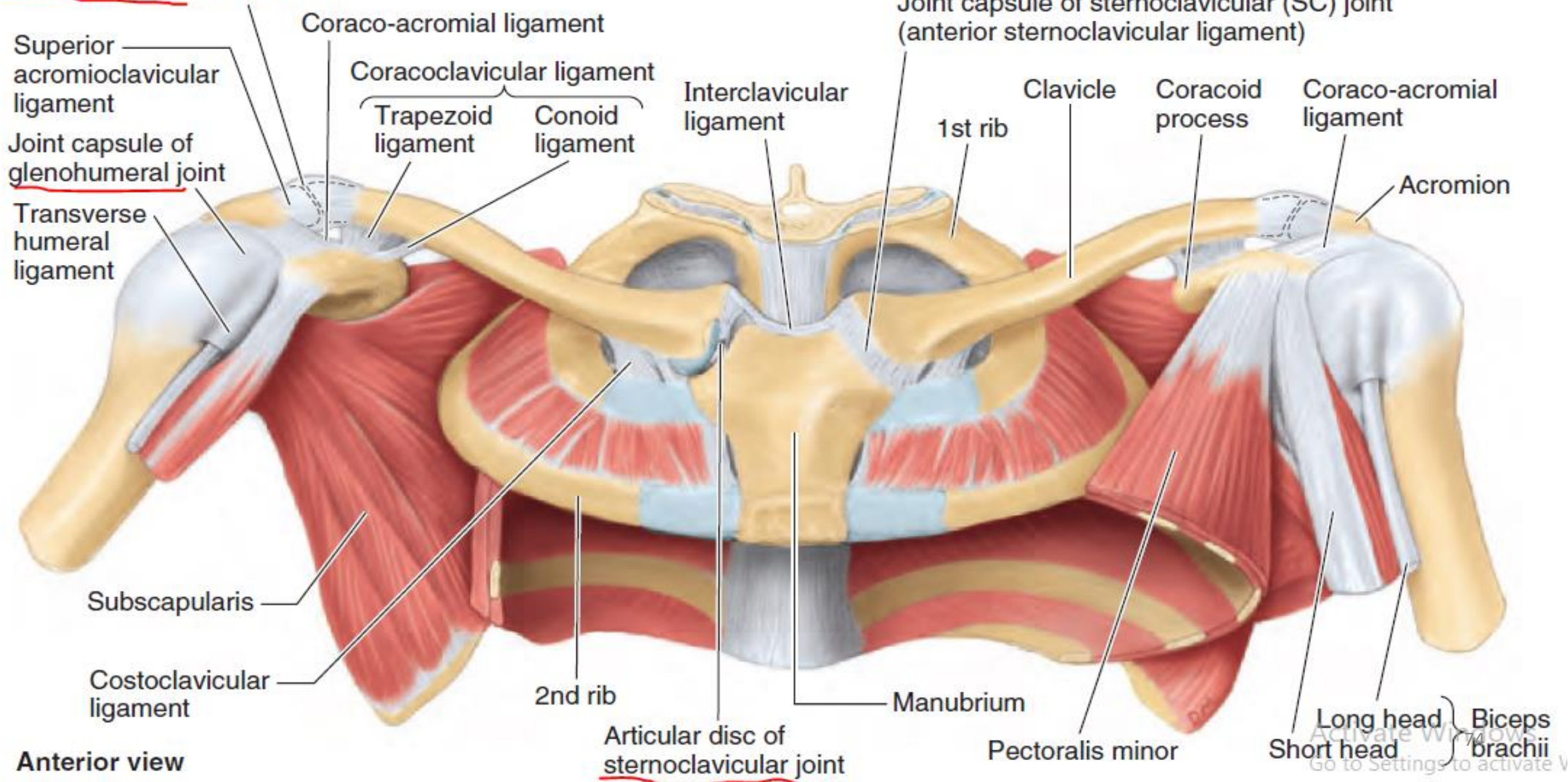


The shoulder joint,,

- The mobility provided by **three** joints working together gives the shoulder complex a wide range of movement.
- The extensive mobility in the shoulder joint complex also results from the flexible support provided by numerous muscles, ligaments, and tendons.
- Functional defects in any of the joints impair movements of the pectoral girdle.
- The bones articulating as part of the shoulder joint complex are the humerus, scapula, clavicle, and sternum.

The shoulder joint,,

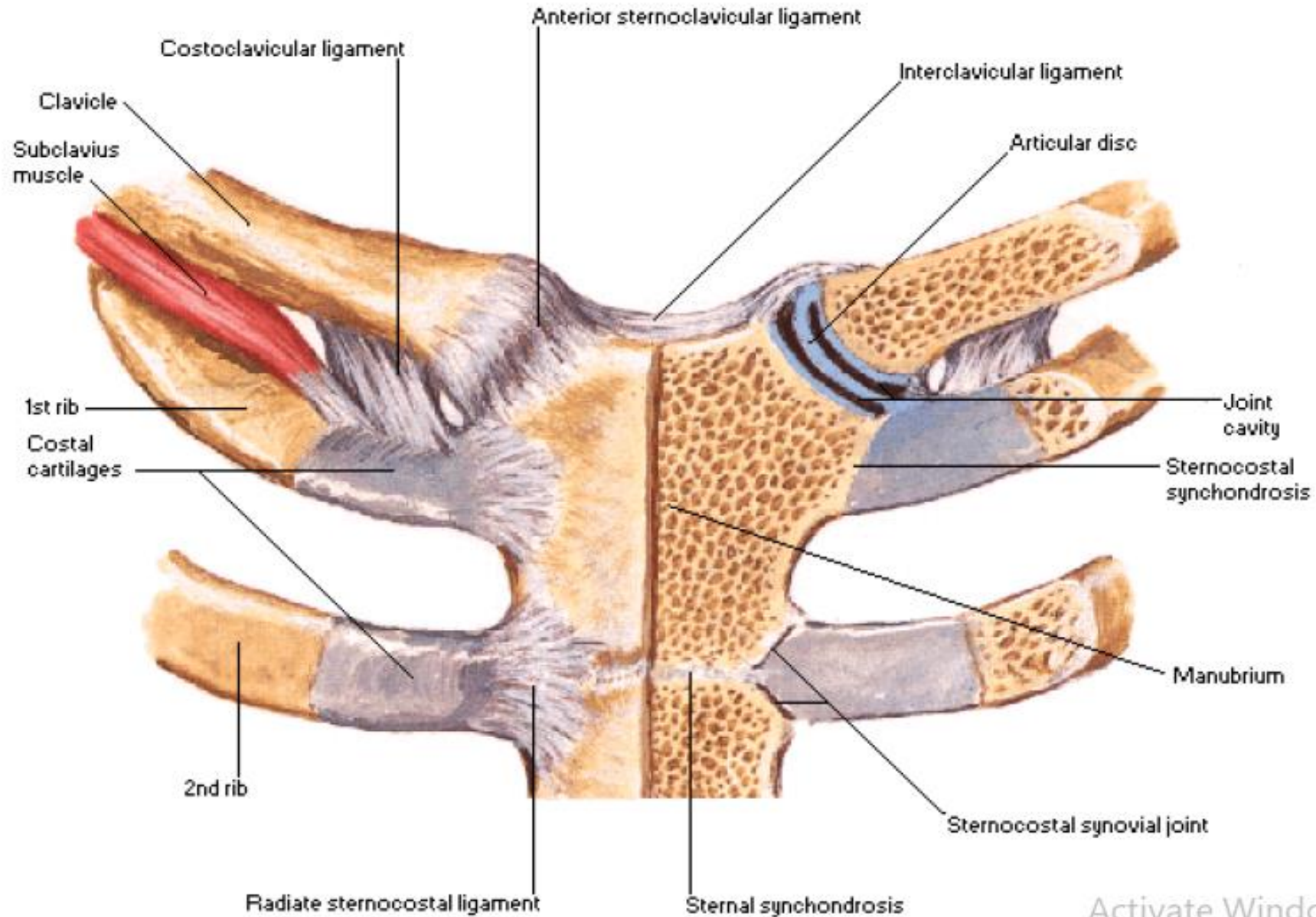
Acromioclavicular (AC) joint



Sternoclavicular Joint

- The presence of an articular disc separates the joint into two cavities, which prevents the clavicle from overriding the sternum.
- ✓ **Type: saddle synovial articulation**
- ✓ It is between sternal end of clavicle and manubrium of sternum and 1st costal cartilage
- ✓ The SC joint is the only articulation between the upper limb and the axial skeleton,
- ✓ divided into two compartments by articular disc
- ❖ **Joint capsule** the joint is surrounded by a capsule and is reinforced by ligaments.

Sternoclavicular Joint,,

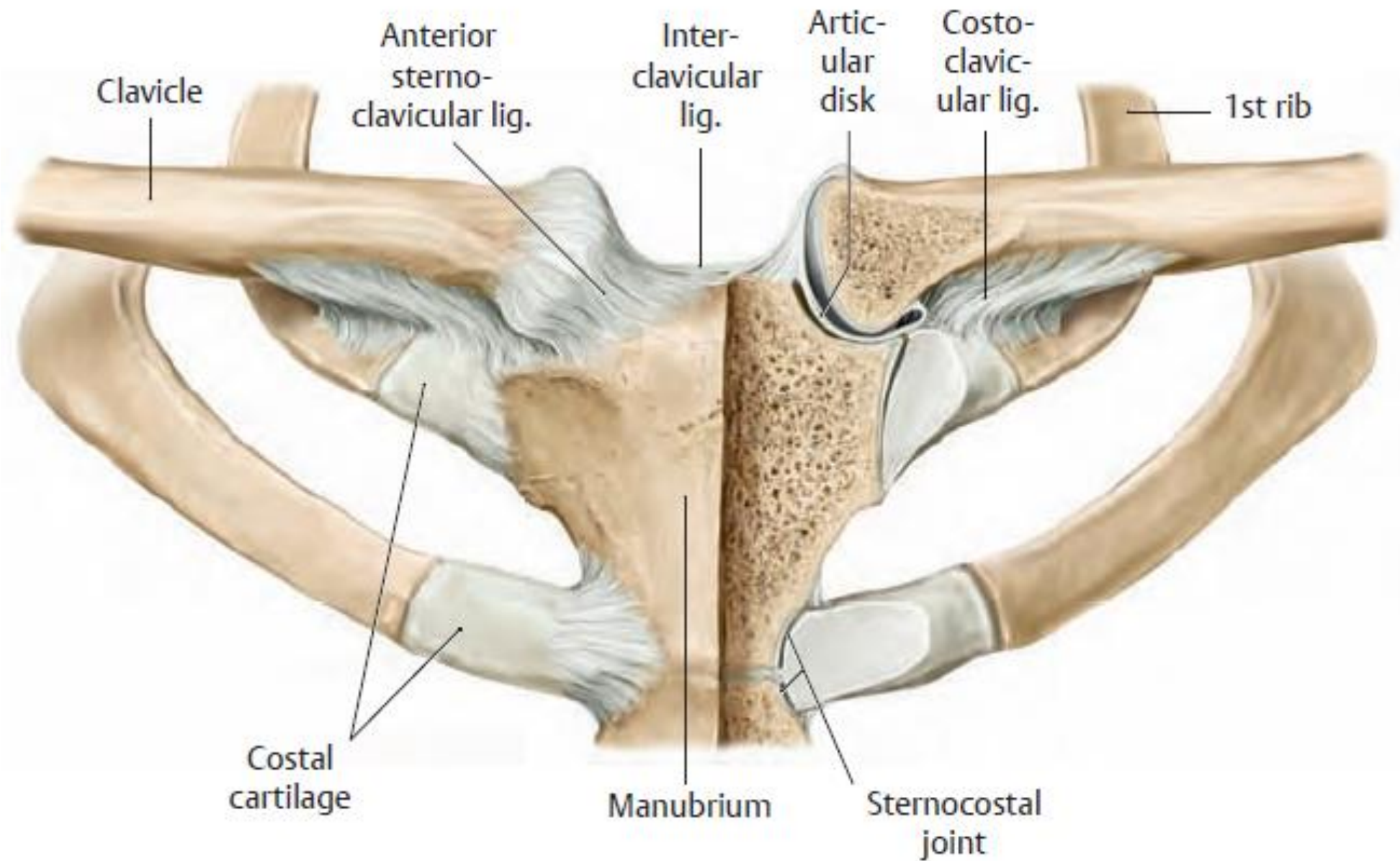


Sternoclavicular Joint ,,

□ Ligaments of Sternoclavicular joint:

- ✓ **Anterior and posterior SC ligaments** reinforce joint capsule anteriorly & posteriorly
- ✓ **Interclavicular ligament** strengthens the capsule superiorly
- ✓ **Costoclavicular ligament** • anchors the inferior surface of the sternal end of the clavicle to the 1st rib and its costal cartilage and limiting elevation of the pectoral girdle
- The strength of the SC joint depends on its ligaments and articular disc.
- ❖ **Sternoclavicular joint** allows elevation and depression, protraction and retraction, and circumduction of the shoulder.

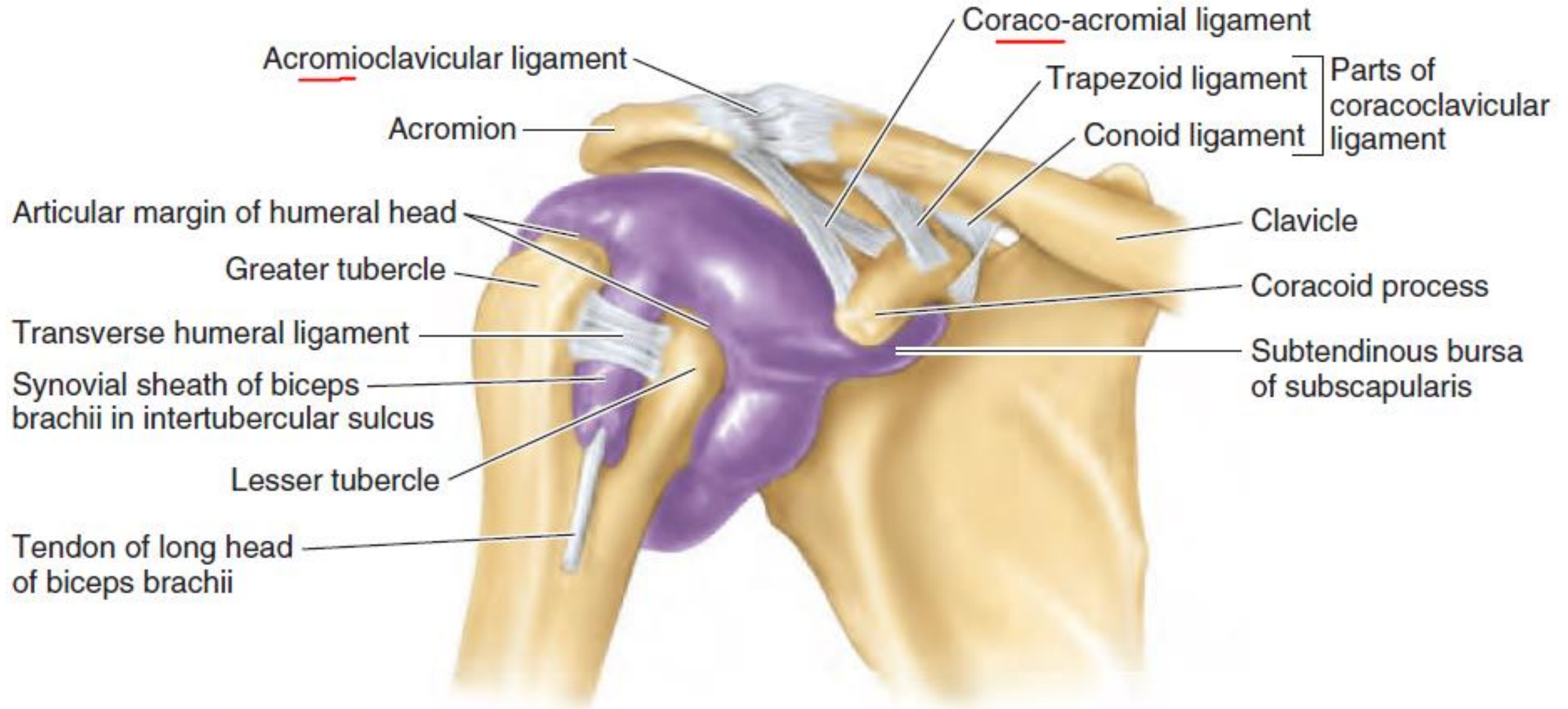
Sternoclavicular Joint ,,



Acromioclavicular Joint

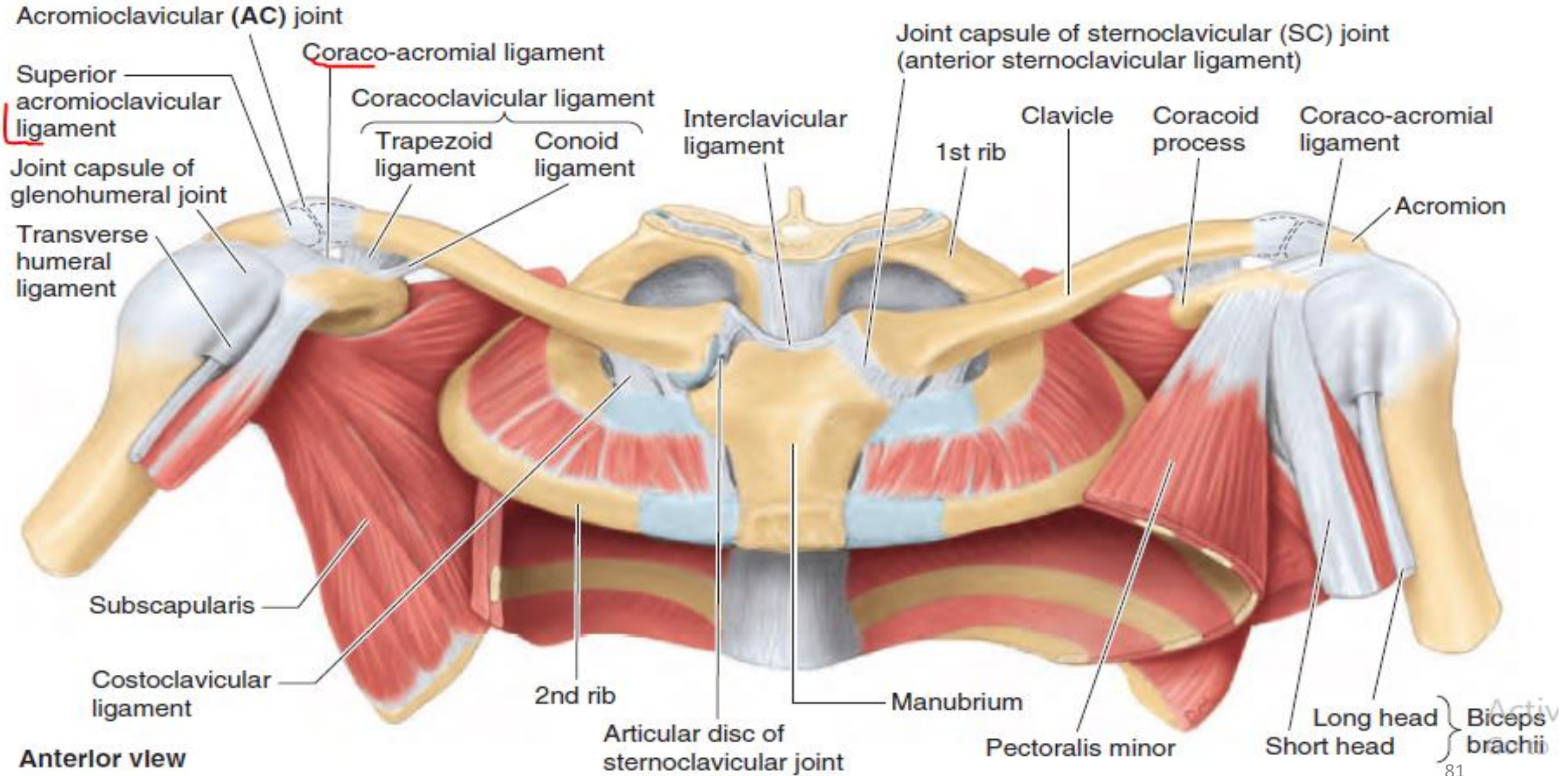
- The **acromioclavicular joint** (AC joint) is a plane type of synovial joint b/n the lateral end of the clavicle and the medial surface of the acromion of the scapula.
- **This joint** allows a gliding movement when the scapula rotates.
- The joint is surrounded by a capsule and is reinforced by several ligaments.
- Superiorly, the capsule is thickened to form the **acromioclavicular ligament**.
- However, the major support for the joint is the **coracoclavicular ligament**.
- This ligament is composed of 2 parts the trapezoid and conoid ligaments, which connect the inferior aspect of the clavicle to the coracoid process of the scapula

Acromioclavicular Joint,,



(A) Anterior view

Acromioclavicular Joint,,



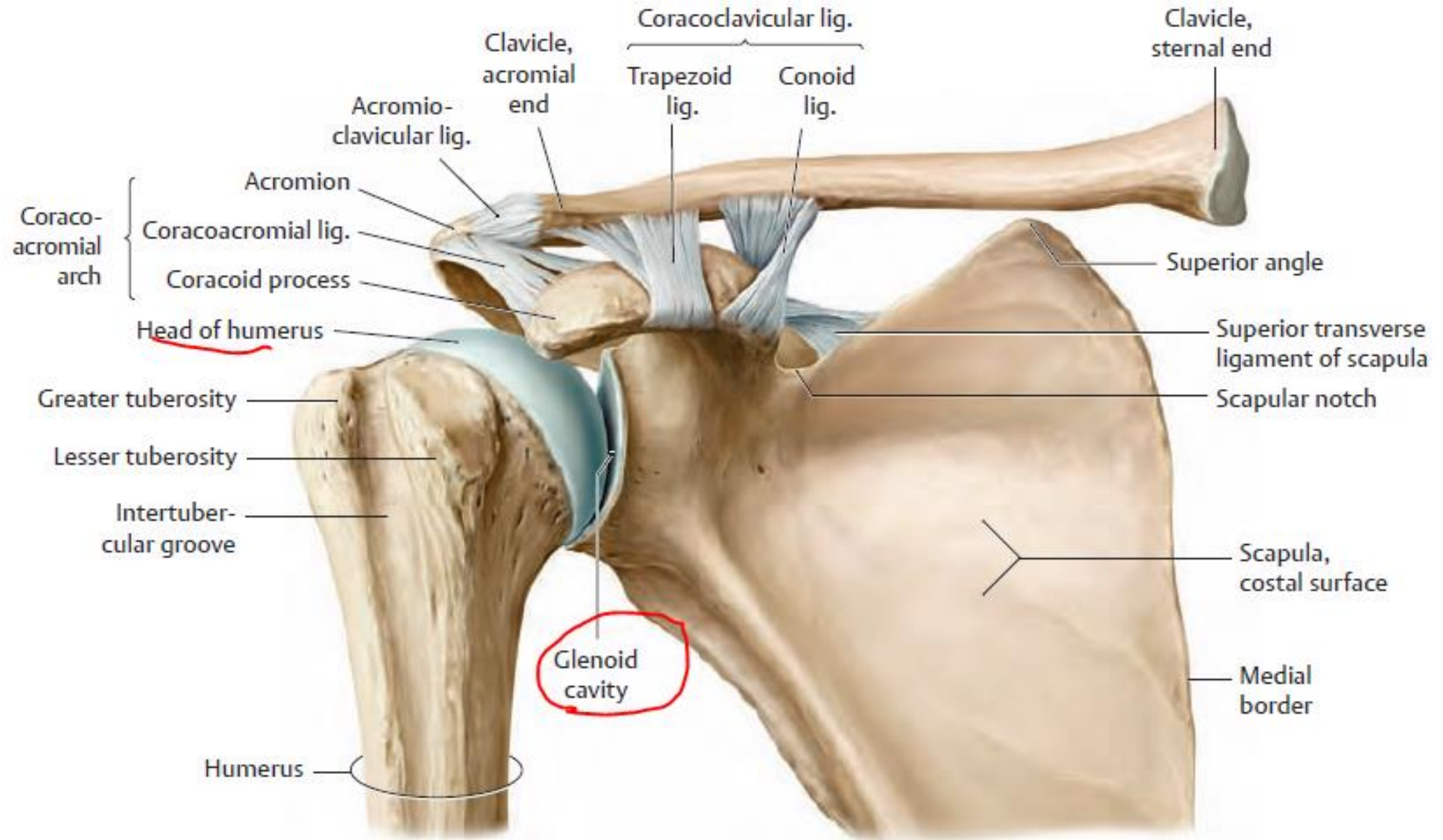
Glenohumeral joint

- The **glenohumeral (shoulder) joint** is a multiaxial, ball-and socket synovial joint which allows abduction & adduction, flexion & extension, and circumduction & rotation.
- This joint has more freedom of movement than any other joint in the body.
- The head of the humerus is large compared with the shallow glenoid cavity, this joint is relatively unstable but several factors help to increase the its stability.
- Is surrounded by the **fibrous capsule** attached to the margin of the glenoid cavity.
- **The capsule** is reinforced by the **rotator cuff muscles**, the **glenohumeral ligaments**, and the **coracohumeral ligaments**.

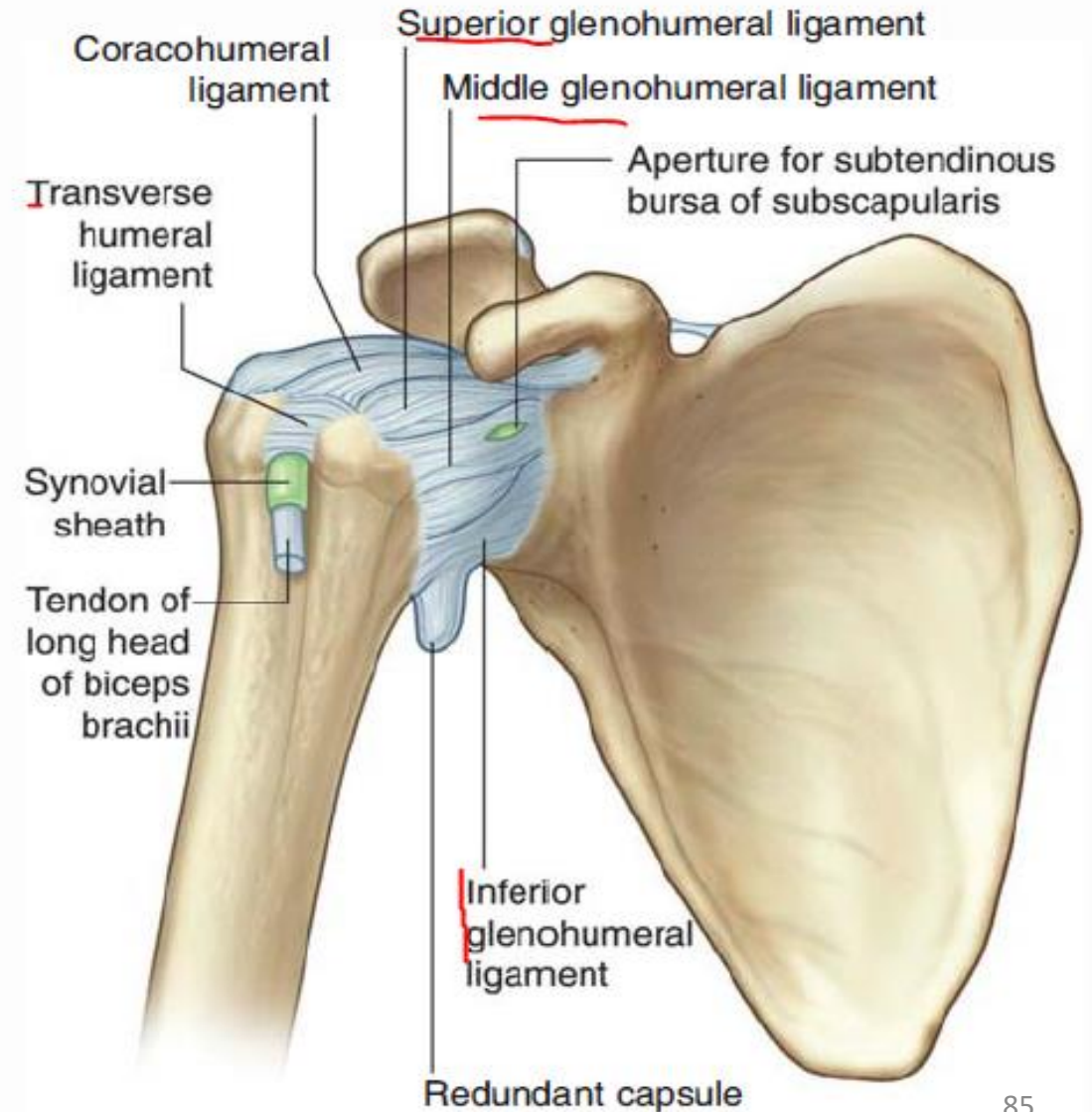
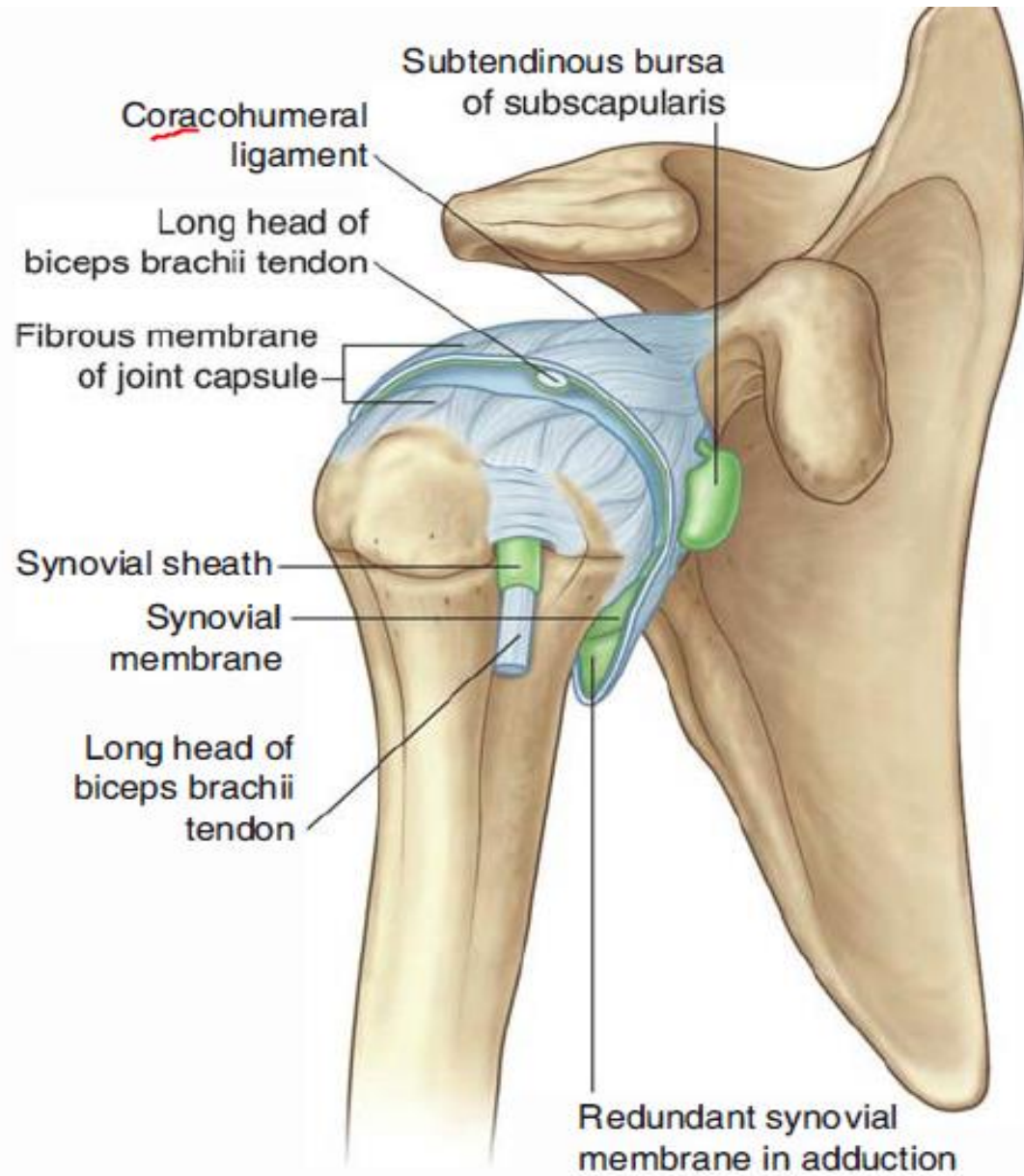
Glenohumeral joint,,

- The joint capsule is reinforced **anteriorly** by the **superior, middle and inferior glenohumeral ligaments**.
- These are referred to as **intrinsic ligaments** due to the fact that they are part of the joint capsule.
- Superiorly, the capsule is reinforced by a further intrinsic ligament – the **coracohumeral** ligament.
- Also reinforcing the superior aspect of the joint is the **coracoacromial** ligament, which forms part of the **coraco-acromial** arch (along with the acromion and coracoid process).

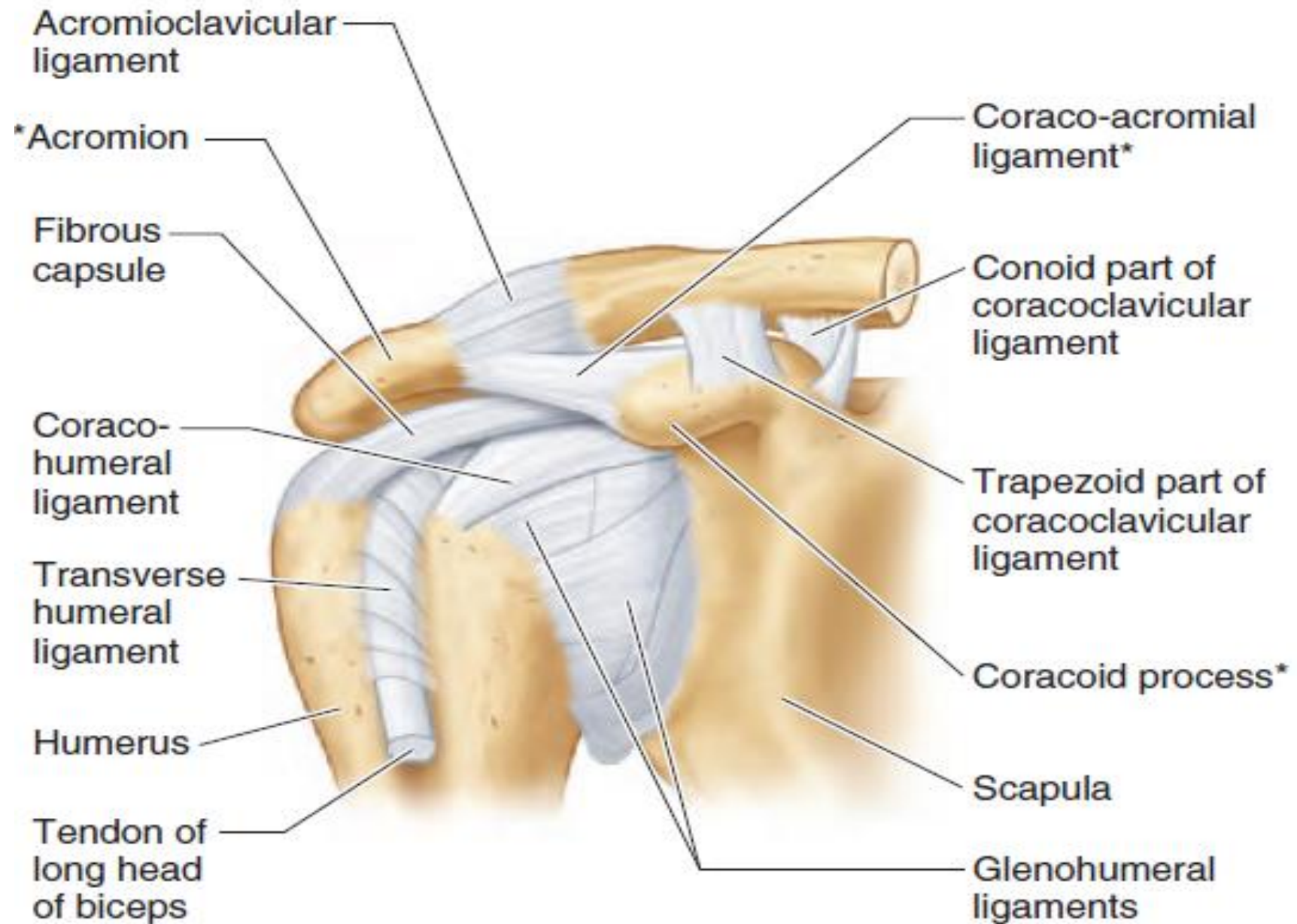
Glenohumeral joint,,



Glenohumeral joint,,

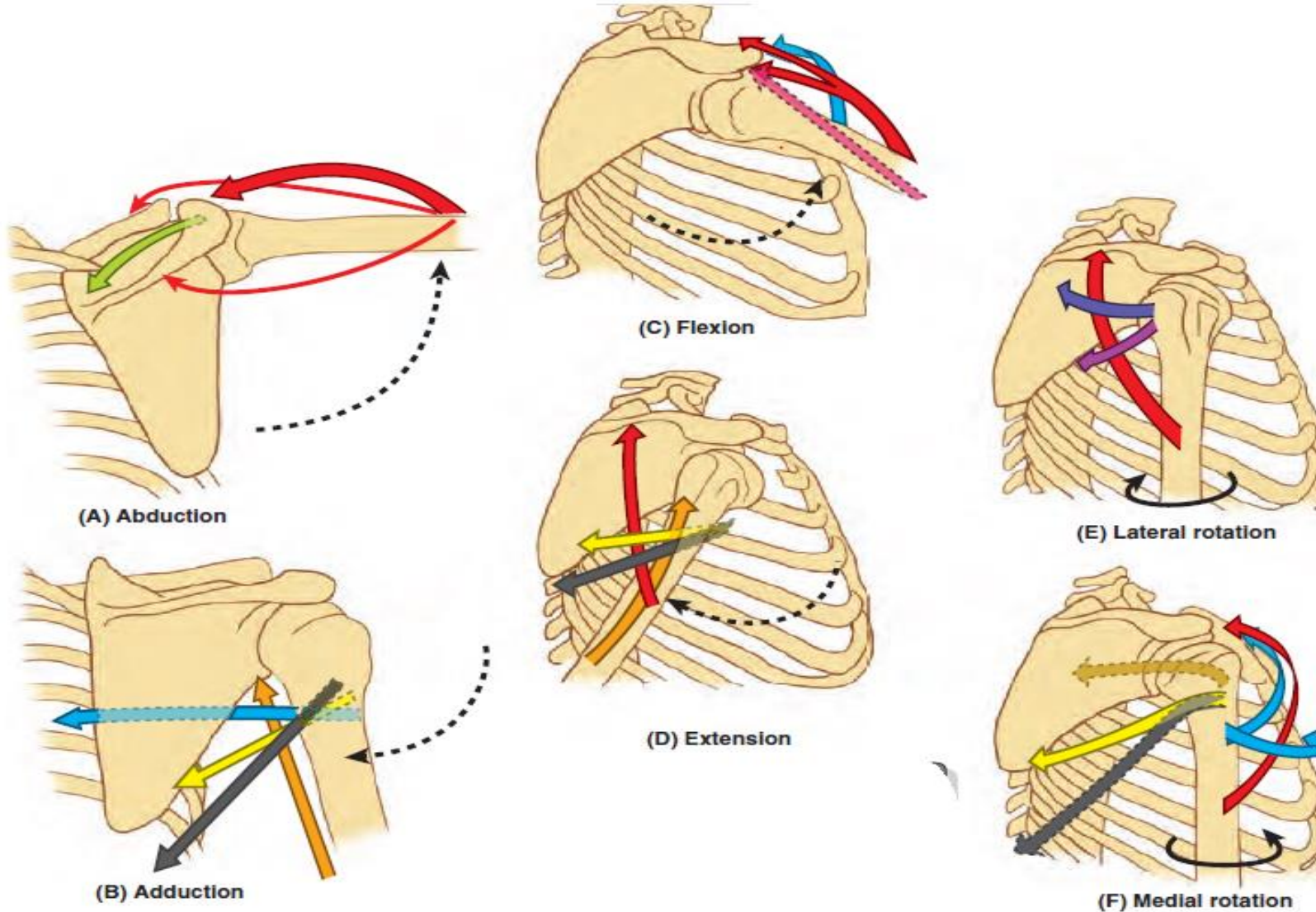


Glenohumeral joint,,,



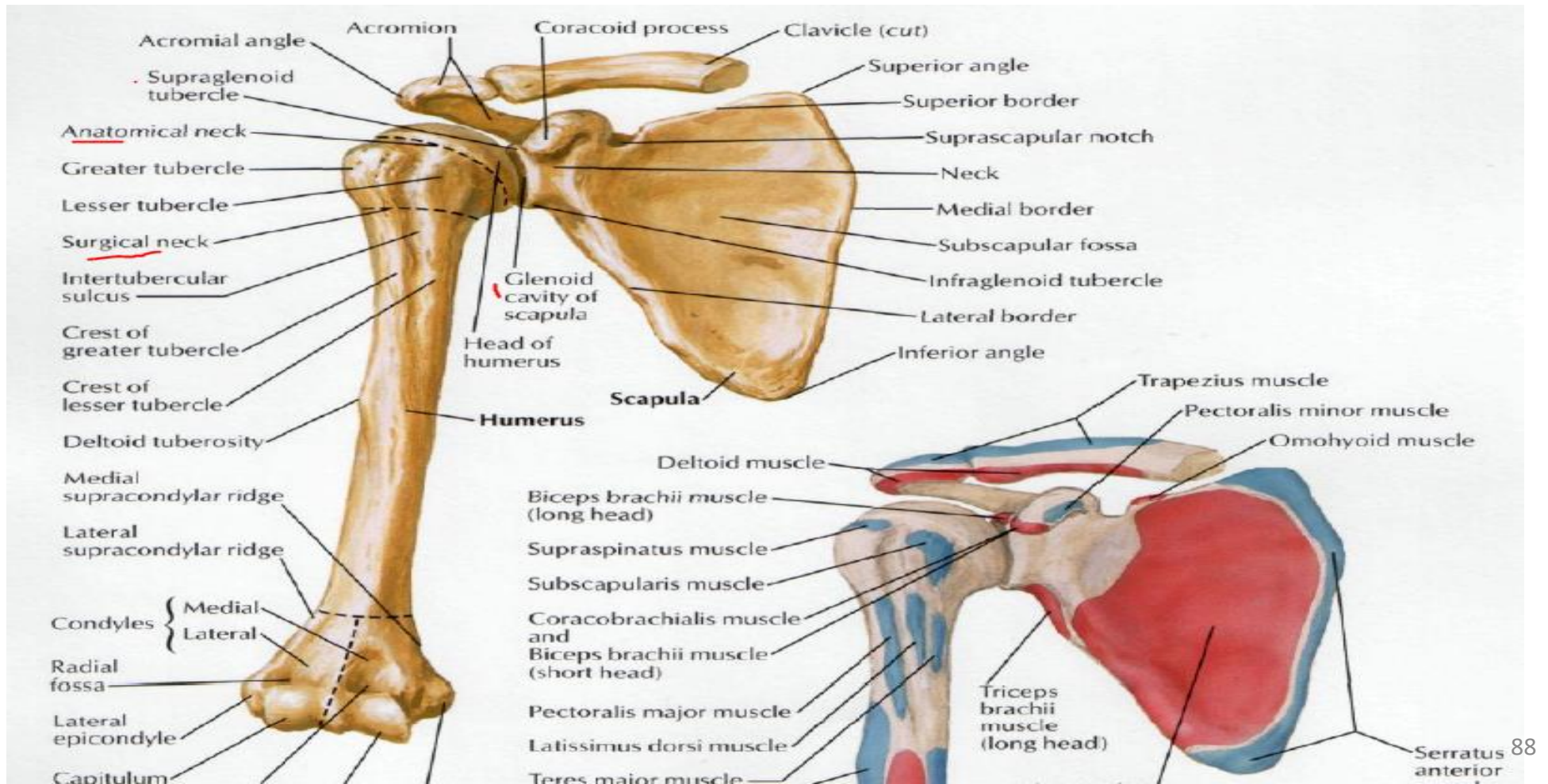
Glenohumeral joint,,

□ Movements of glenohumeral joint.



Glenohumeral joint,,

- Its position prevents superior dislocation of the joint.
- Muscles that act on the **glenohumeral joint** include *axio-appendicular muscles, and scapulohumeral muscles,*

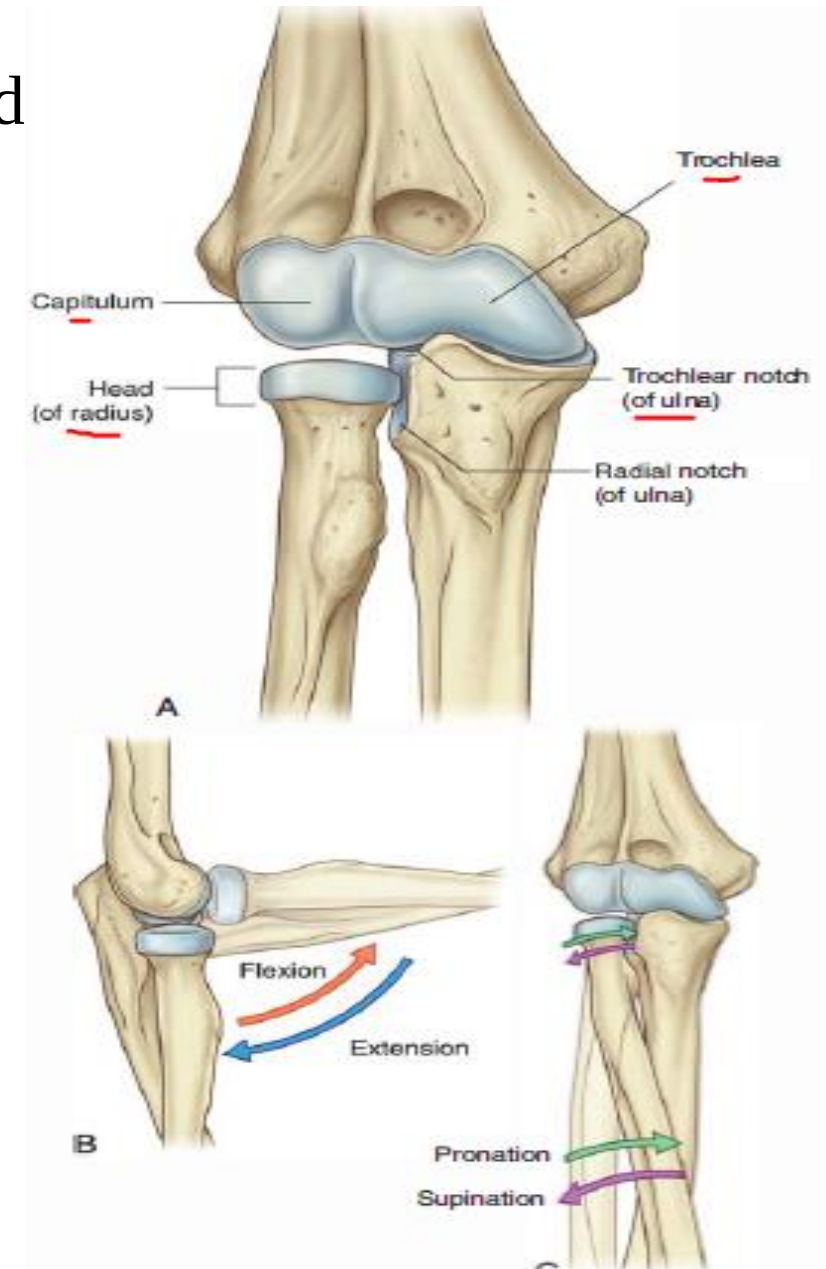


Elbow Joint

- The elbow is a synovial joint at the junction of the arm and forearm.
- It joins the distal end of the humerus to the proximal ends of the radius & ulna.

□ The elbow joint comprises 3 distinct articulations:

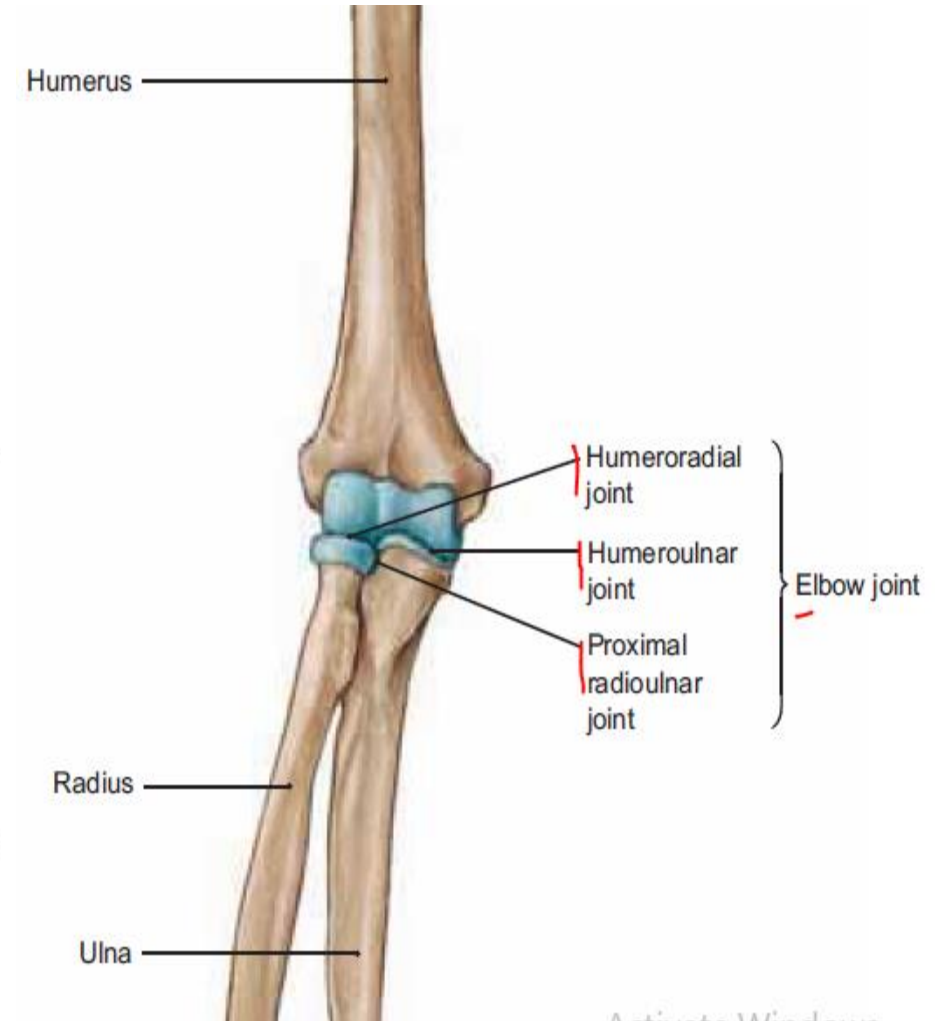
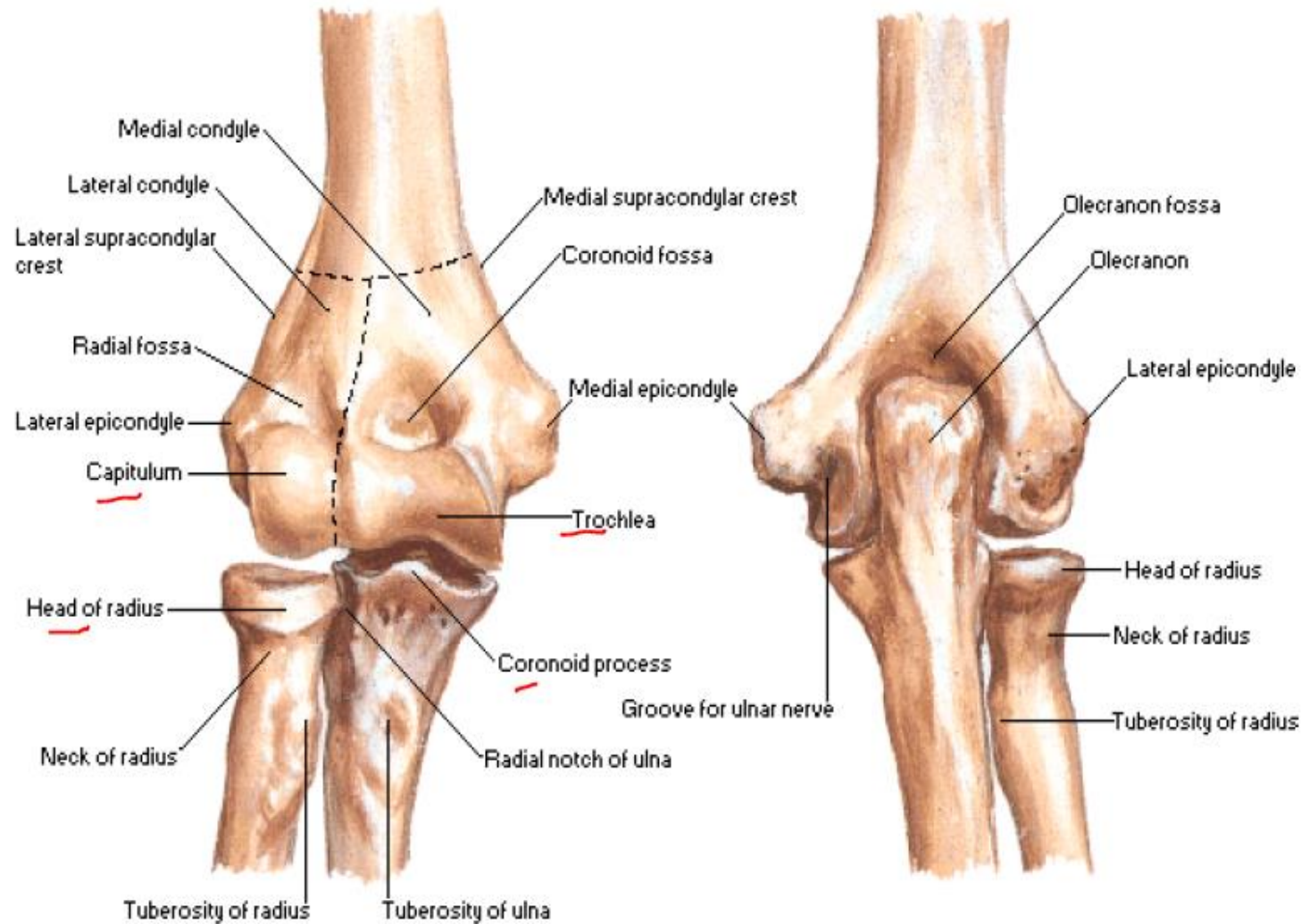
- I. The **humero-ulnar joint** b/n the trochlea of the humerus and the trochlear notch of the ulna (a hinge joint)
- II. The **humero-radial joint** b/n the capitulum of the humerus & the head of the radius (a ball-and-socket joint)
- III. The **proximal radio-ulnar joint** between the head of the radius and the radial notch of the ulna



Elbow Joint,,

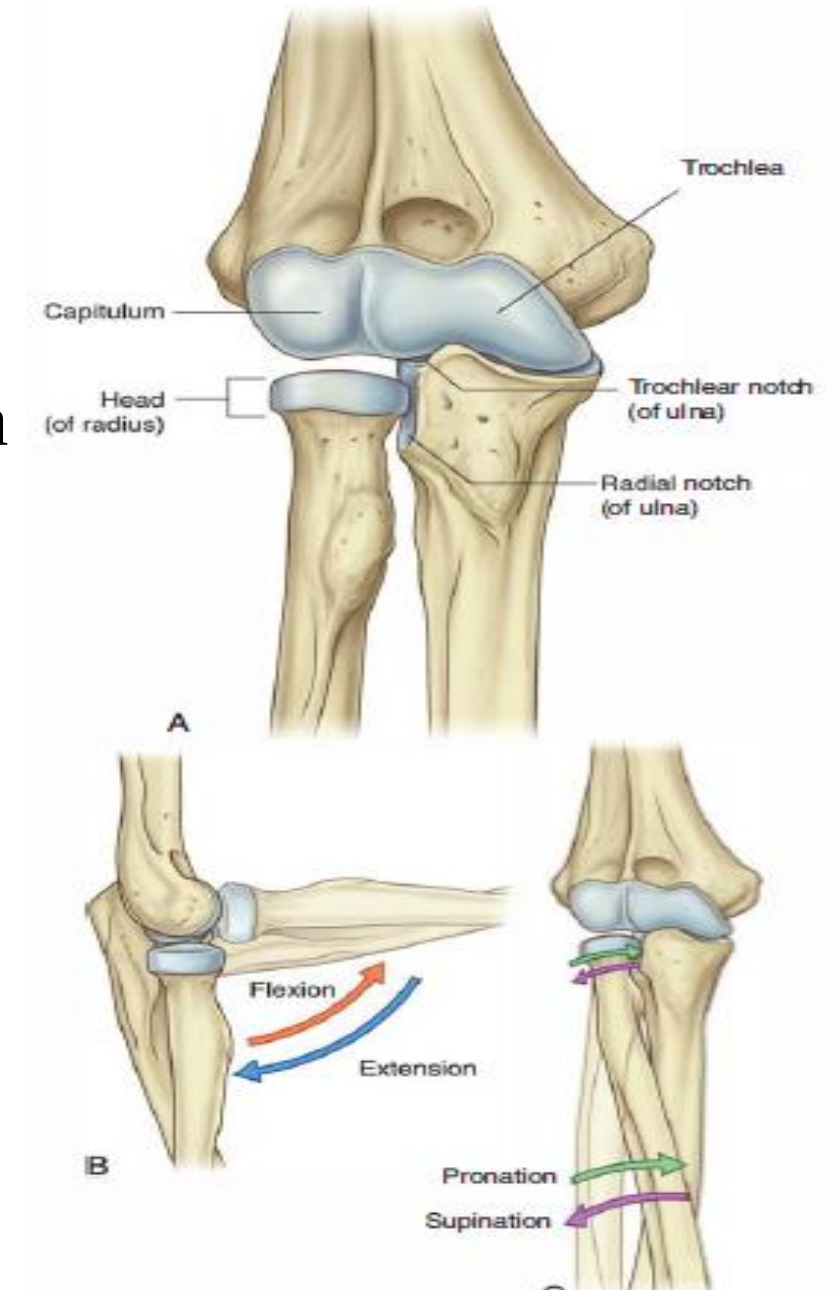
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Anterior and Posterior Views



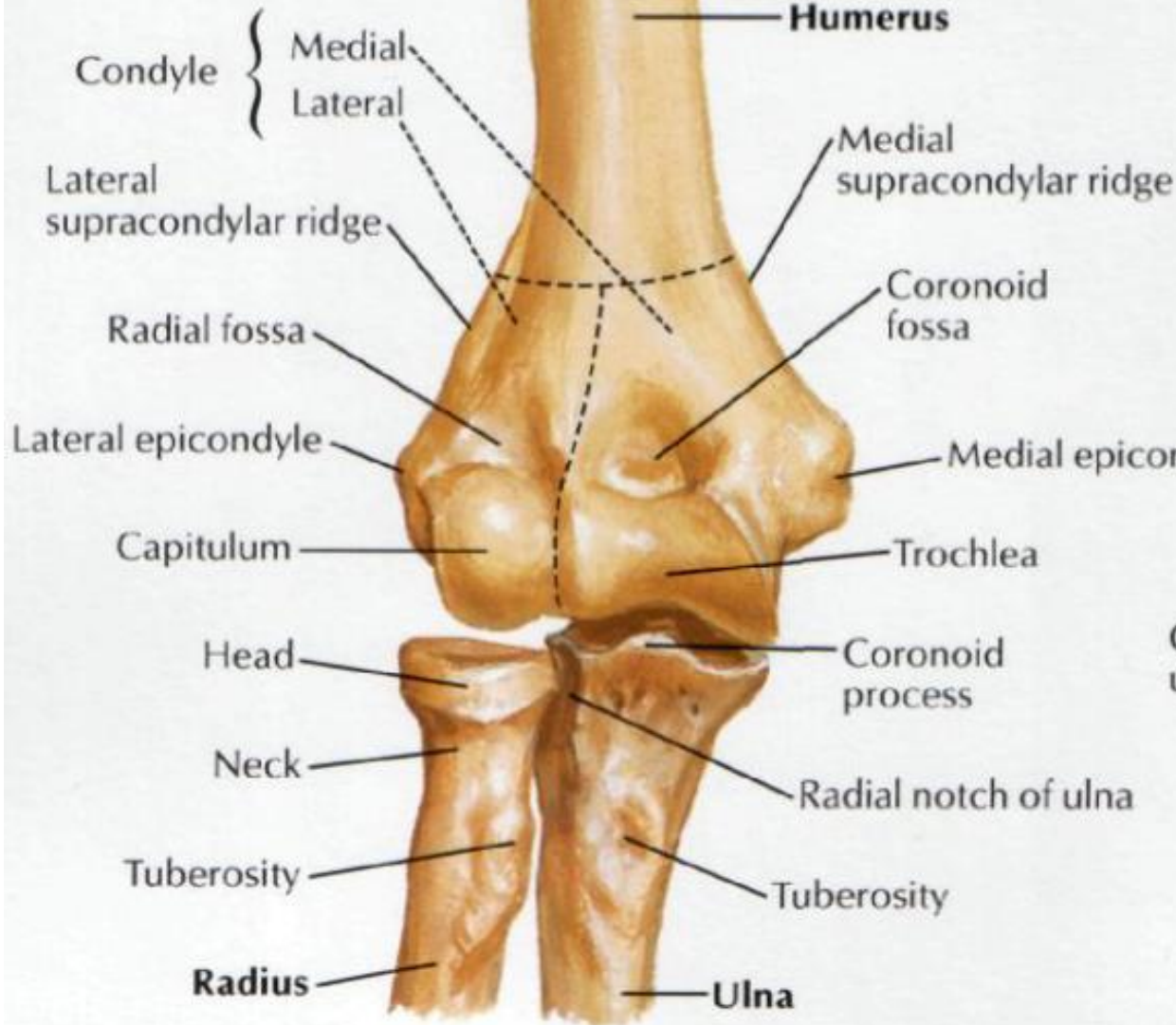
Elbow Joint,,

- The elbow joint primarily mobile in only one axis, which allows active **flexion and extension**
- The **humero-ulnar and humeroradial joints** act as a single hinge joint during **extension and flexion** of the forearm; the **proximal radio-ulnar joint** allows rotation of the head of the radius during **pronation & supination** of the forearm.
- The **proximal radio-ulnar joint** is a pivot type of synovial joint.

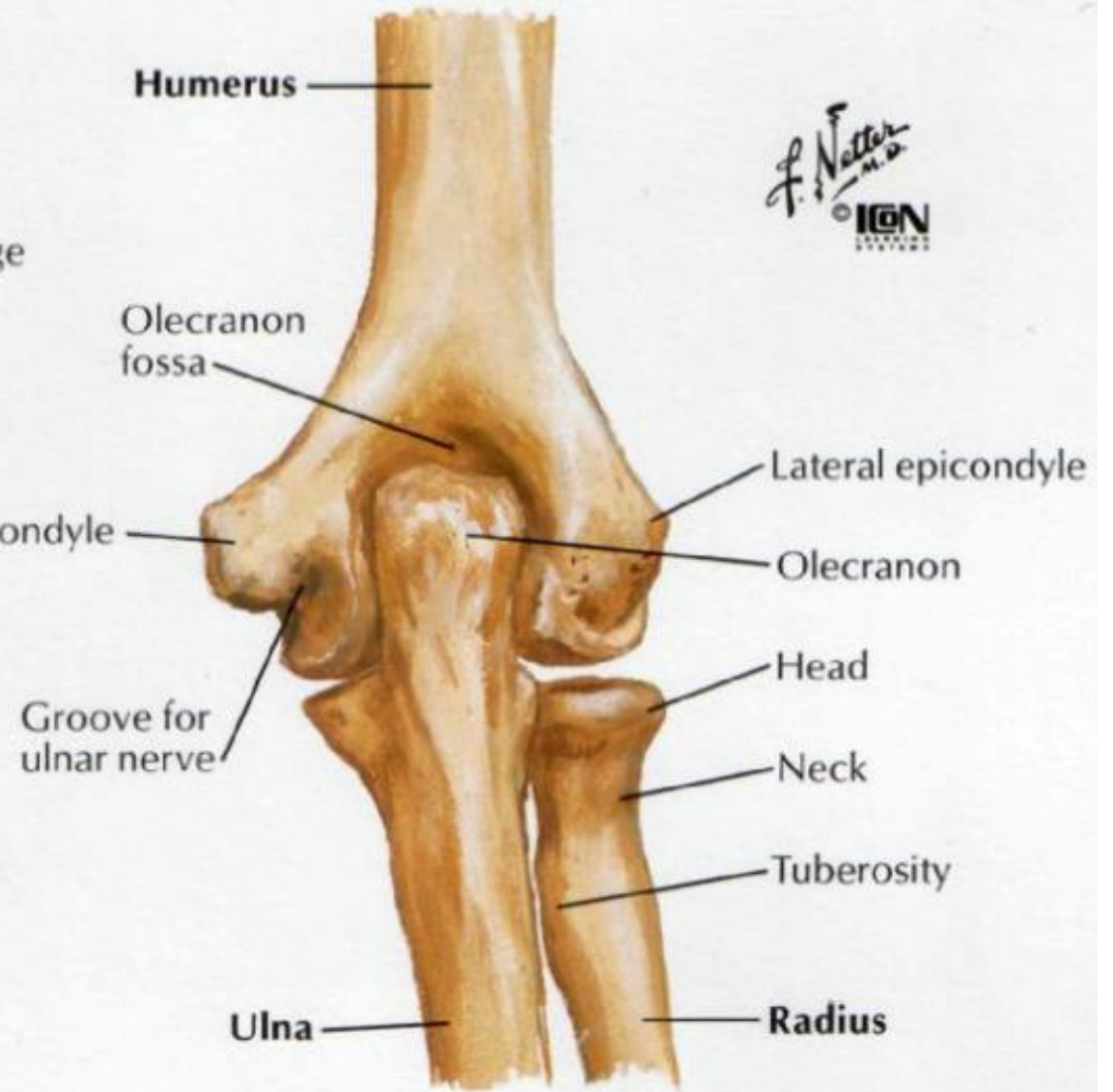


Elbow Joint,,

Right elbow



In extension: anterior view

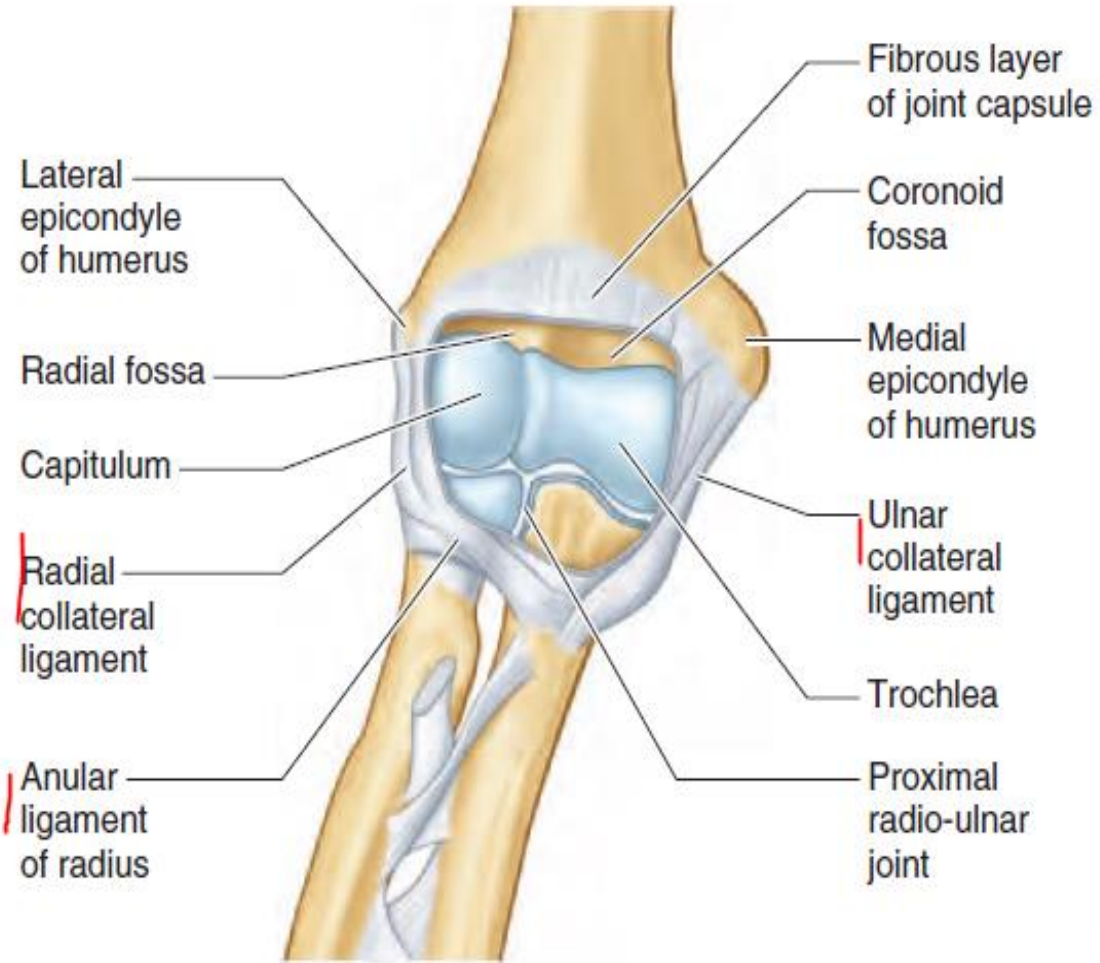


In extension: posterior view

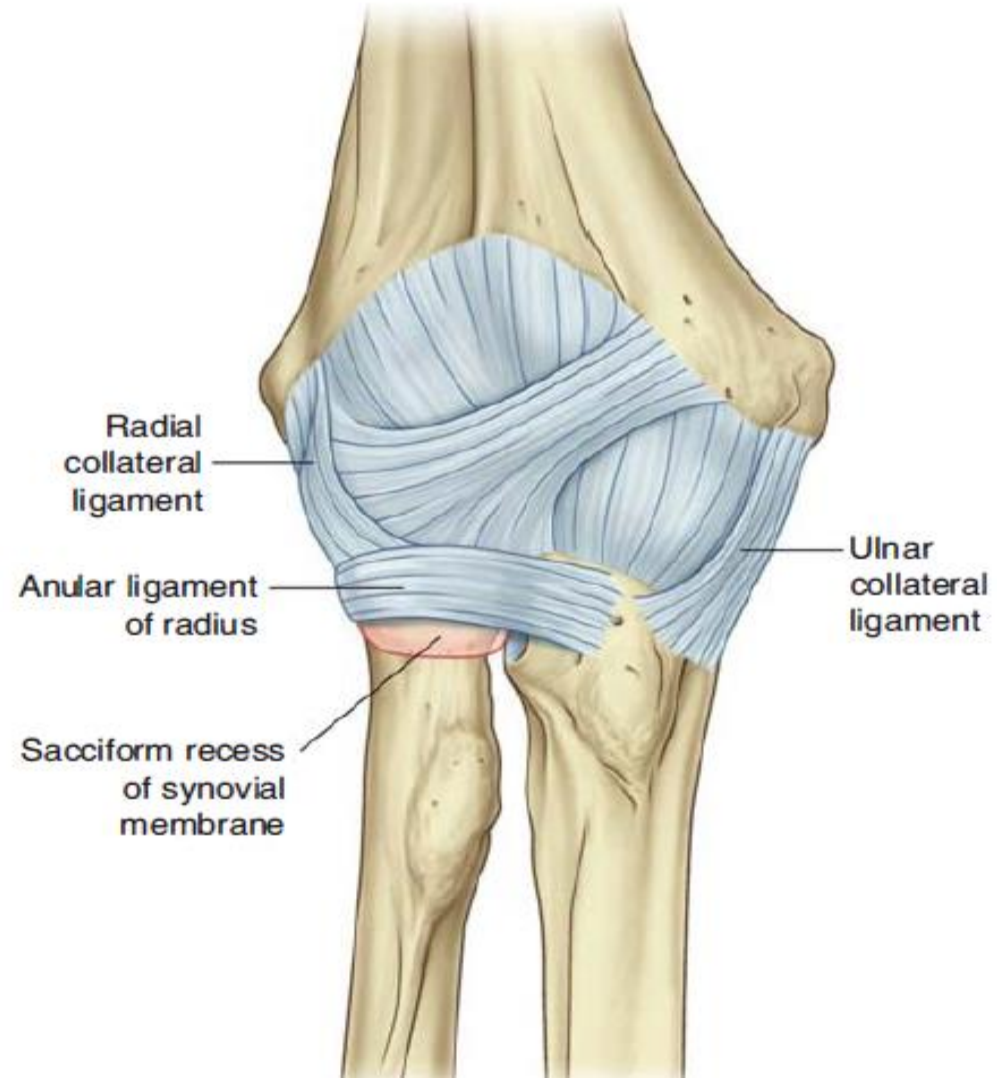
Elbow Joint,,

- ❖ **These three articulations of the elbow joint are enclosed within the joint capsule of the elbow.**
- The joint capsule thickened to form medially and laterally collateral ligament.
- ✓ The radial collateral ligament is fan-like and joins the lateral epicondyle of the humerus to the anular ligament of the radius.
- ✓ The ulnar collateral ligament is triangular and extends from the medial epicondyle of the humerus to the coronoid process and olecranon of the ulna.
- ❖ **The anular ligament** of the radius is attached to the margins of the radial notch of the ulna. It holds the head of the radius in the radial notch of the ulna

Elbow Joint,,

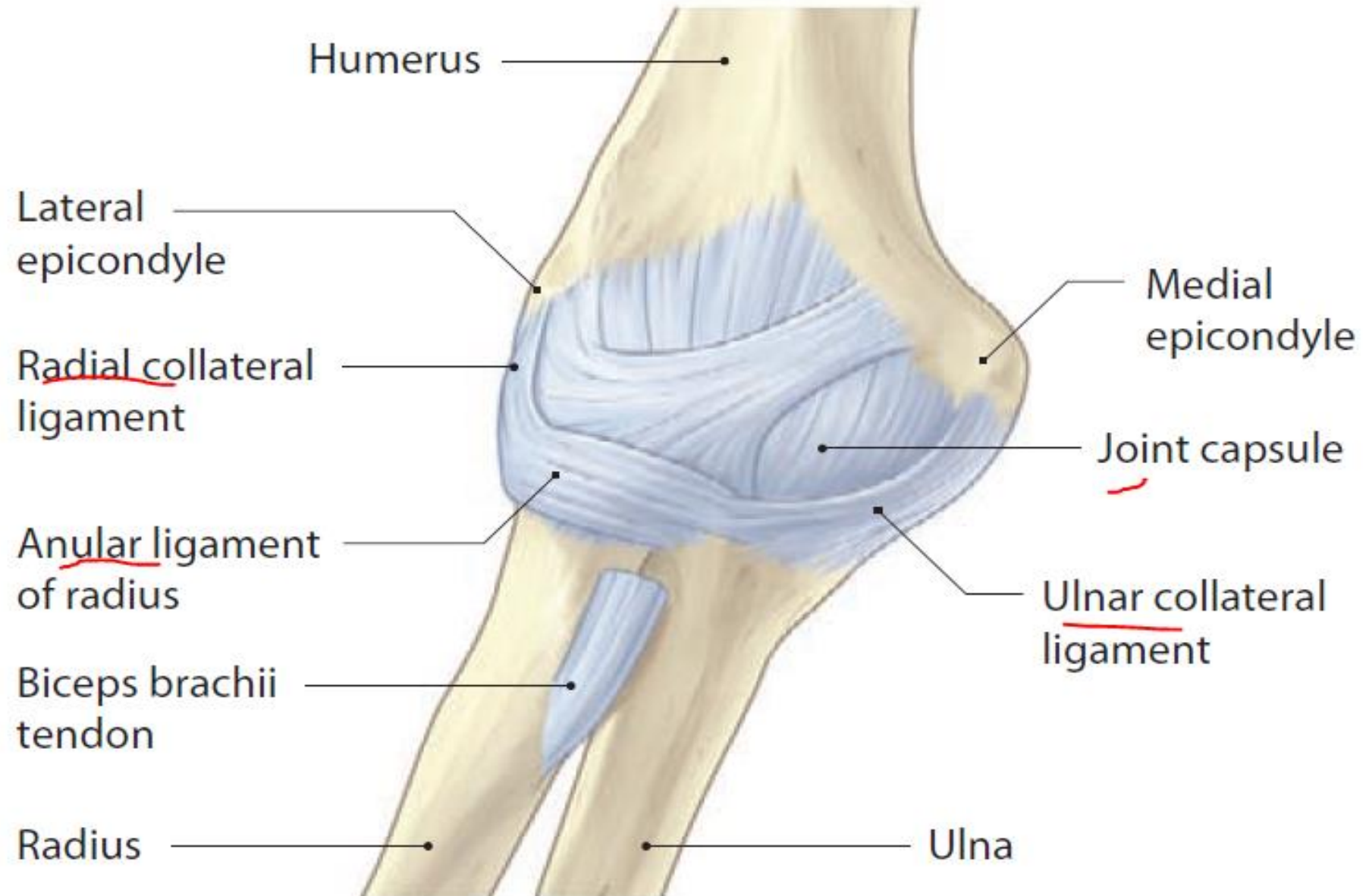


(A) Anterior view



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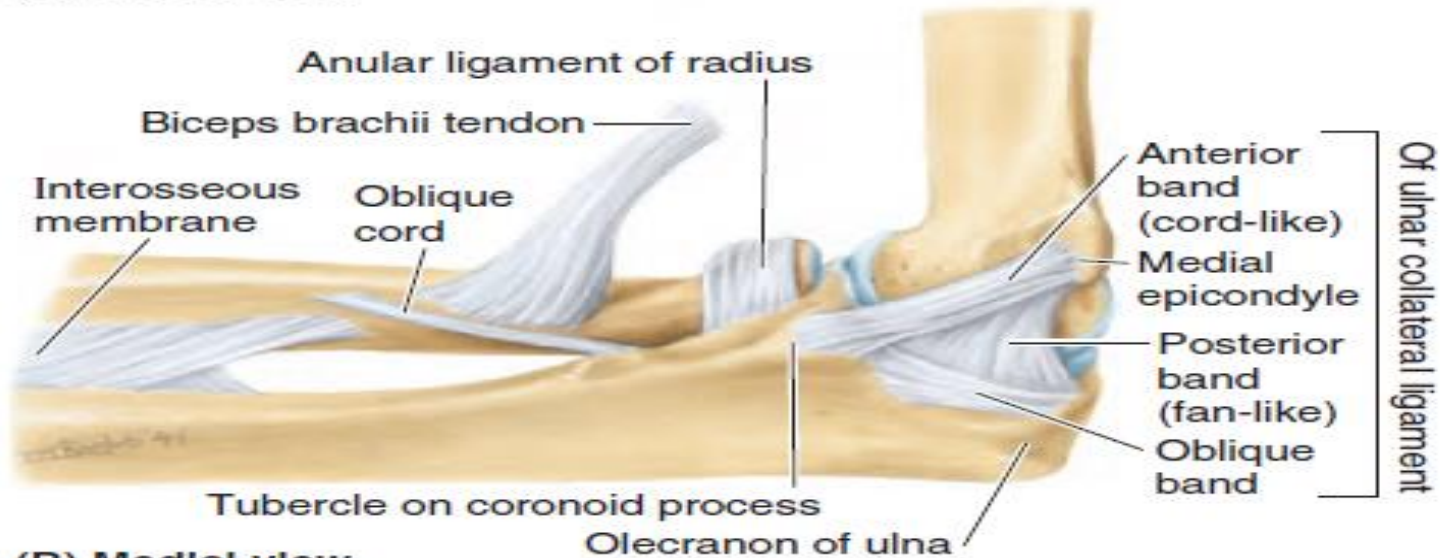
Elbow Joint,,



Elbow Joint,,



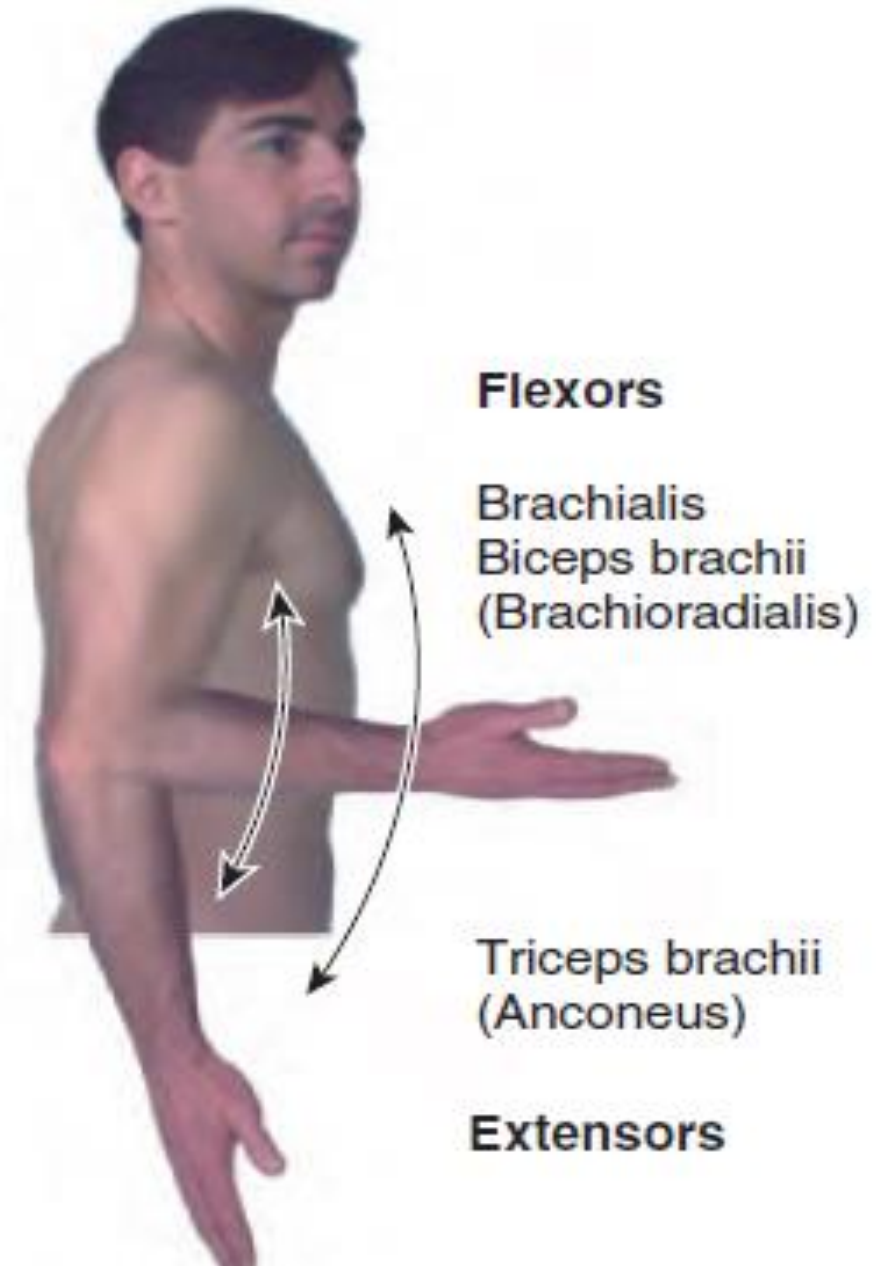
(A) Lateral view



(B) Medial view

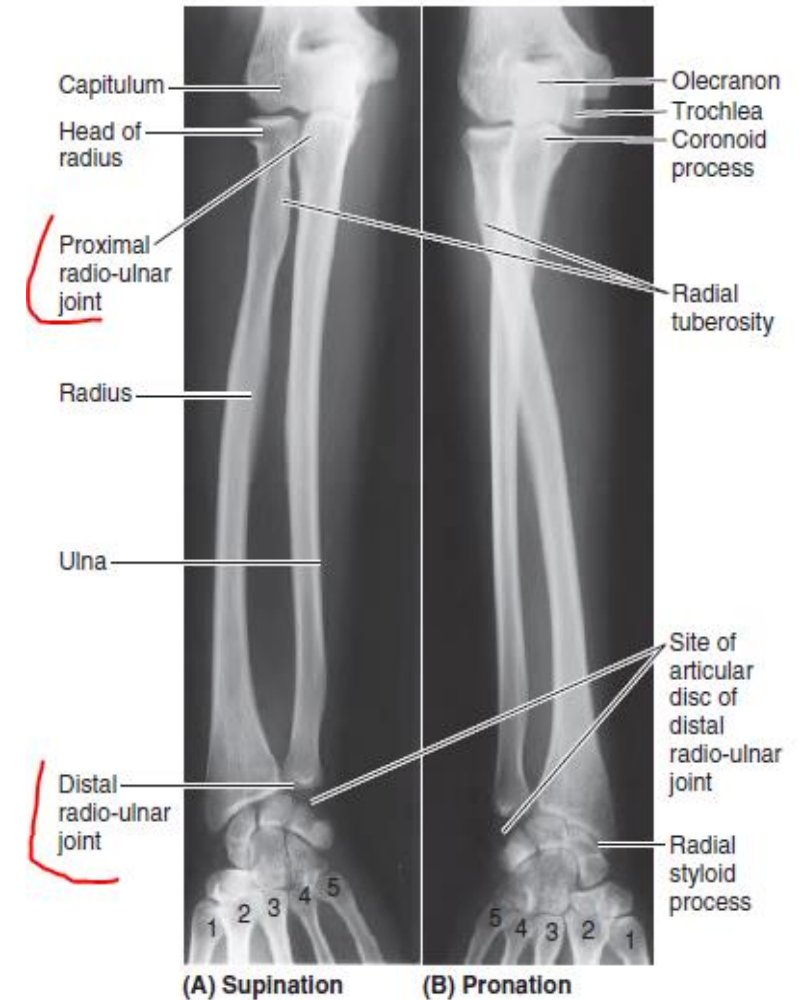
Elbow Joint,,

- There are many muscles that affect elbow joint movement.
- *The chief flexors of the elbow joint are the brachialis and biceps brachii.*
- *The chief extensor of the elbow joint is the triceps brachii*, especially the medial head, weakly assisted by the anconeus.



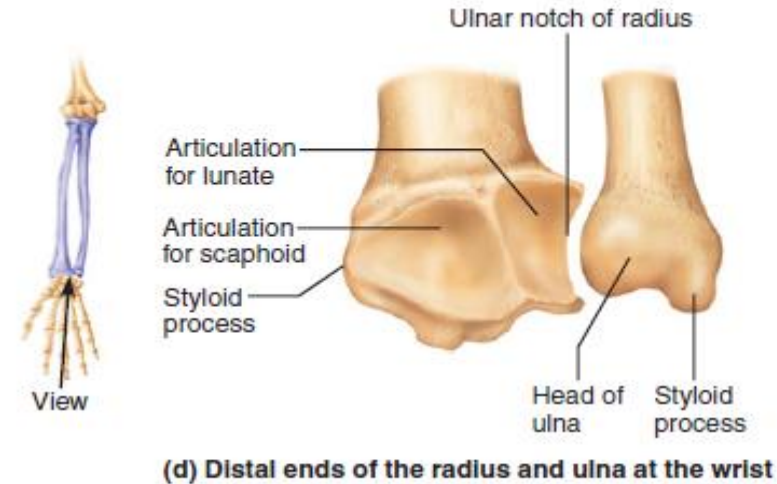
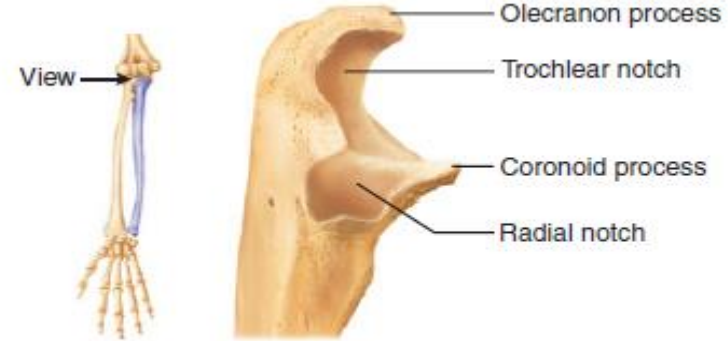
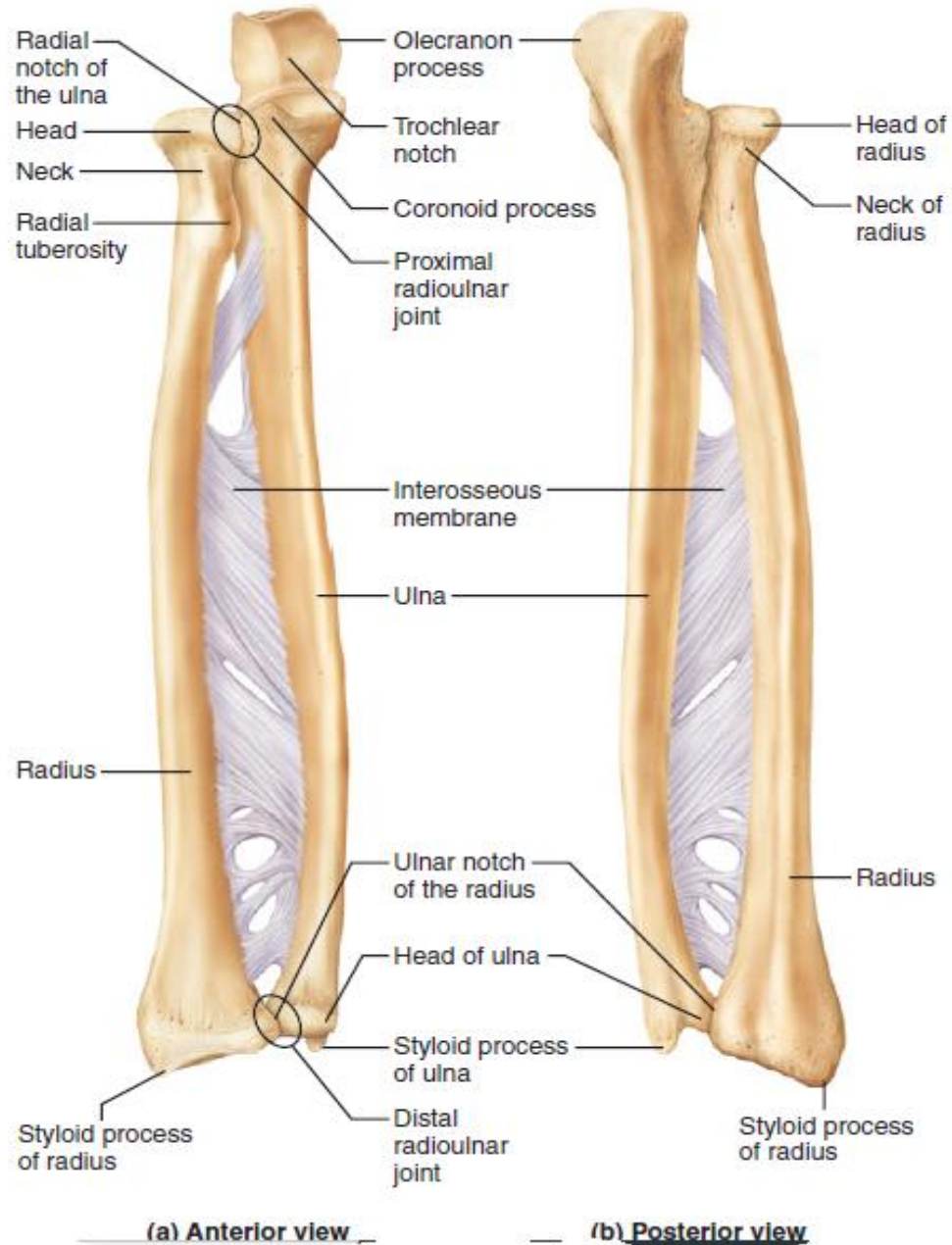
Radioulnar joints

- Pronation and supination occur at the proximal and distal radioulnar joints.
- At the proximal radioulnar joint, the head of the radius is held within the anular ligament, which is attached to the anterior and posterior edges of the radial notch of the ulna.
- The head of the radius rotates within the anular ligament during pronation and supination.

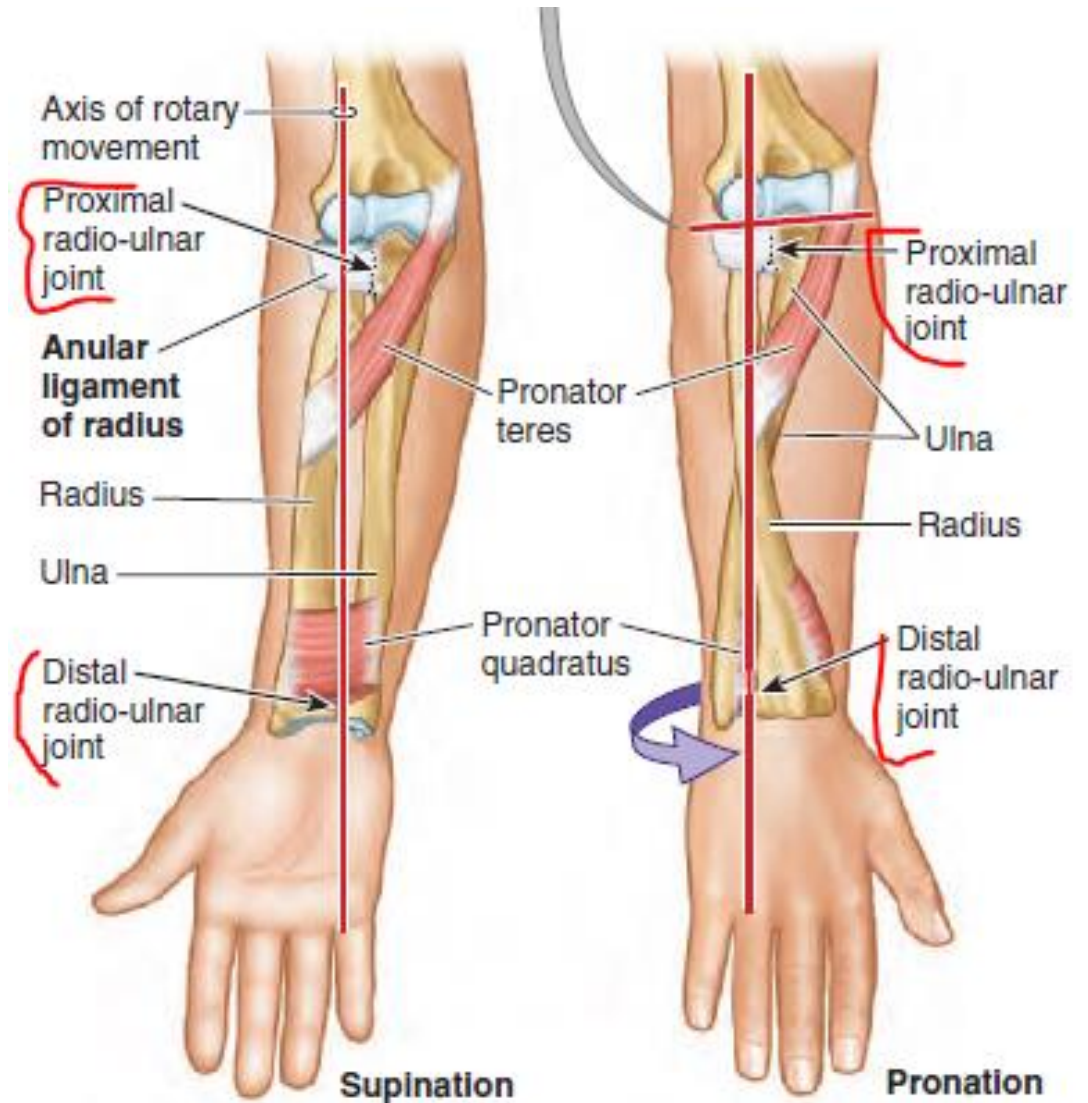
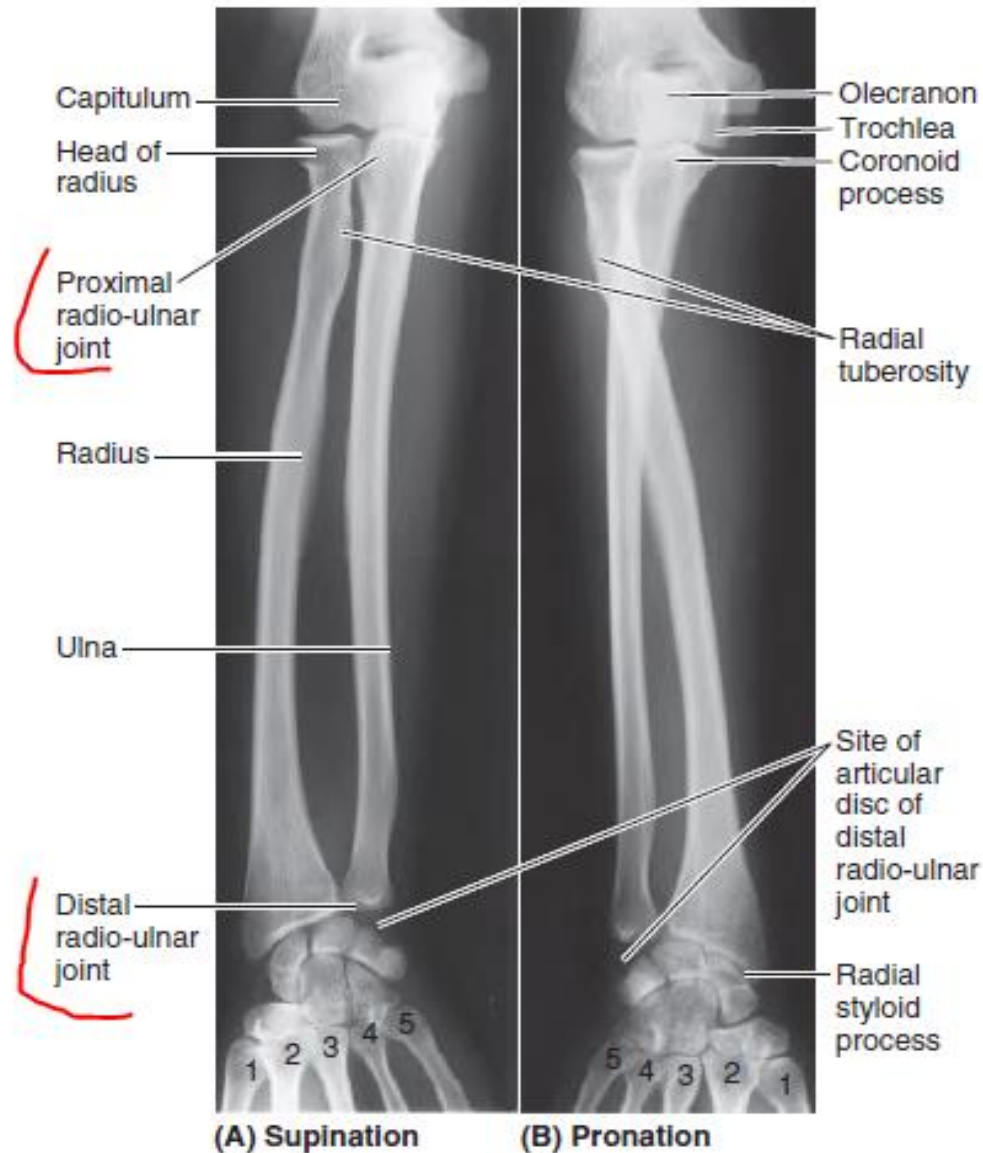


- **At the distal radioulnar** joint the head of the ulna articulates with the ulnar notch of the radius.
- The distal ends of the radius and ulna are held together by an articular disc (known as the triangular ligament).

Radioulnar joints,,,



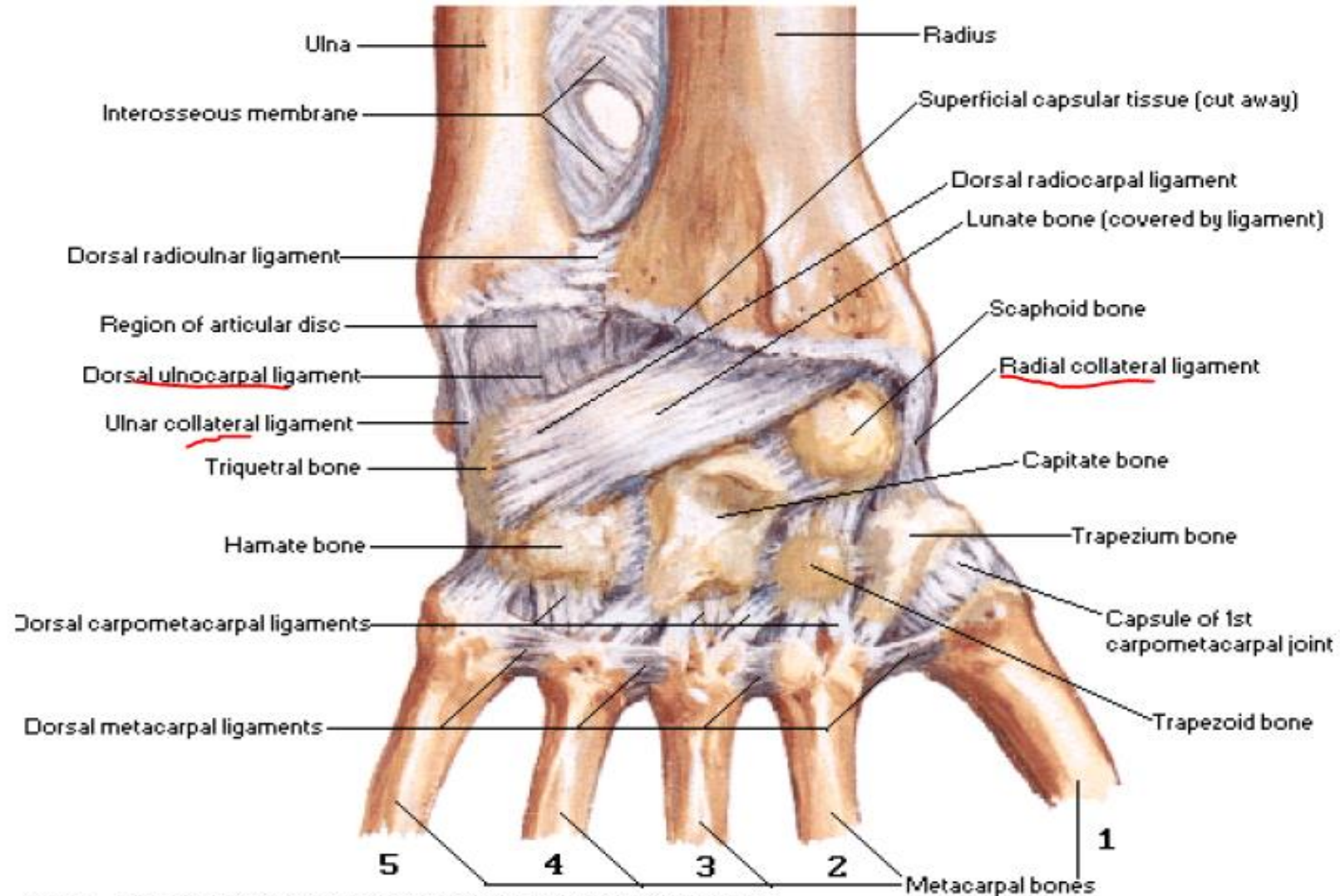
Radioulnar joints,,



Radioulnar joints,,

≡

Posterior [Dorsal] View



Note: dorsal ligaments weaker than palmar ligaments

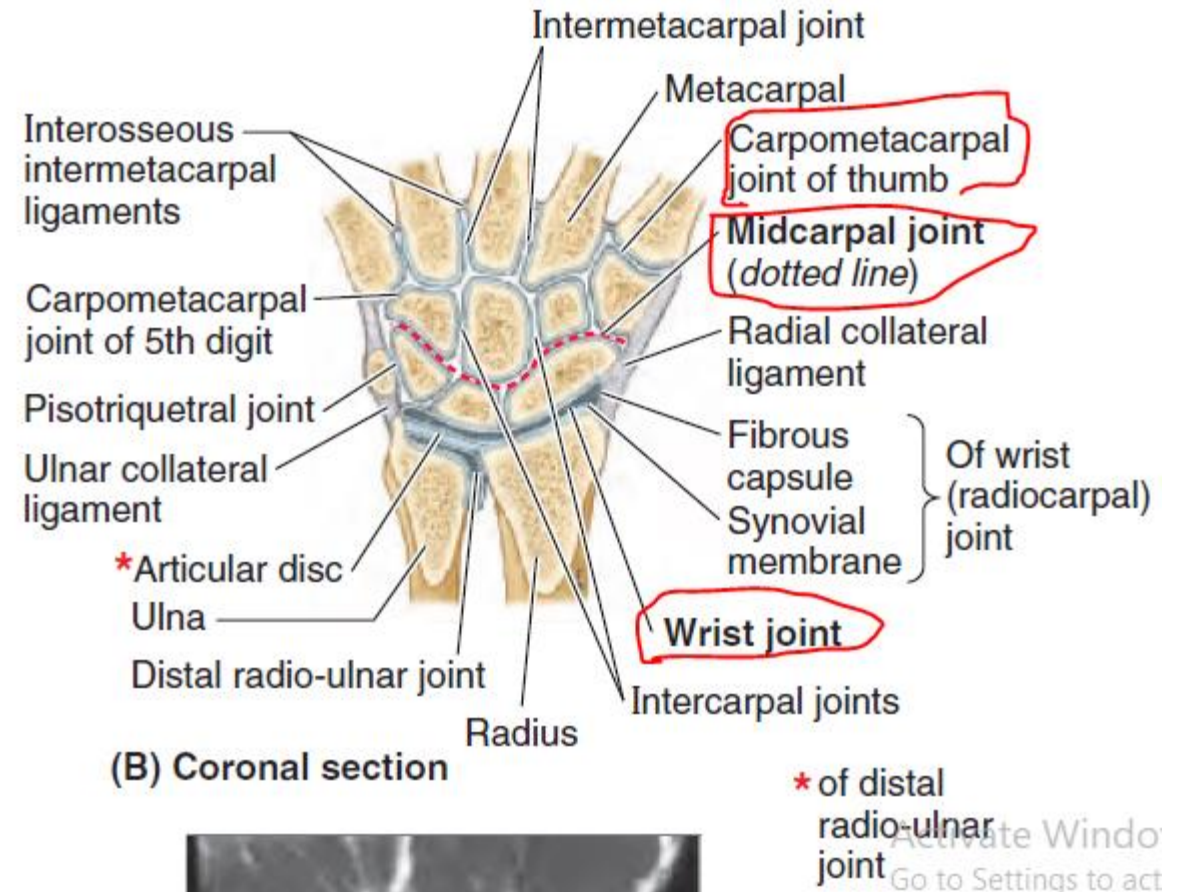
The wrist (radiocarpal) joint

- The anatomy of wrist joint is probably the most complex joints of the body that connects hand and forearm.
- It is actually a collection of multiple bones and joints.
- The bones comprising the wrist include the distal ends of the radius and ulna, eight carpal bones, and the proximal portions of the five metacarpal bones (see the images below).
- The **wrist (radiocarpal) joint** is a condyloid (ellipsoid) type of synovial joint.
- **Note! The ulna does not participate in the wrist joint.**

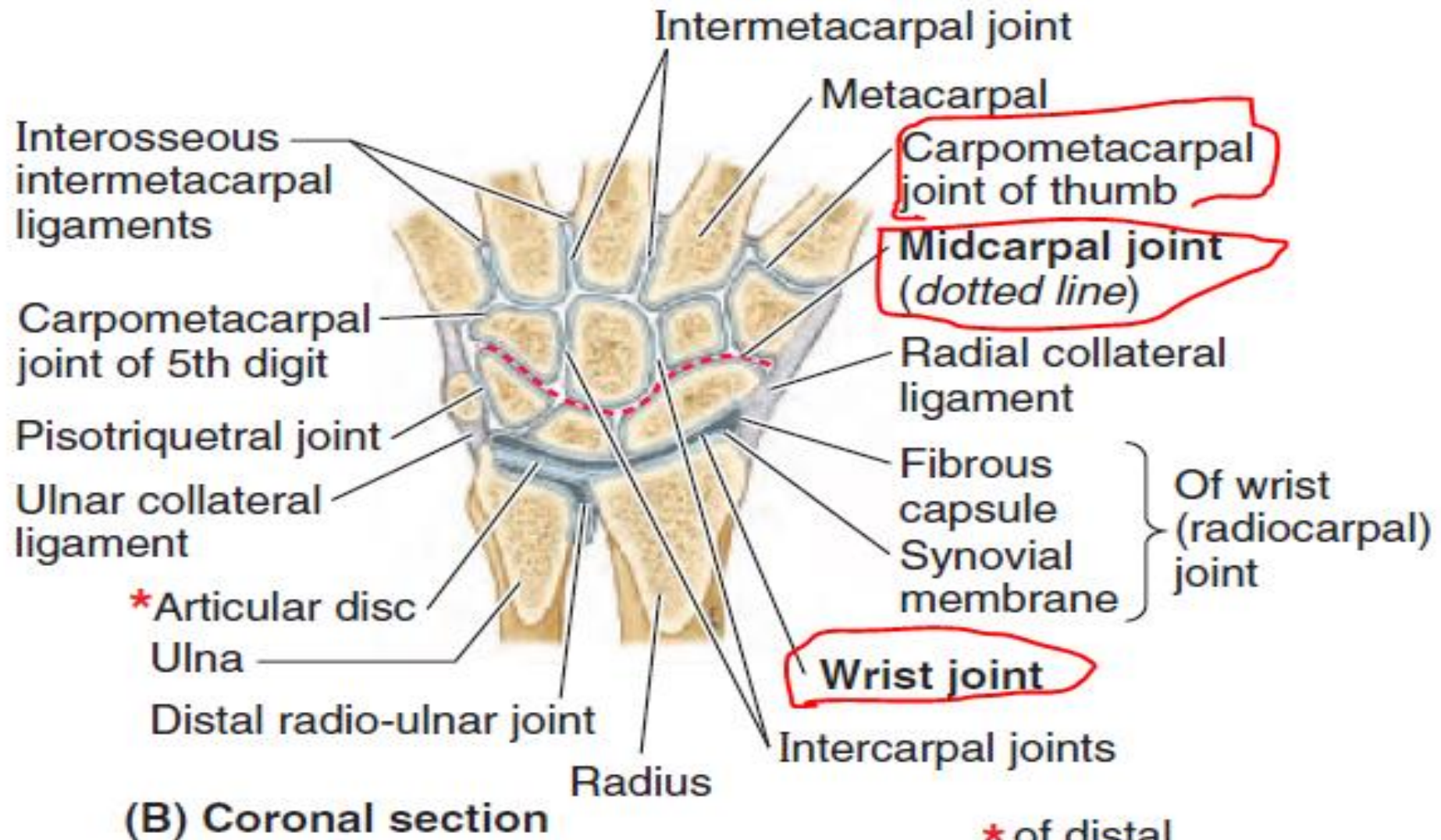
The wrist (radiocarpal) joint,,

❖ The carpal bones form three primary articulations:

- ✓ wrist joint (**radiocarpal joint**) between the forearm and wrist
- ✓ **midcarpal joints** between the proximal and distal rows of carpal bones
- ✓ **carpometacarpal joints** between the distal row of carpal bones and the hand



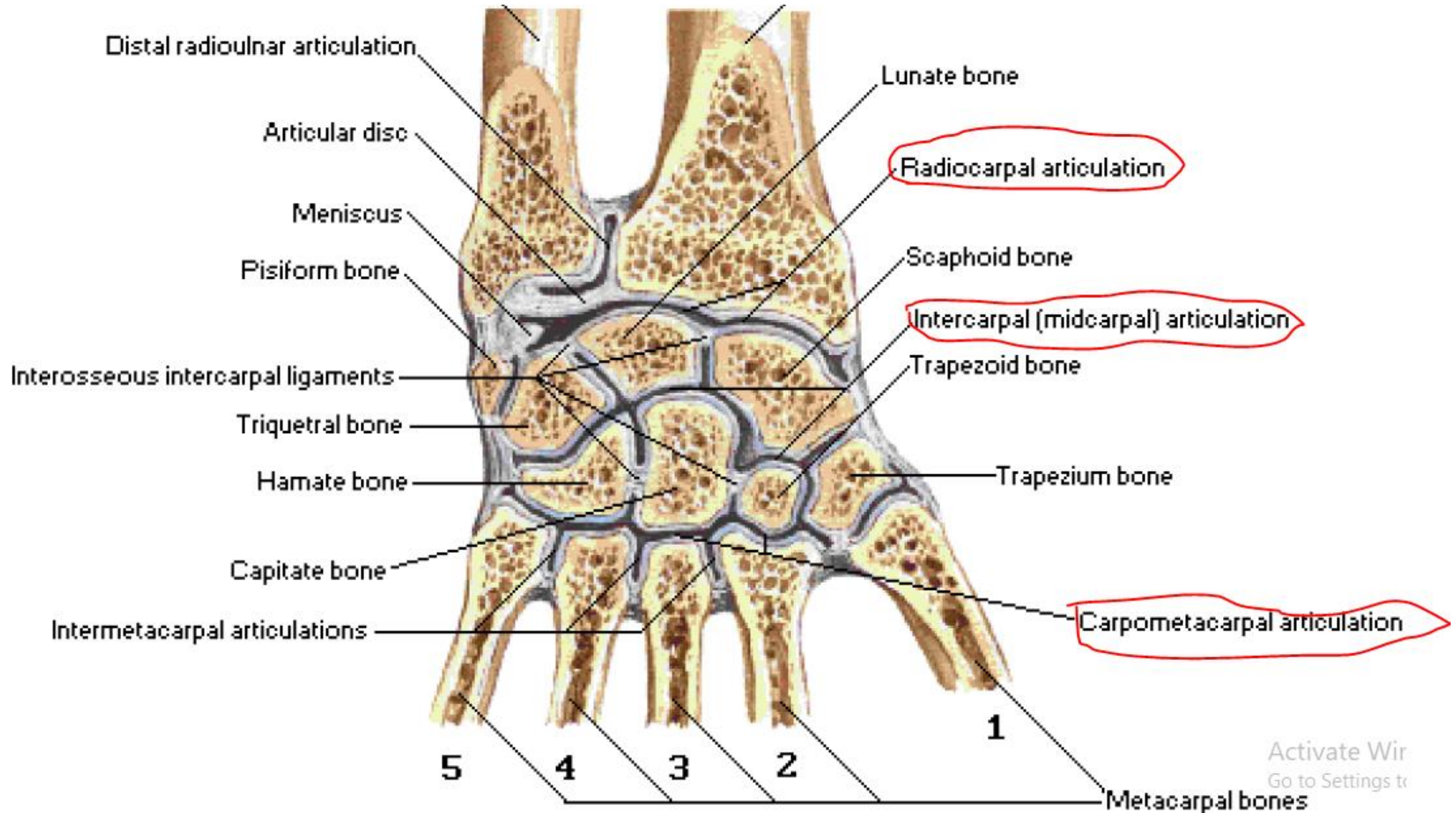
The wrist (radiocarpal) joint,,



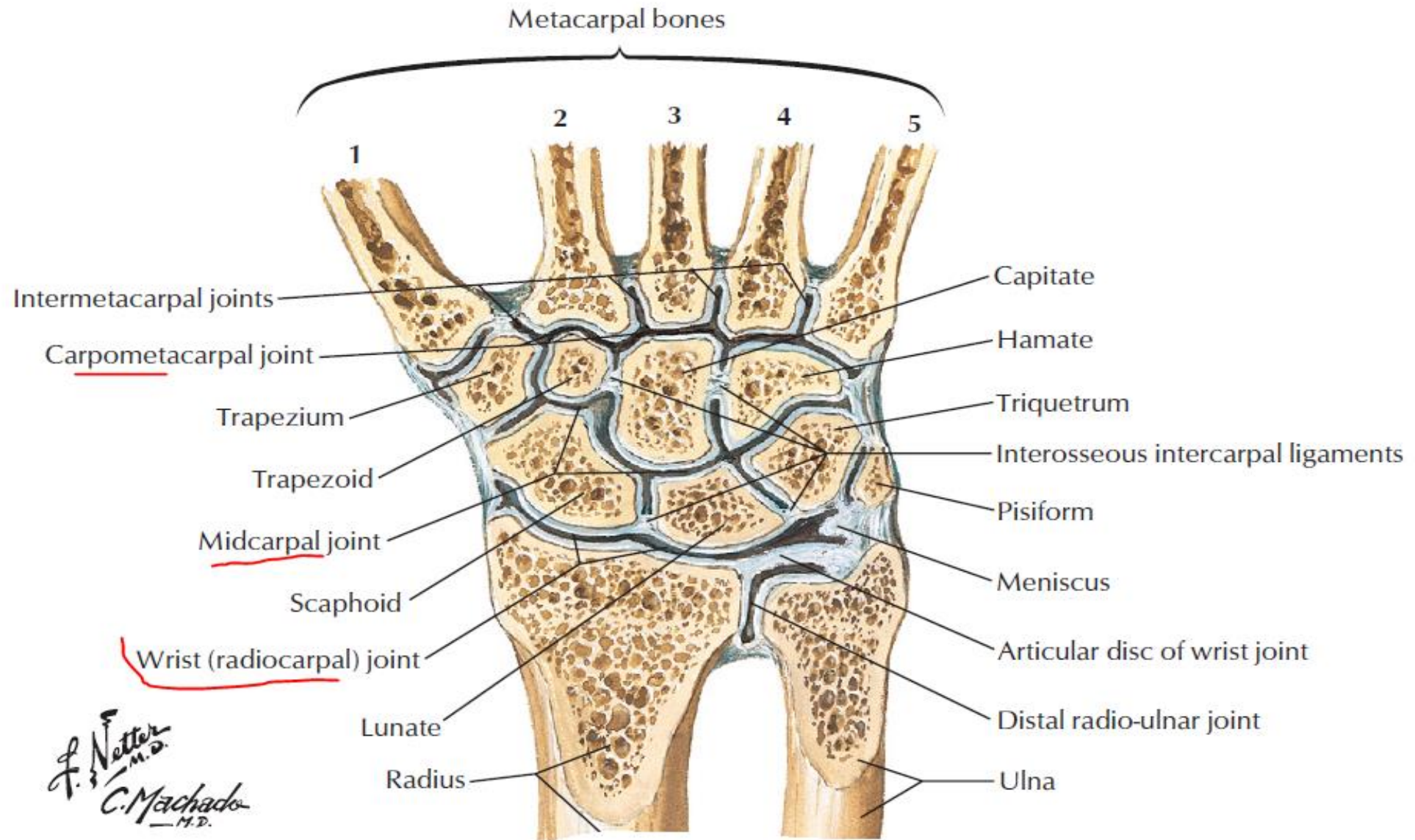
* of distal
radio-ulnar
joint

Activate Windows
Go to Settings to activate Windows.

The wrist (radiocarpal) joint,,



The wrist (radiocarpal) joint,,

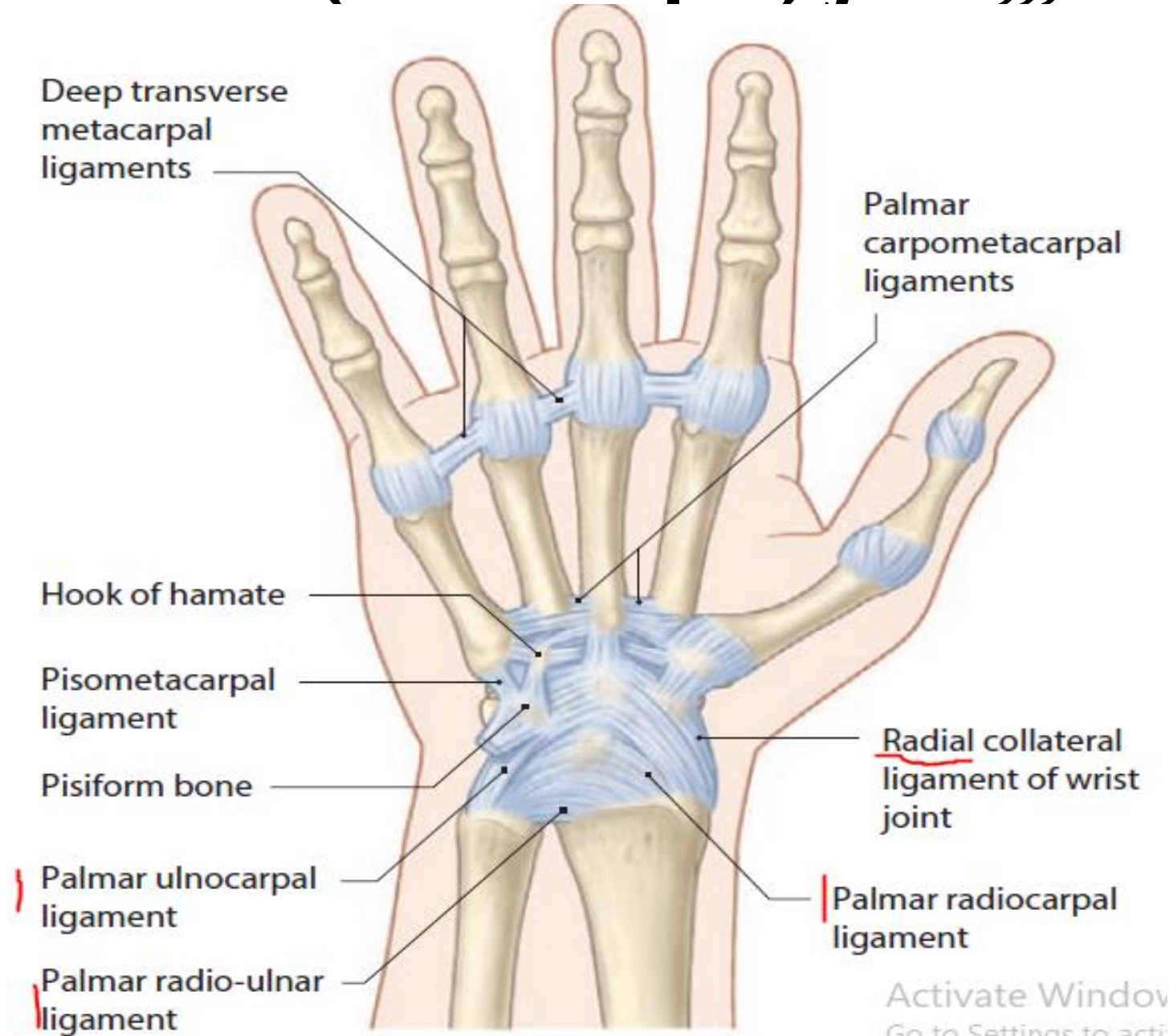


The wrist (radiocarpal) joint,,

❑ **Ligaments of wrist joint:** *fibrous layer of the joint capsule* surrounds the wrist joint and is attached to the distal ends of the radius & ulna and the proximal row of carpals.

- The joint capsule has many thickenings, which are **named as ligaments** (e.g., the **palmar radiocarpal** and **ulnocarpal**, and **dorsal radiocarpal** and **ulnocarpal ligaments** and the **ulnar** and **radial collateral ligaments of the wrist joint**).
- ✓ The joint capsule is also strengthened medially by the **ulnar collateral ligament**, which is attached to the ulnar styloid process and triquetrum.
- ✓ The joint capsule is also strengthened laterally by the **radial collateral ligament**, which is attached to the radial styloid process and scaphoid.

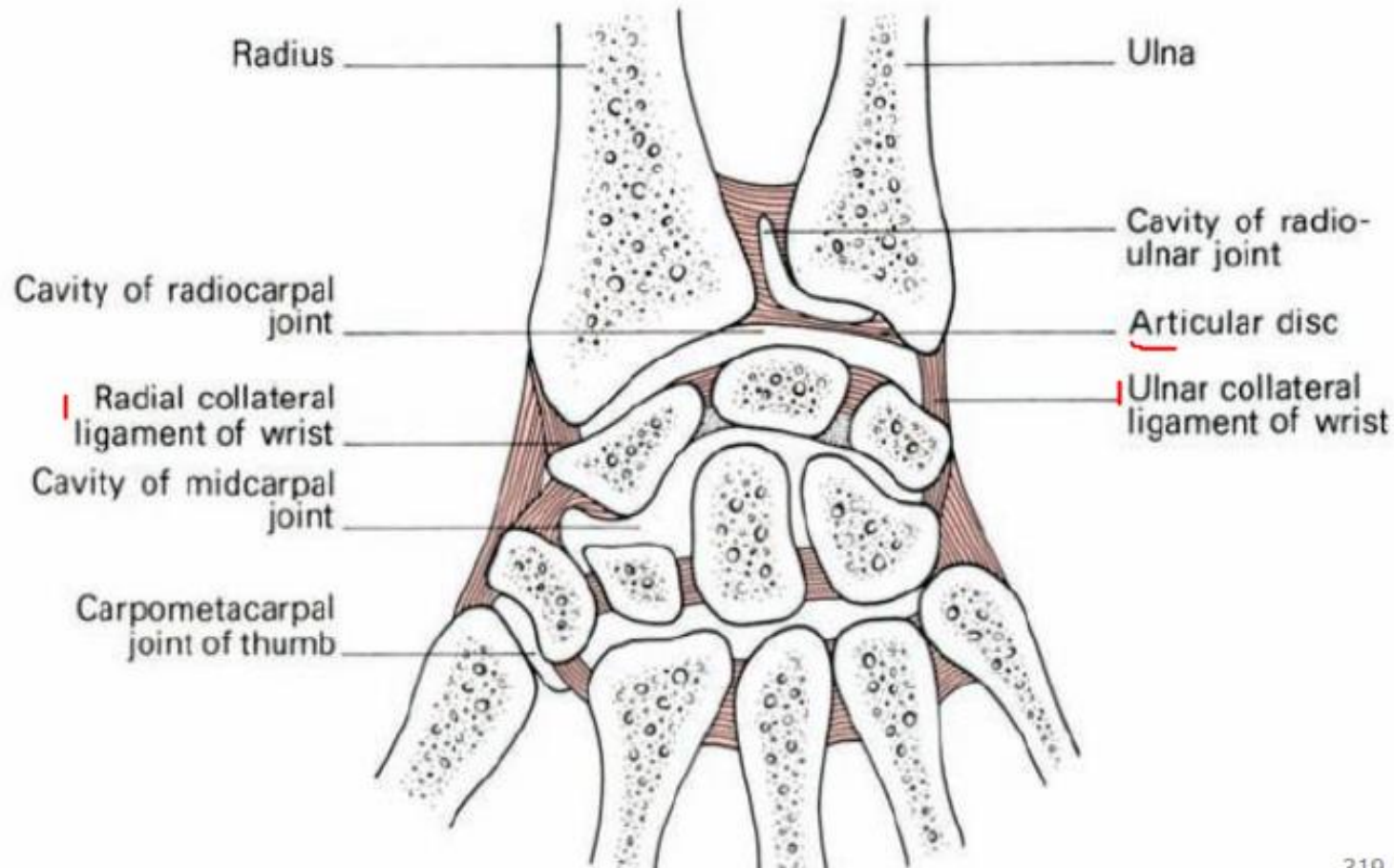
The wrist (radiocarpal) joint,,,



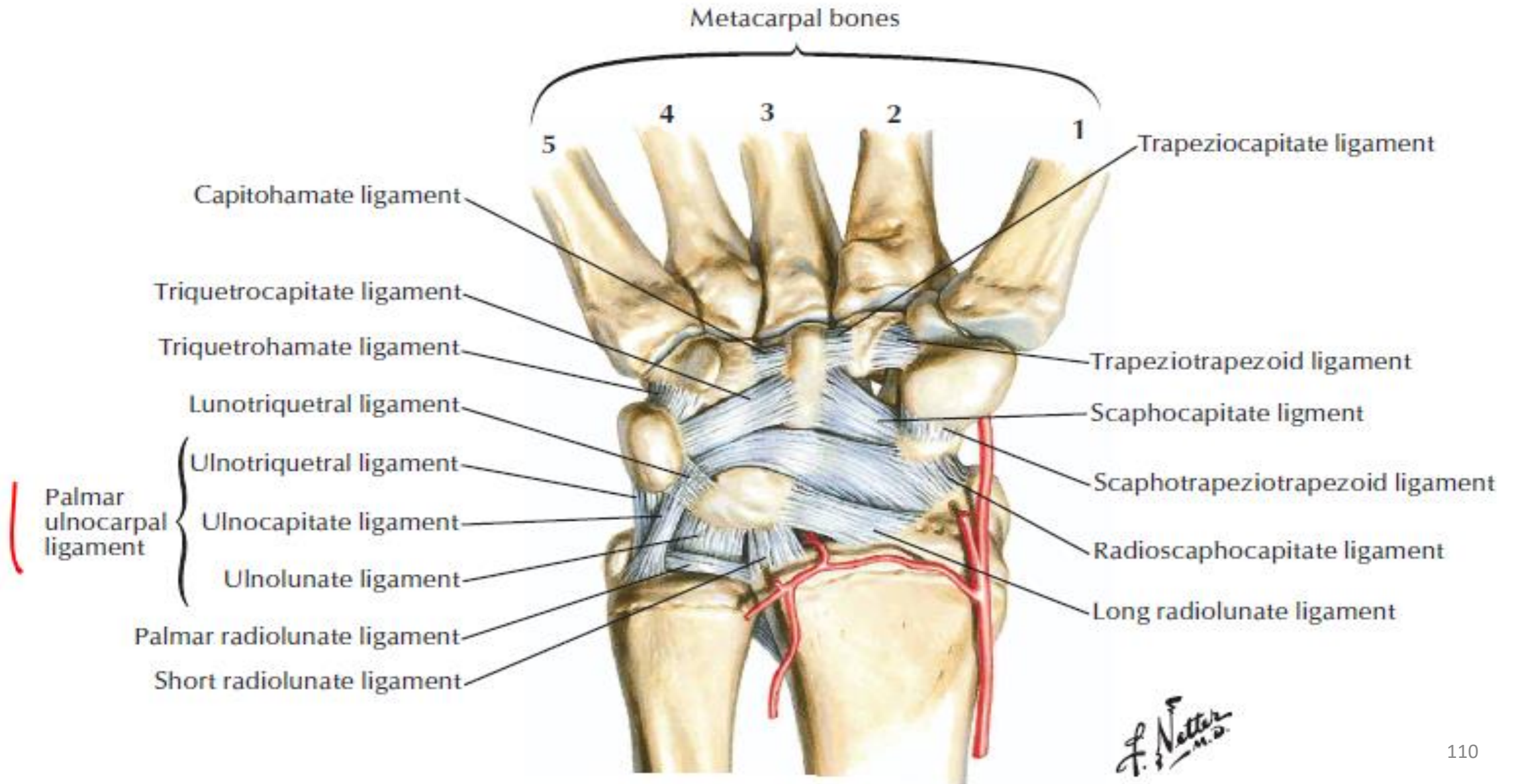
The wrist (radiocarpal) joint,,

□Note!

- ✓ Anterior and posterior ligaments strengthen fibrous capsule
- ✓ Ulnar collateral ligament attaches to styloid process of ulna and triquetrum
- ✓ Radial collateral ligament attaches to styloid process of radius and scaphoid

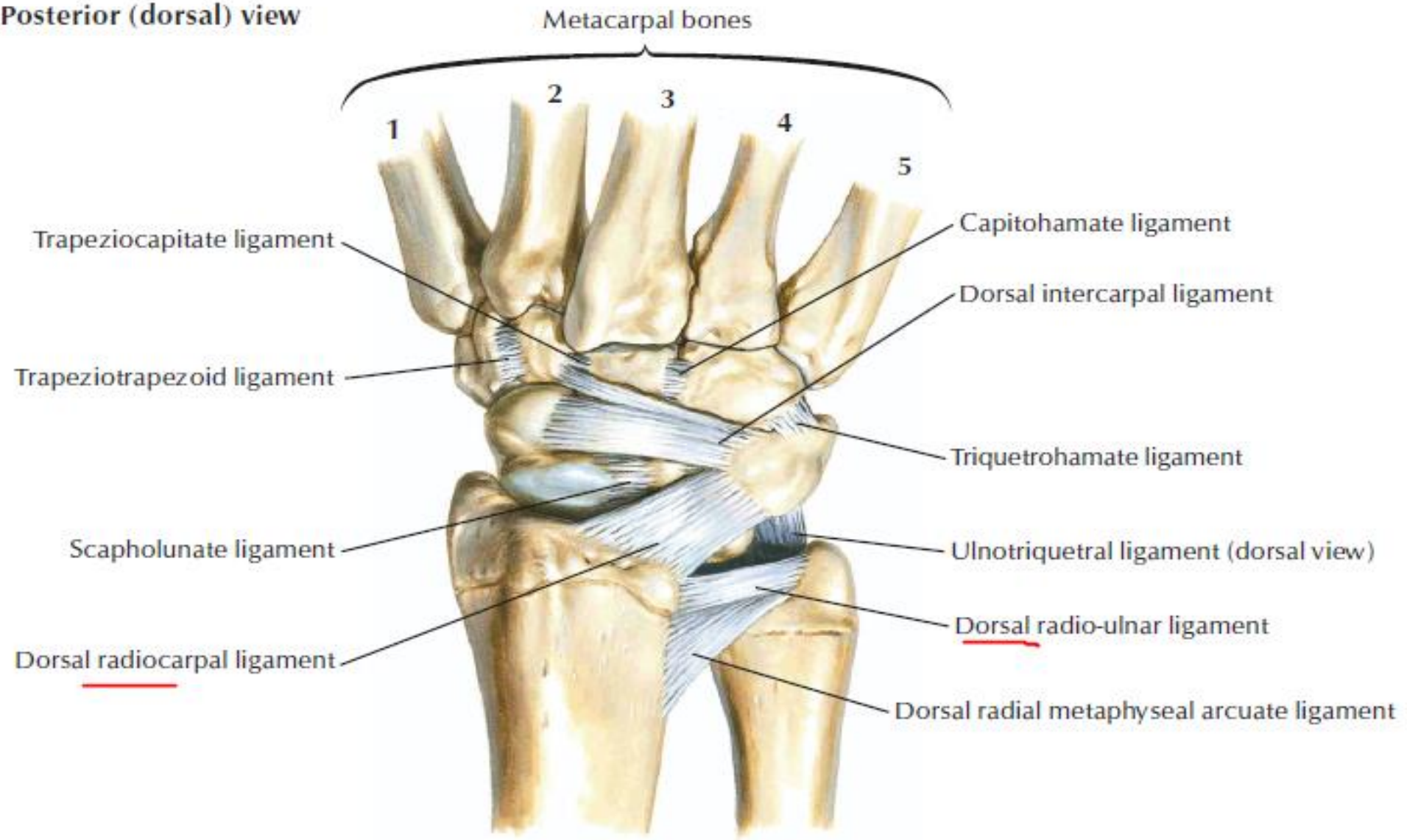


The wrist (radiocarpal) joint,,



The wrist (radiocarpal) joint,,

Posterior (dorsal) view

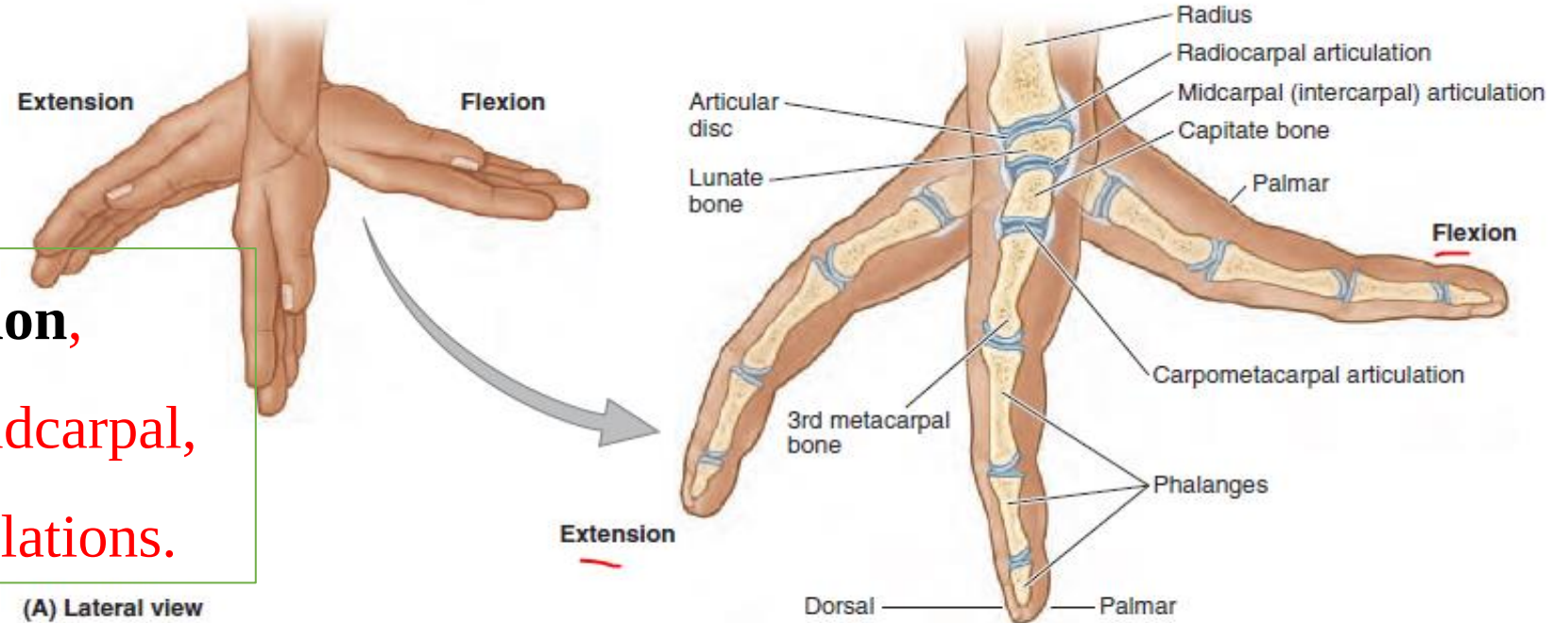


Muscles moving wrist joint

- Movement at the wrist is produced primarily by the “carpi” muscles of the forearm,(carpi ulnaris, flexors &extensors) and (carpi radialis, flexors &extensors).
- *Flexion of the wrist joint* is produced by flexors FCR and FCU which are assisted by flexors of the fingers and thumb, the palmaris longus and the APL.
- *Extension of the wrist joint* is produced by the ECRL, ECRB, and ECU, with assistance from the extensors of the fingers and thumb.
- *Abduction of the wrist joint* is produced by the APL, FCR, ECRL, and ECRB.
- *Adduction of the wrist joint* is produced by simultaneous contraction of the ECU and FCU.

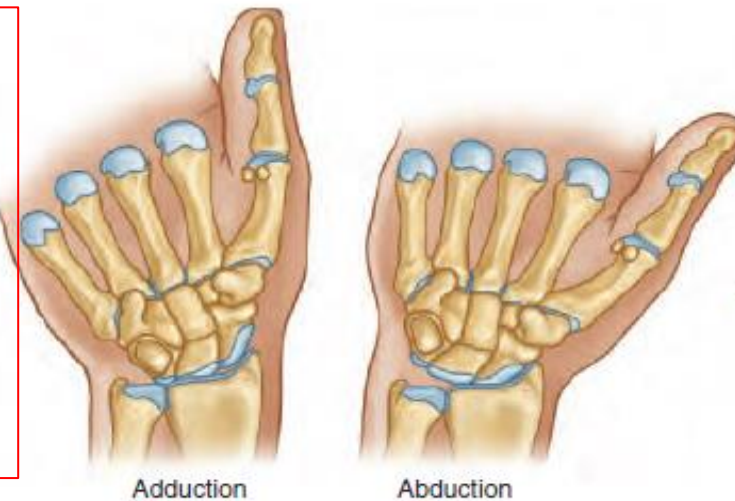
Muscles moving wrist joint,,

- During **extension** and **flexion**, observe the radiocarpal, midcarpal, and carpometacarpal articulations.

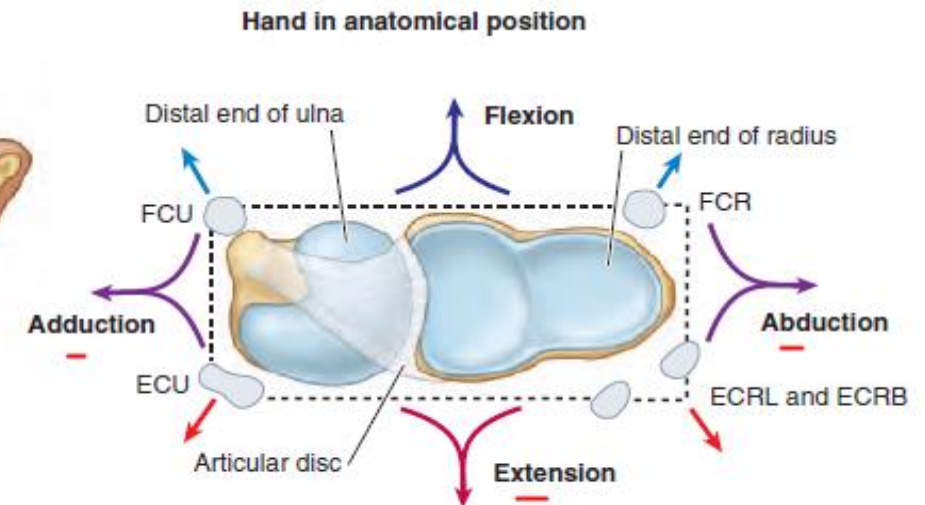


(A) Lateral view

- Movement at the radiocarpal and midcarpal joints during **adduction** and **abduction**.



(B) Anteroposterior view



(C) Inferior view

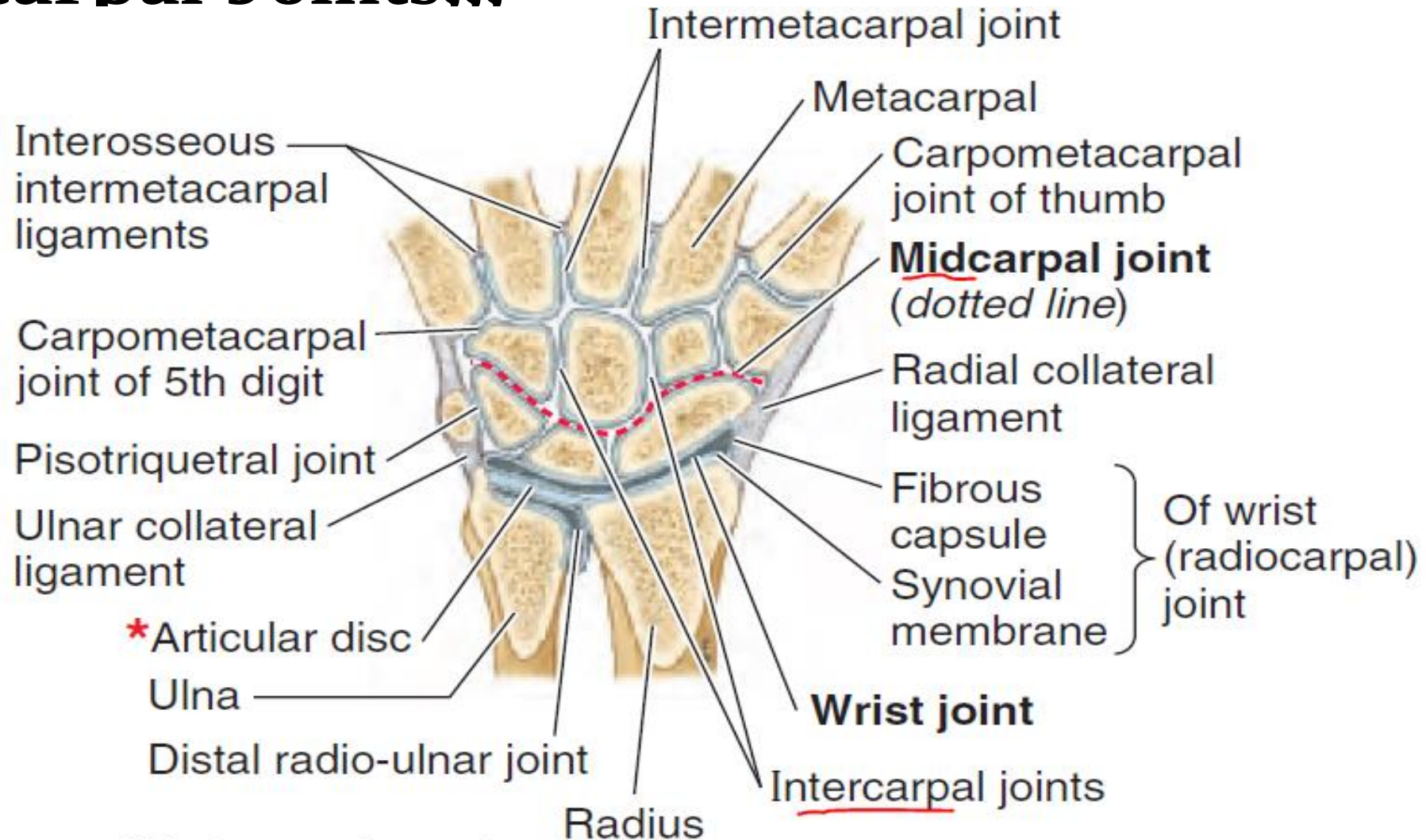
Intercarpal Joints

- The **intercarpal joints**, interconnecting the carpal bones, are plane synovial joints.

❑ **In the intercarpal joints articulation sites can be summarized as follows:**

- ✓ Joints between the carpal bones of the proximal row.
- ✓ Joints between the carpal bones of the distal row.
- ✓ The **midcarpal joint**, a complex joint between the proximal and distal rows of carpal bones.
- ✓ The **pisotriquetral joint**, formed from the articulation of the **pisiform** with the palmar surface of the **triquetrum**.

Intercarpal Joints...



(B) Coronal section

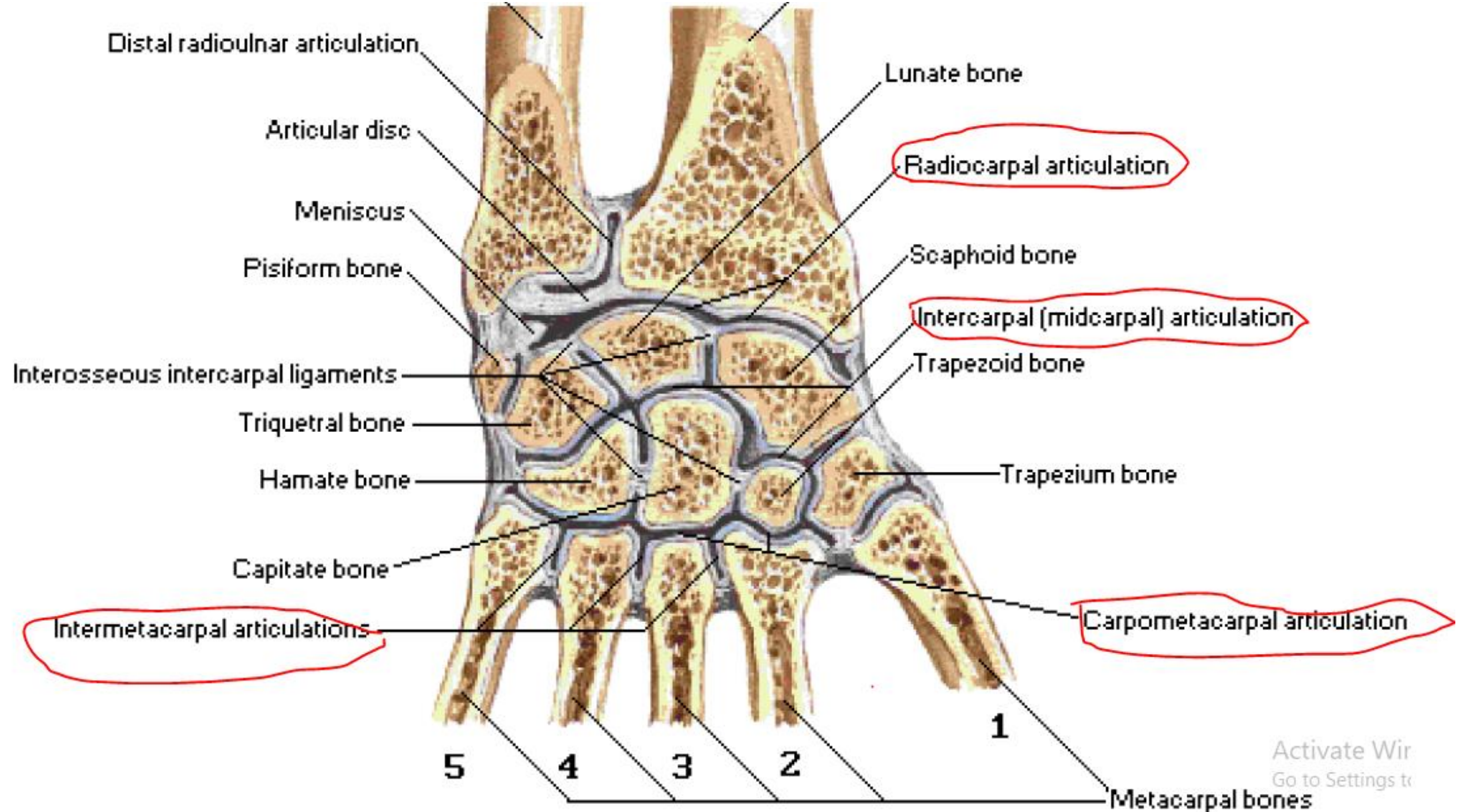


* of distal radio-ulnar joint

Carpometacarpal and Intermetacarpal Joints

- The **carpometacarpal (CMC)** and **intermetacarpal (IM)** joints are the plane type of **synovial** joint, except for the **CMC joint of the thumb**, which is a **saddle** joint, which is a saddle joint
- The distal surfaces of the carpals of the distal row articulate with the carpal surfaces of the bases of the metacarpals at the CMC joints.
- The important CMC joint of the thumb is between the trapezium and the base of the 1st metacarpal; it has a separate articular cavity.
- Like the carpals, adjacent metacarpals articulate with each other; IM joints occur between the radial and ulnar aspects of the bases of the metacarpals.¹¹⁶

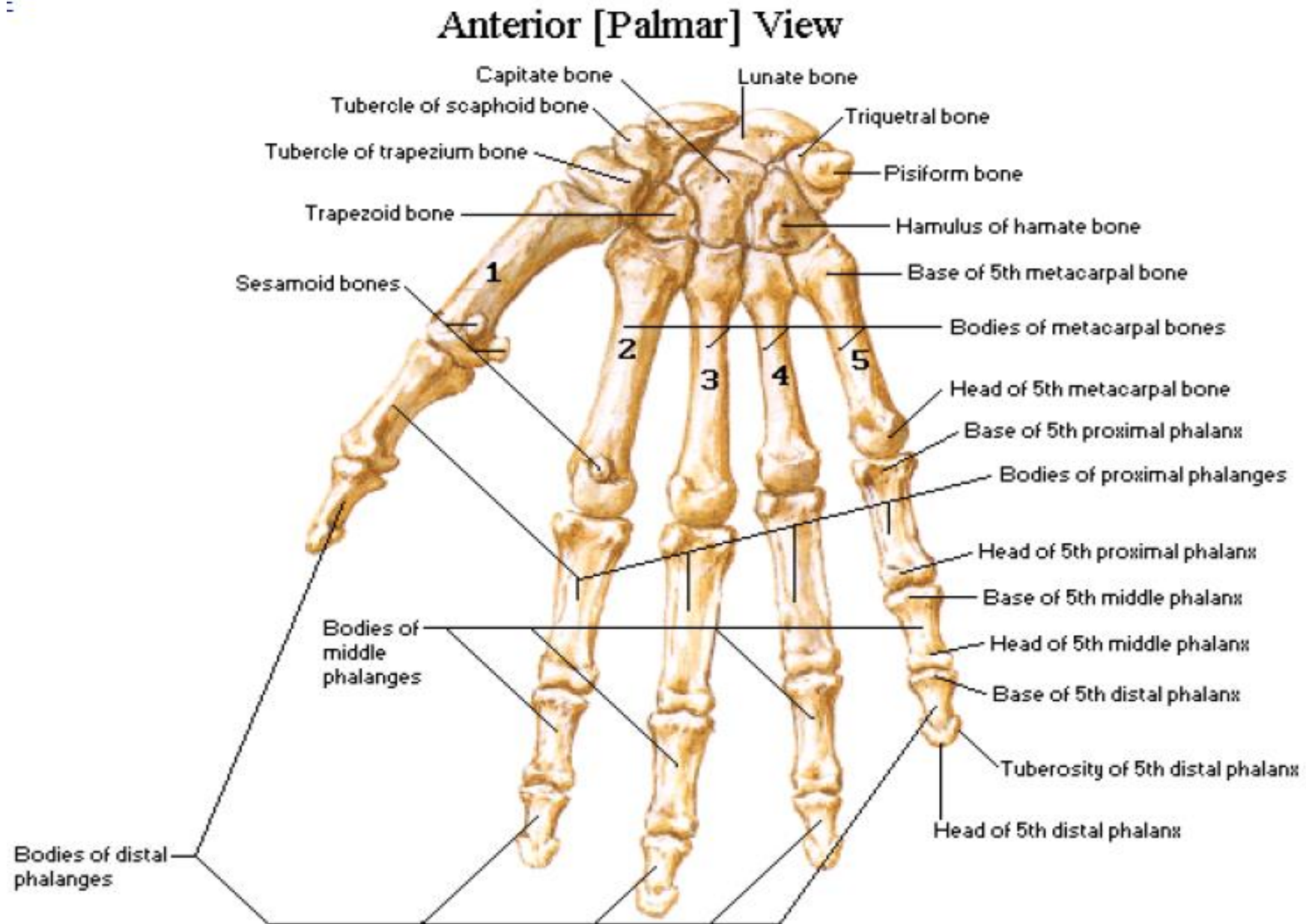
Carpometacarpal and Intermetacarpal Joints,,



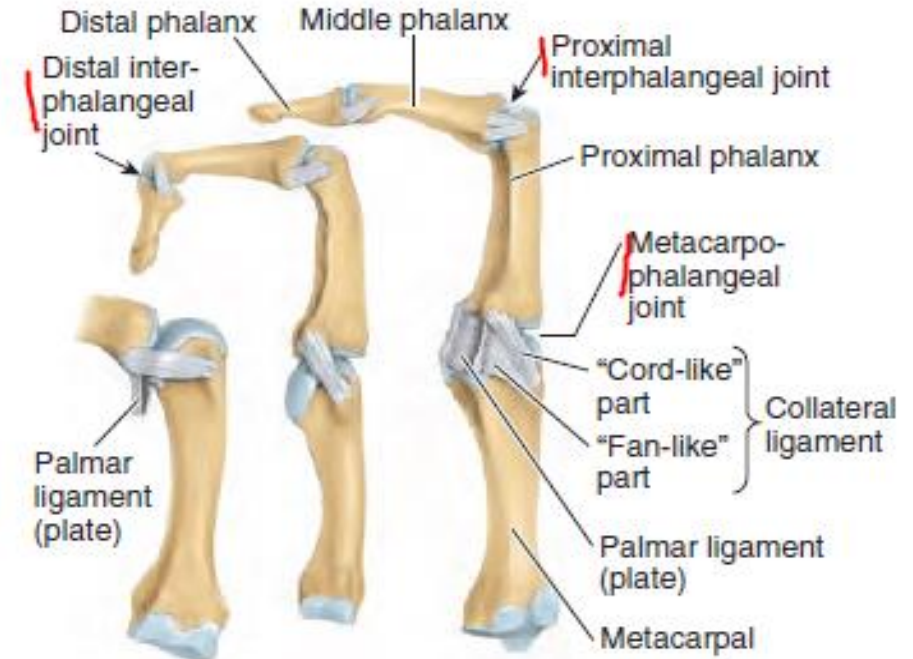
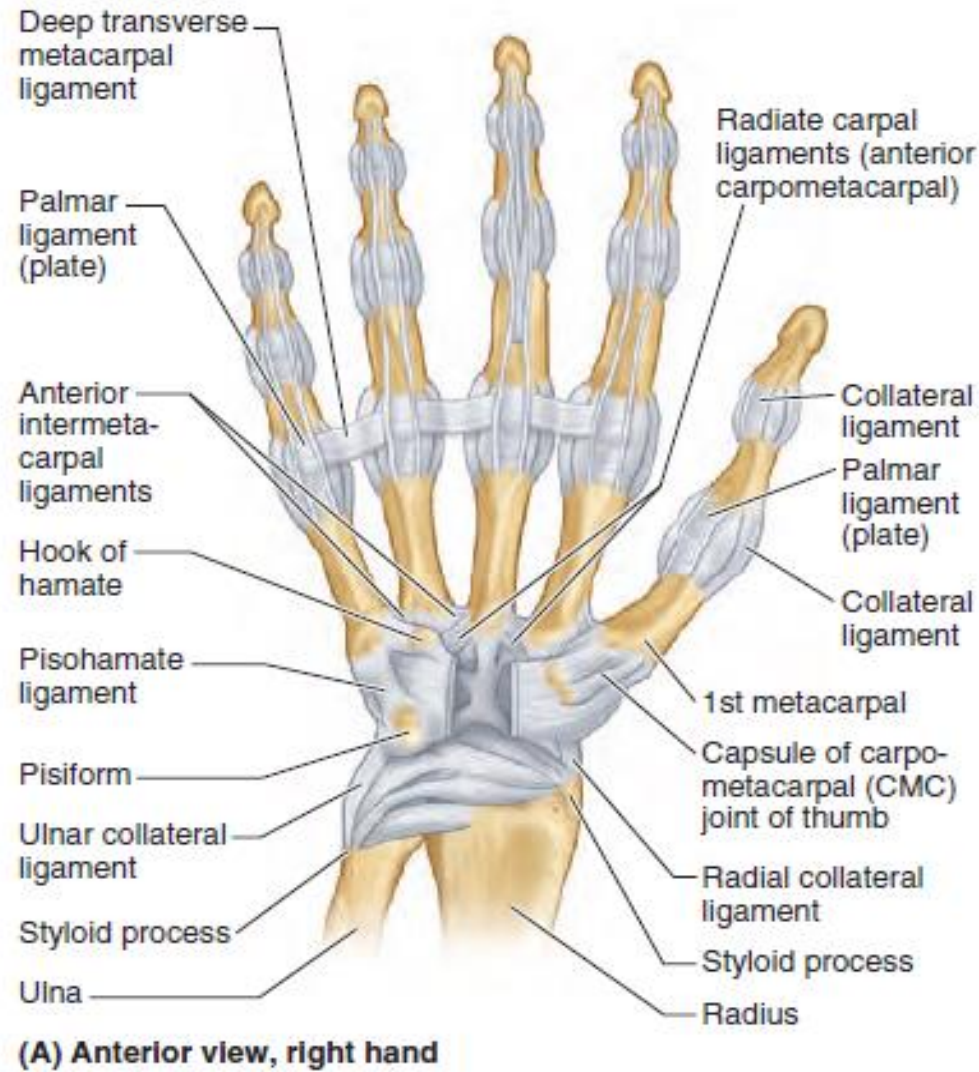
Metacarpophalangeal and Interphalangeal Joints

- The **metacarpophalangeal joints** are the condyloid type of synovial joint that permit movement in two planes: **flexion–extension** and **adduction–abduction**.
- The **interphalangeal joints** are the hinge type of synovial joint that permit **flexion–extension only**.
- The heads of the metacarpals articulate with the bases of the proximal phalanges in the MP joints, and the heads of the phalanges articulate with the bases of more distally located phalanges in the IP joints.

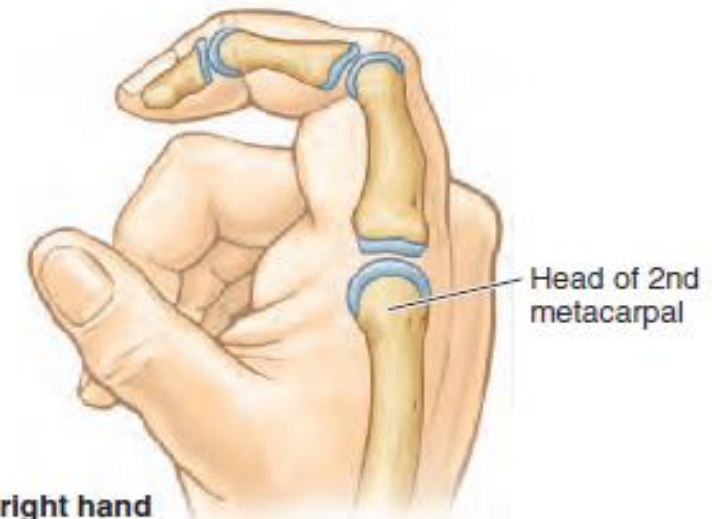
Metacarpophalangeal and Interphalangeal Joints,,



Metacarpophalangeal and Interphalangeal Joints,,,

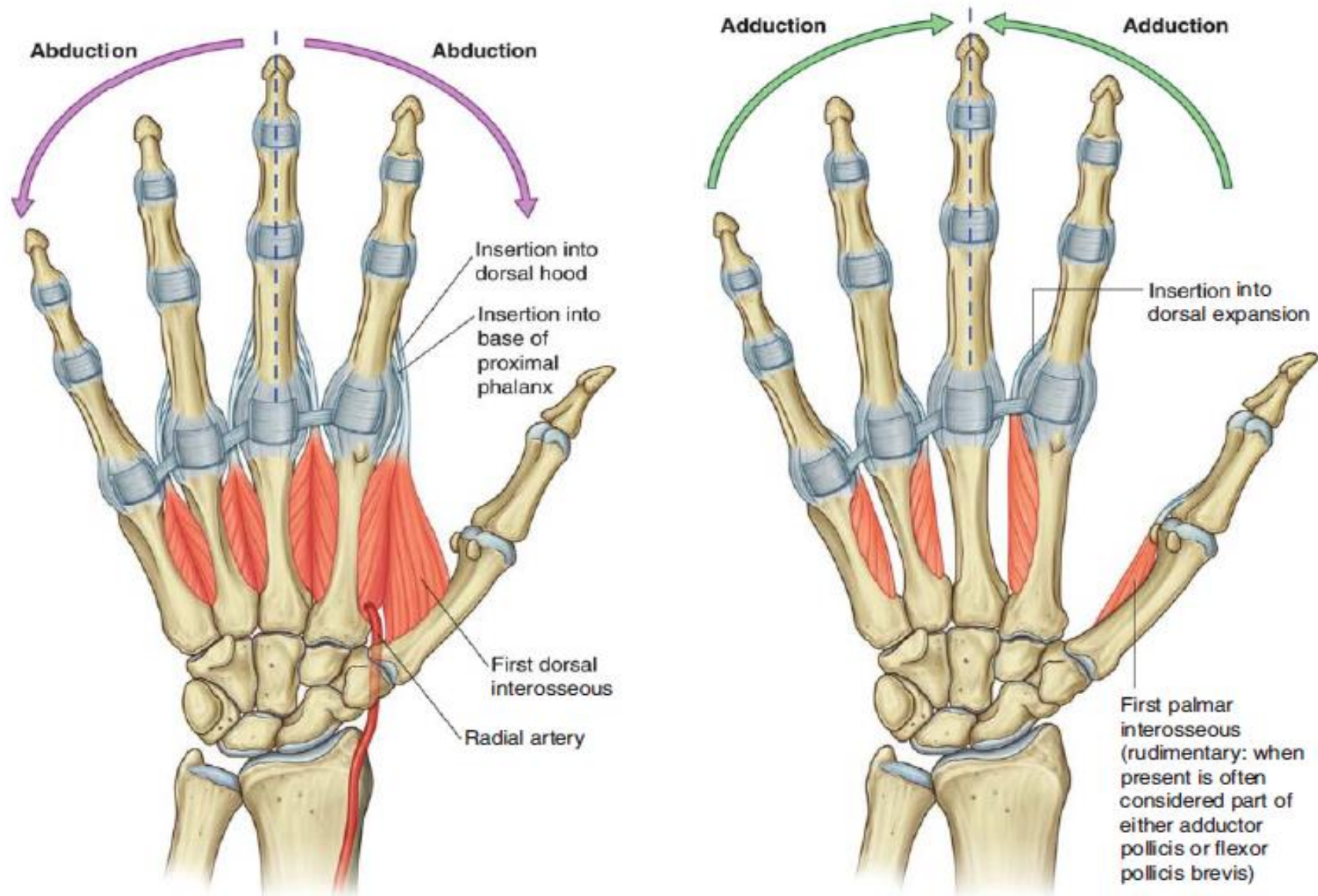


(B) Lateral view, right hand



(C) Lateral view, right hand

Metacarpophalangeal and Interphalangeal Joints,,,





**Thank you
for your attention**